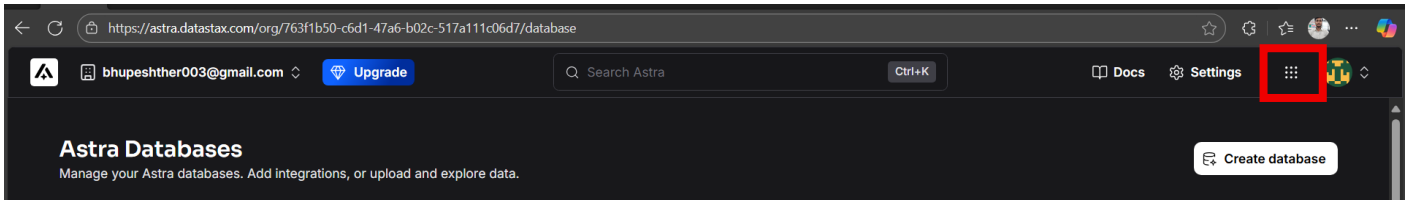
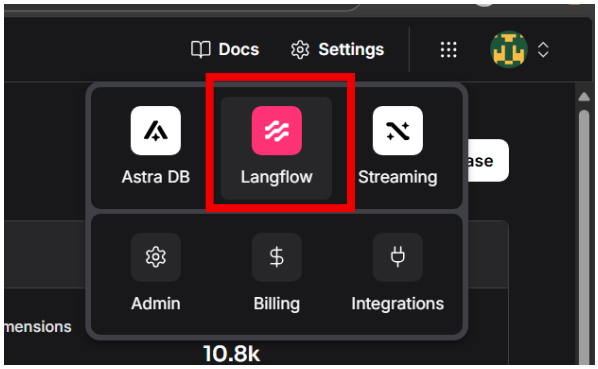
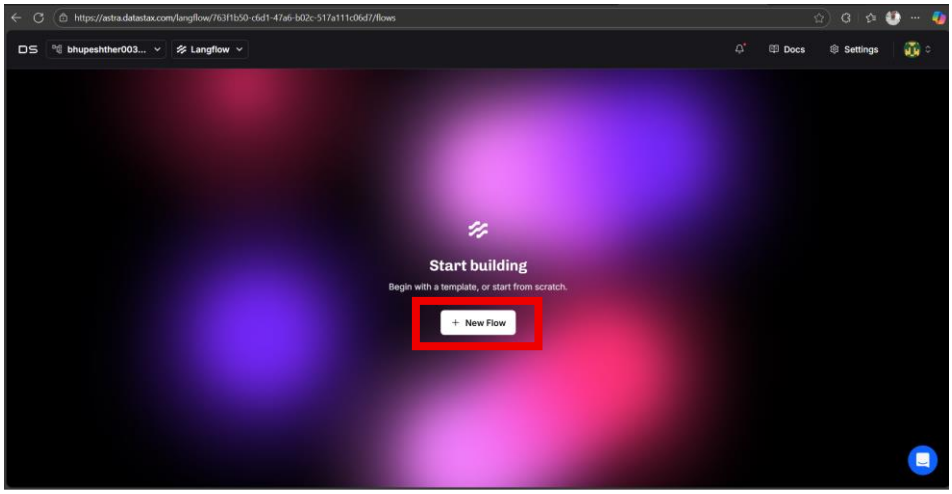
RAG agent with Astra DB using watsonx.ai

Create the astra DB account and click on right side square box and select langflow

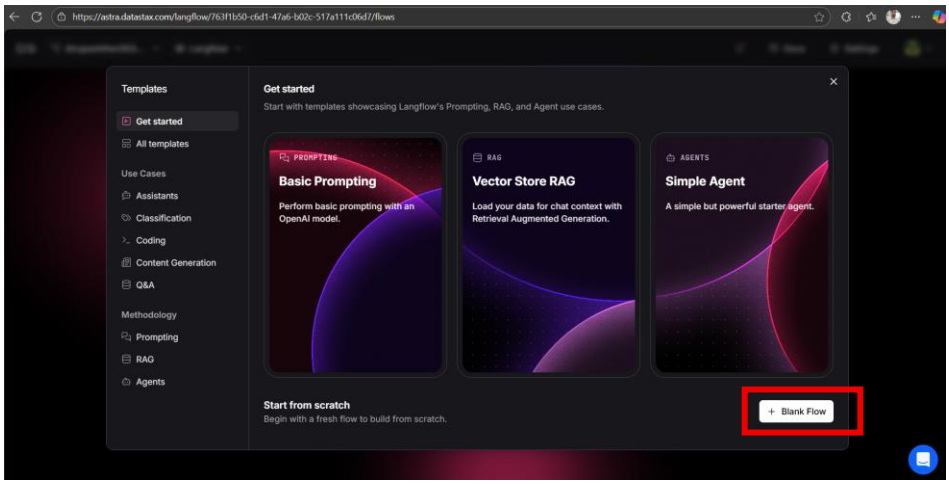




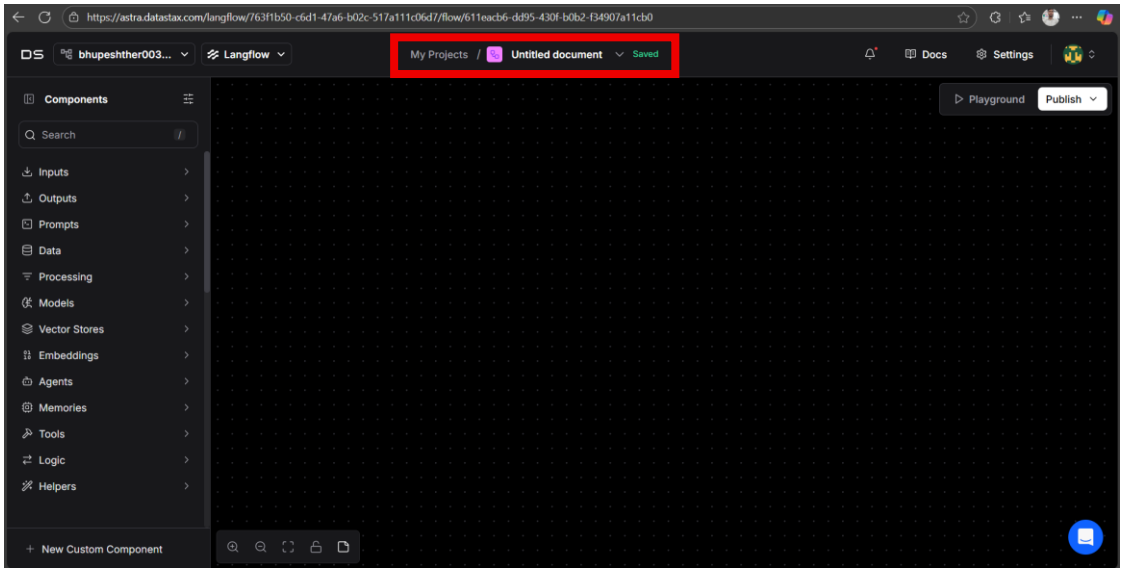
1. Open the langflow online.

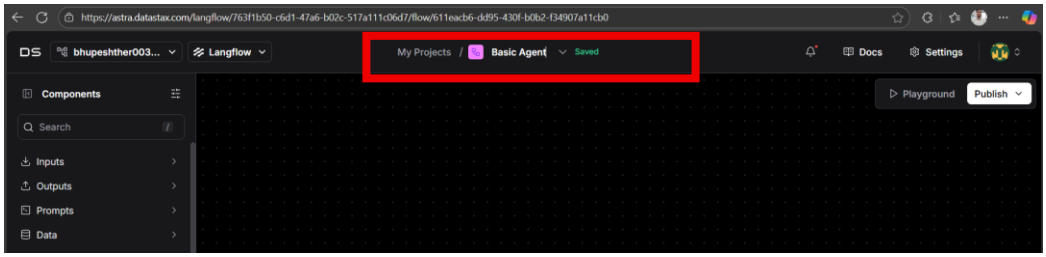


2. We need to create one project for this click on the **blank flow**

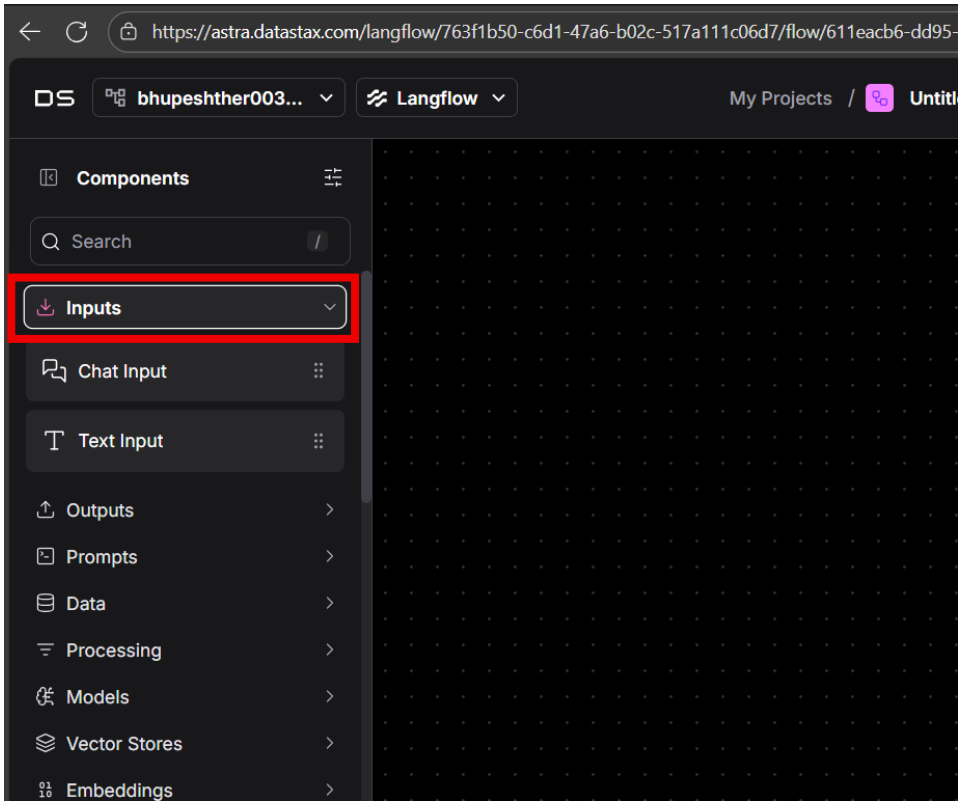


3. Now you are in the workspace where we are creating the our Agent. Change the agent name for this click one untitled document and change the name of your project

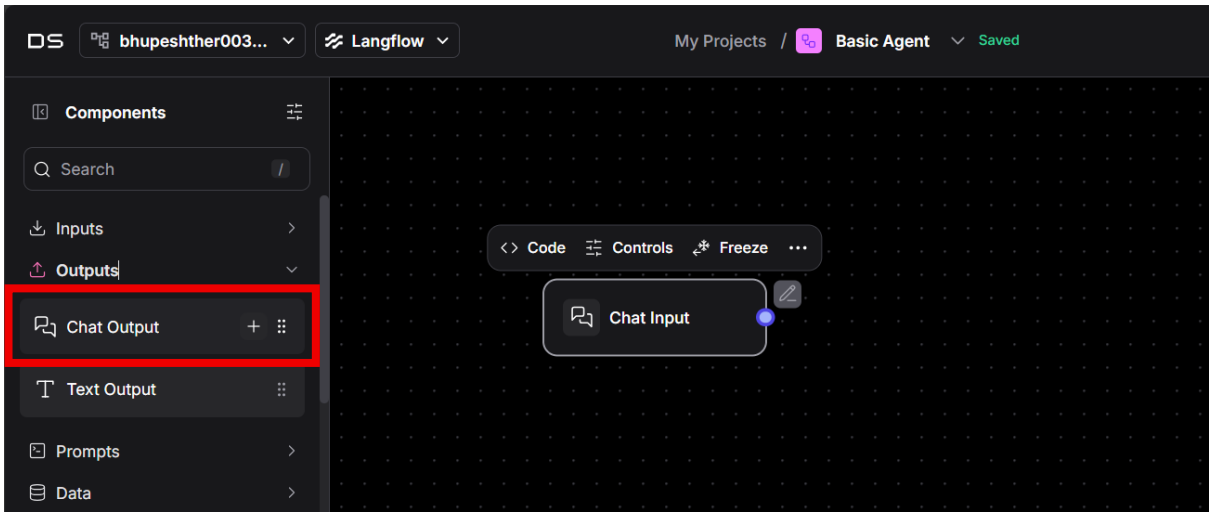


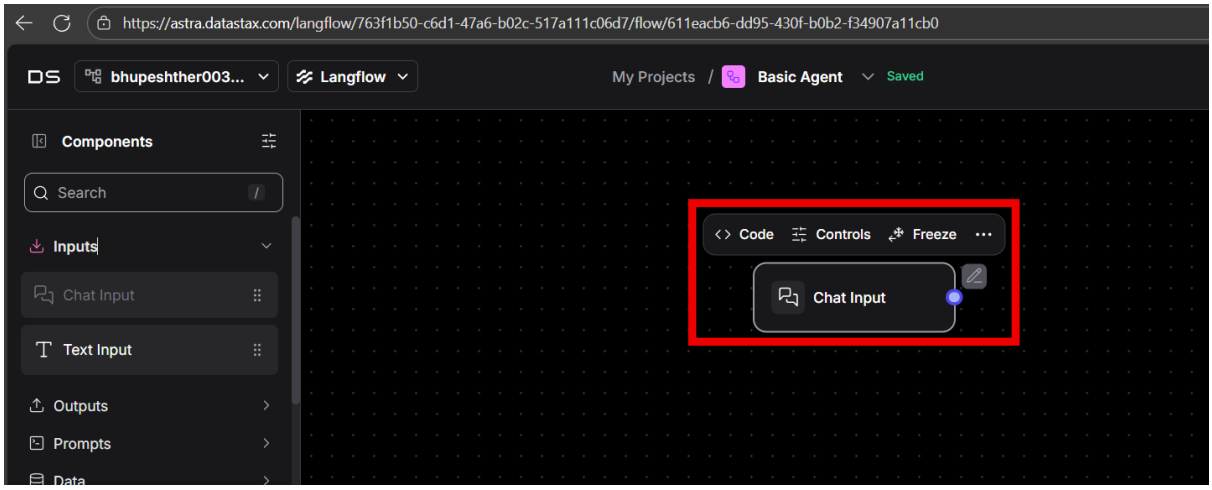


4. Before the starting we need to add input chat so click on the inputs and form that select the chat input.

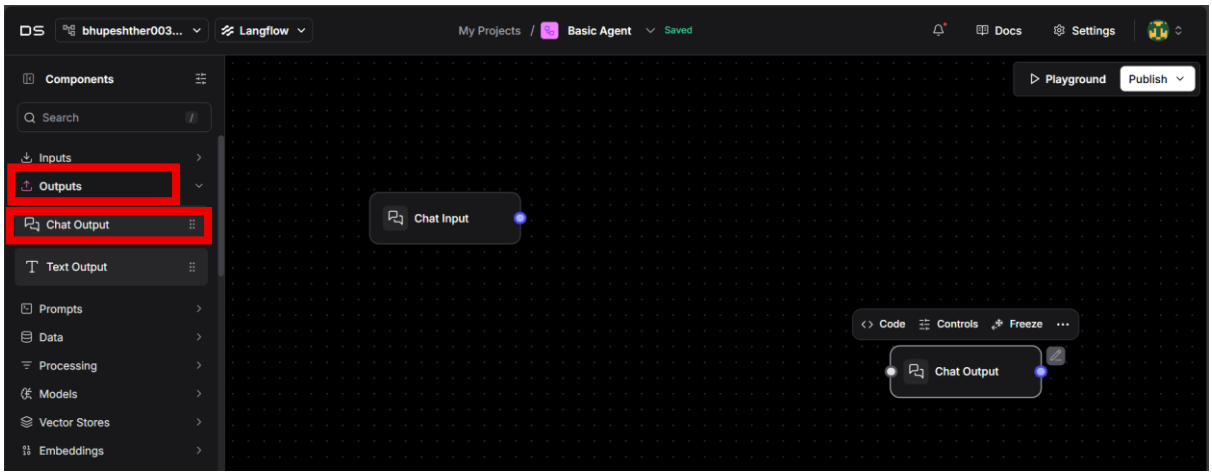


5. Click in + icon to add in the workspace (also you can do drag and drop)

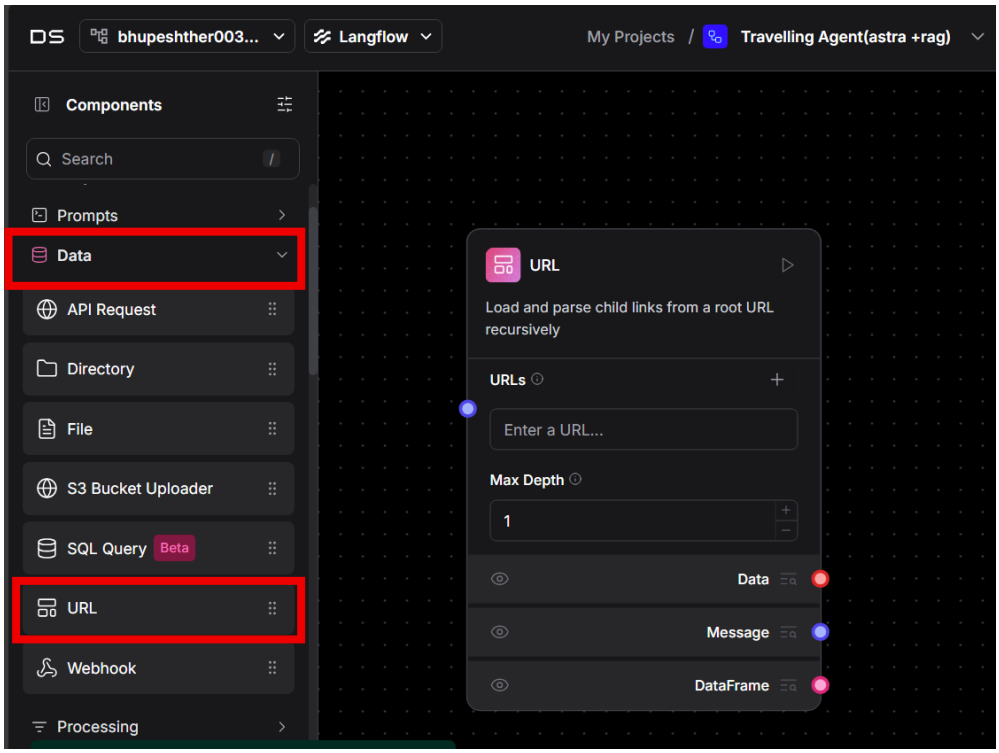




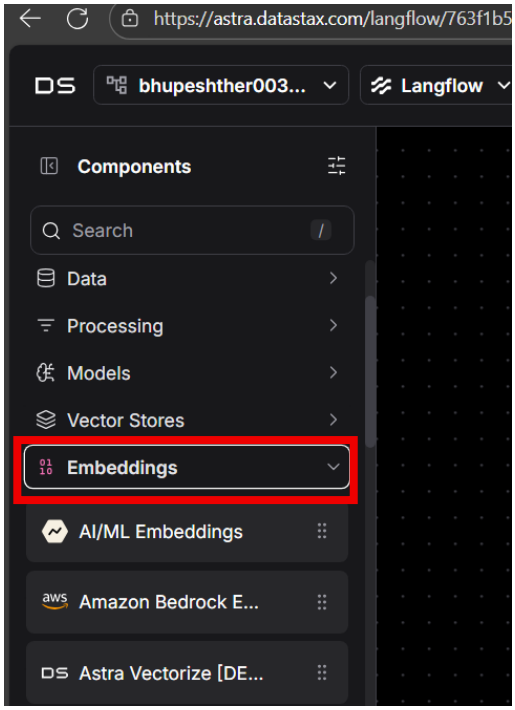
6. Same we need to add output as well so click on the output and select the chat output .



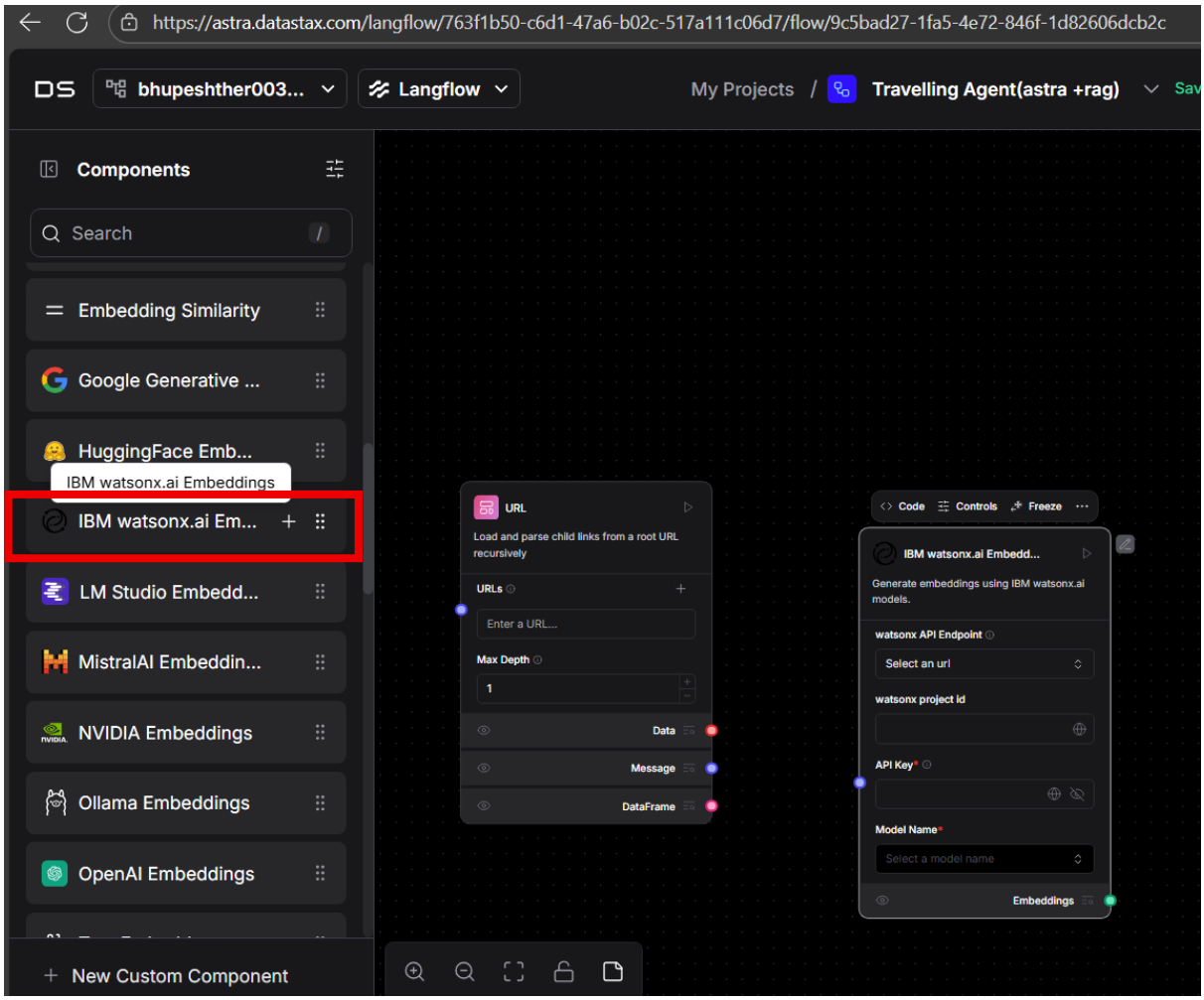
7. So u need to add url for adding the website to get up-to date information.



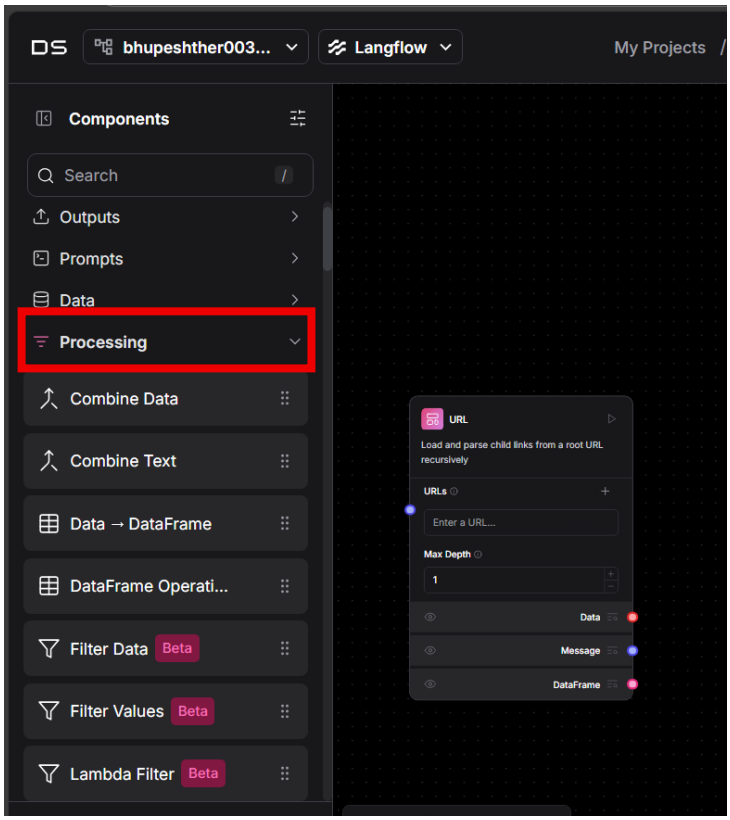
8. So we need to add embedding For that need to add embedding model.



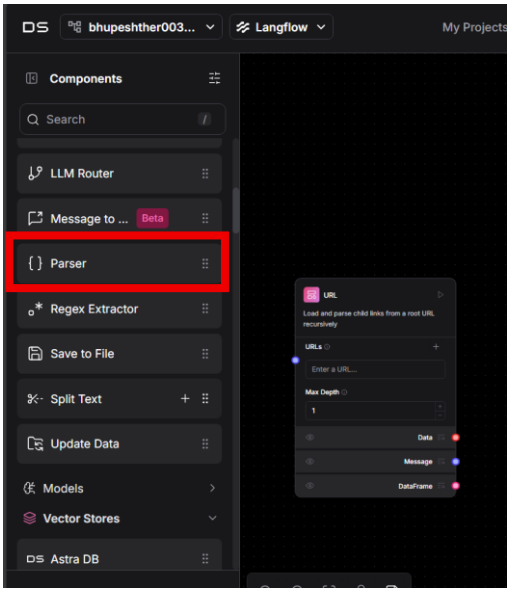
9. Form the list From the list we need to select IBM Watson.ai Embeddings



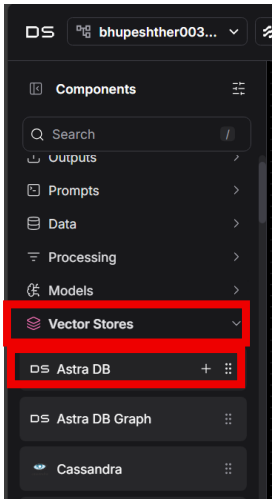
10. Now next for the parse go to the Preprocessing



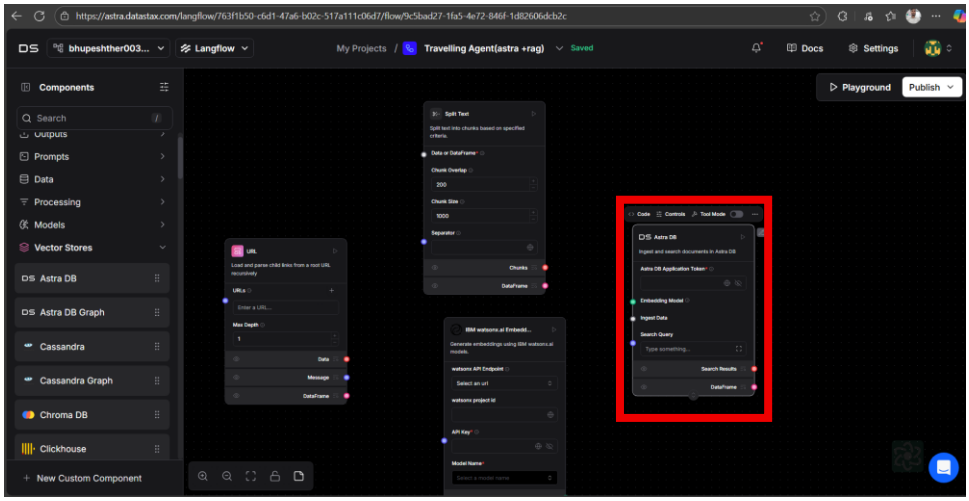
11. And from this select the parse



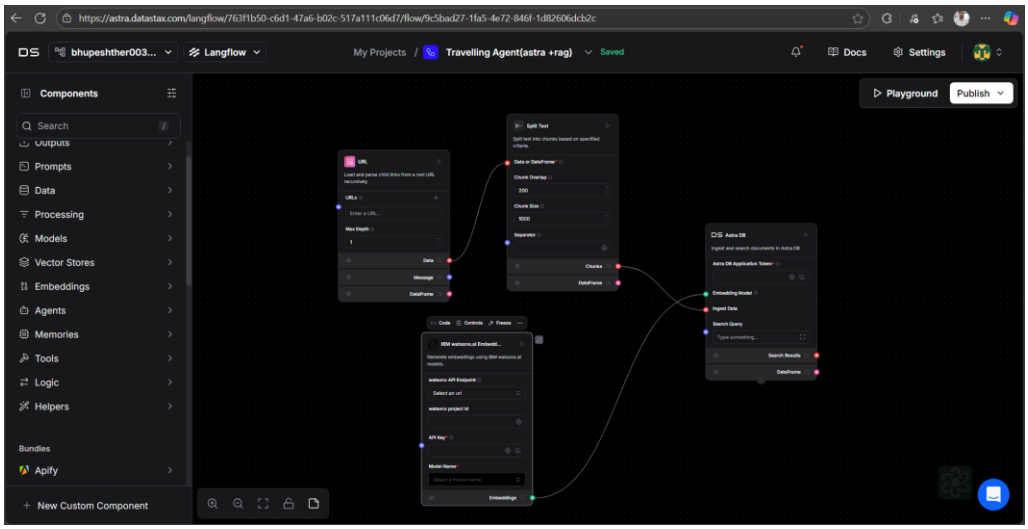
12. Now we need to select the Astra DB to store our information for this go to the vector stores and select Astra DB



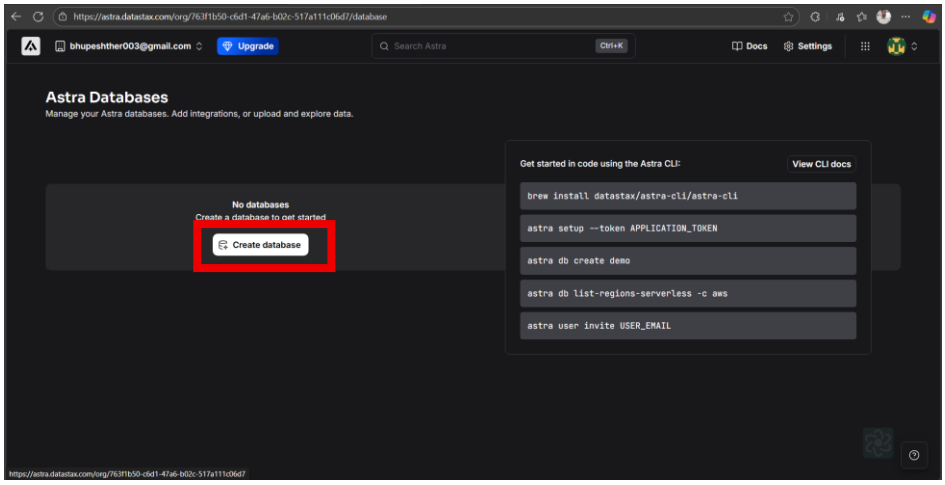
13. Now using drag and drop add in the workspace



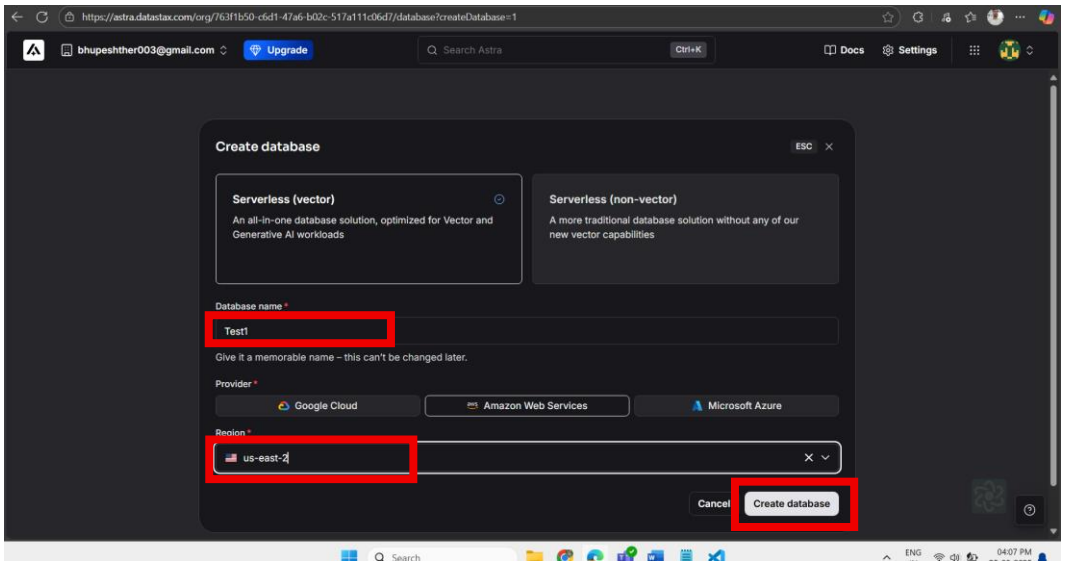
14. Now make connection for this follow below diagram



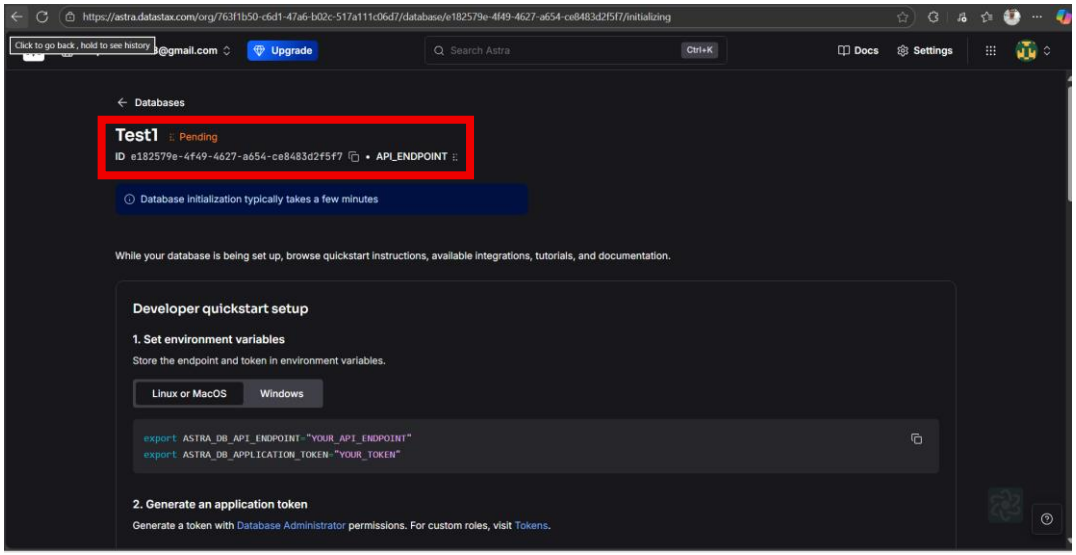
15. Once its done now go to the astra db to create Vector database and click on the create database option



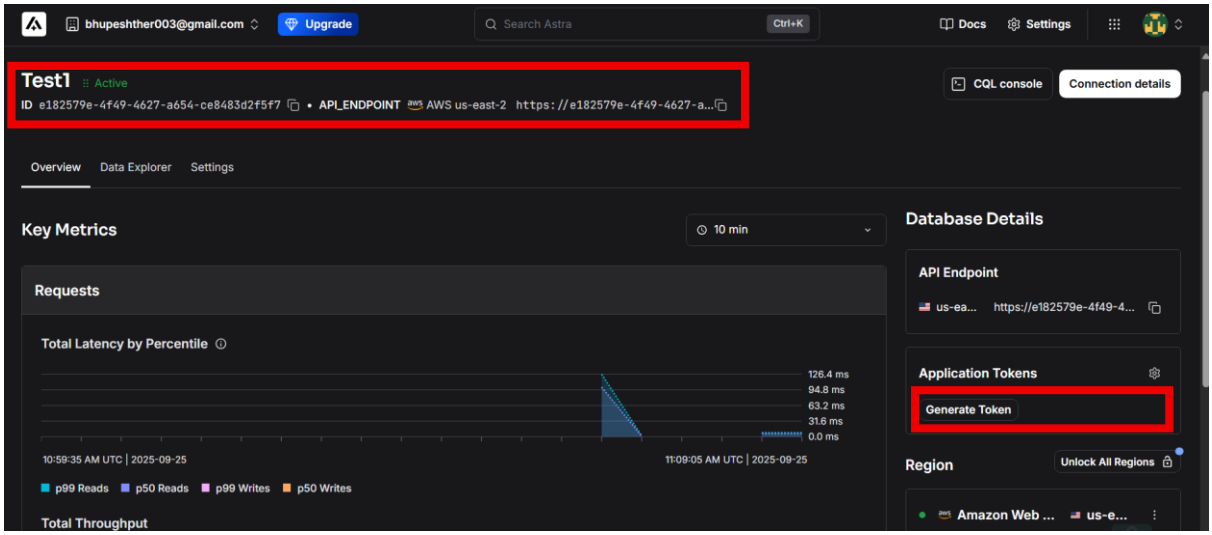
16. Give the name of the database and select region as **us-east-2** and click on the create



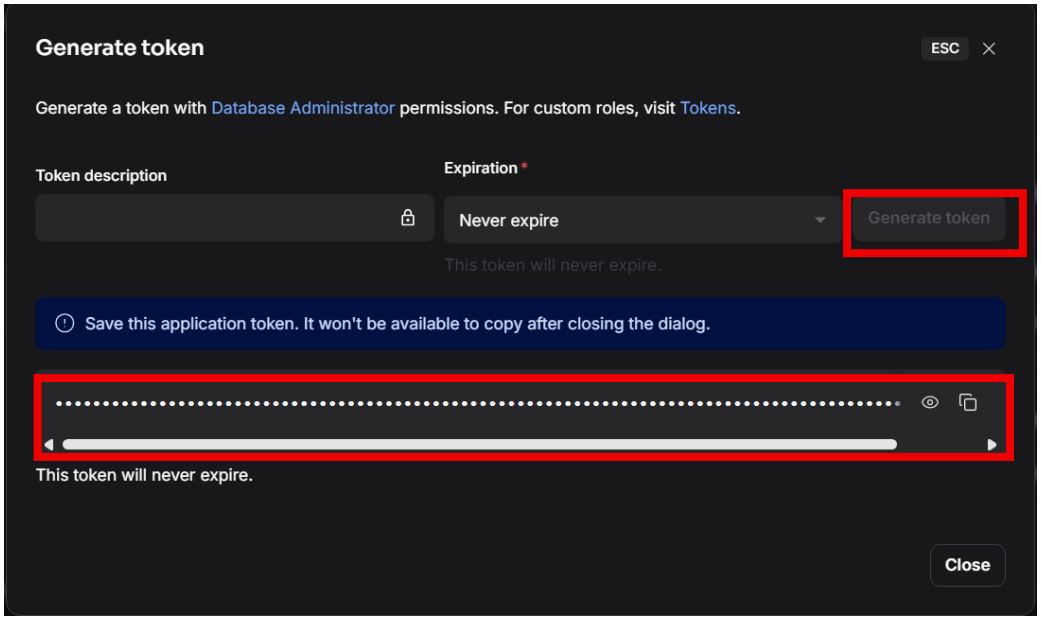
17. Now its will take couple to time to create the database wait for while



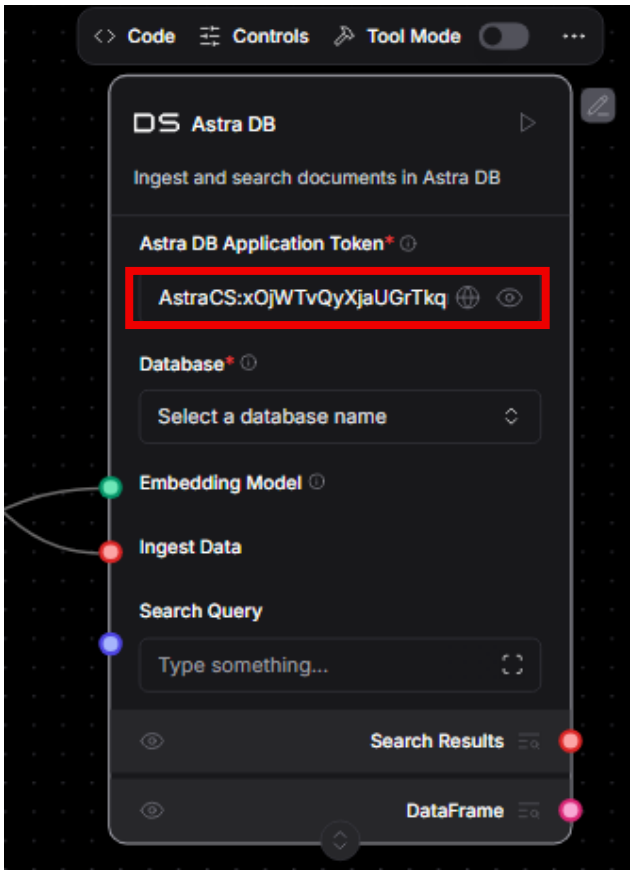
18. Once its done in your right side you getting the option for **Generate Token** click on that



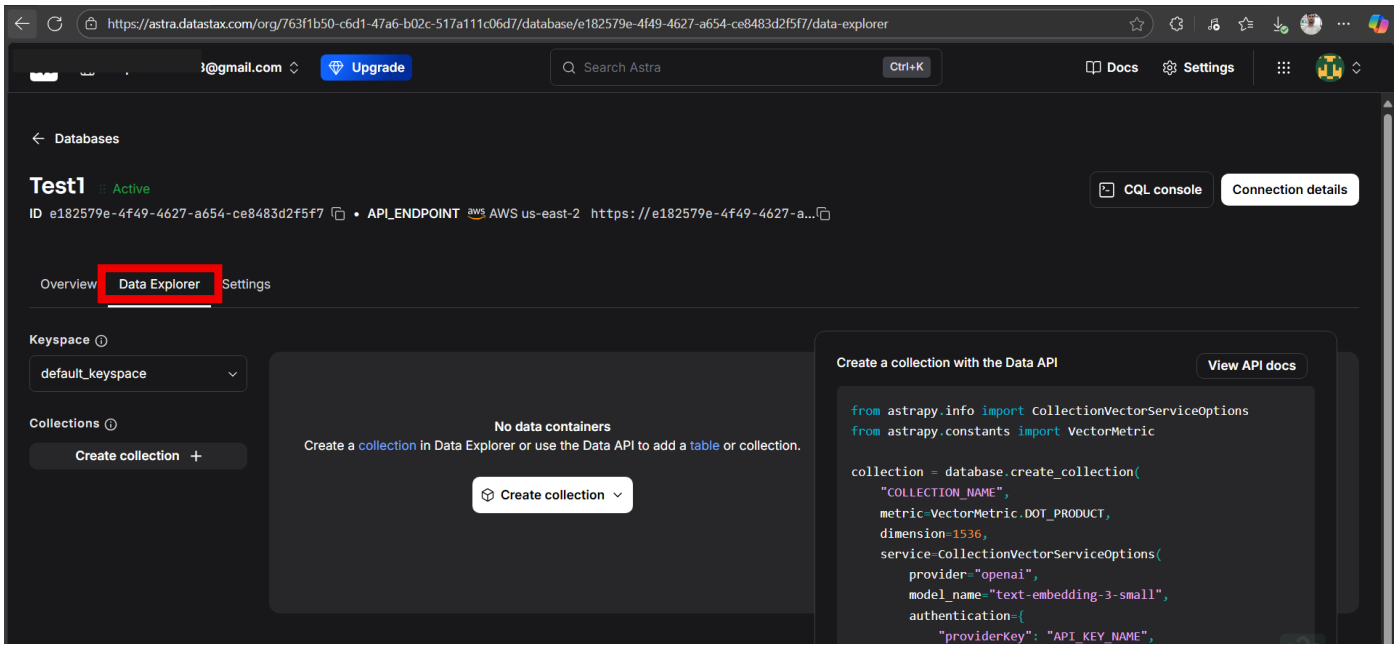
19. Click on generate token and your token will be generated of that one and paste somewhere because in hole project we need that one



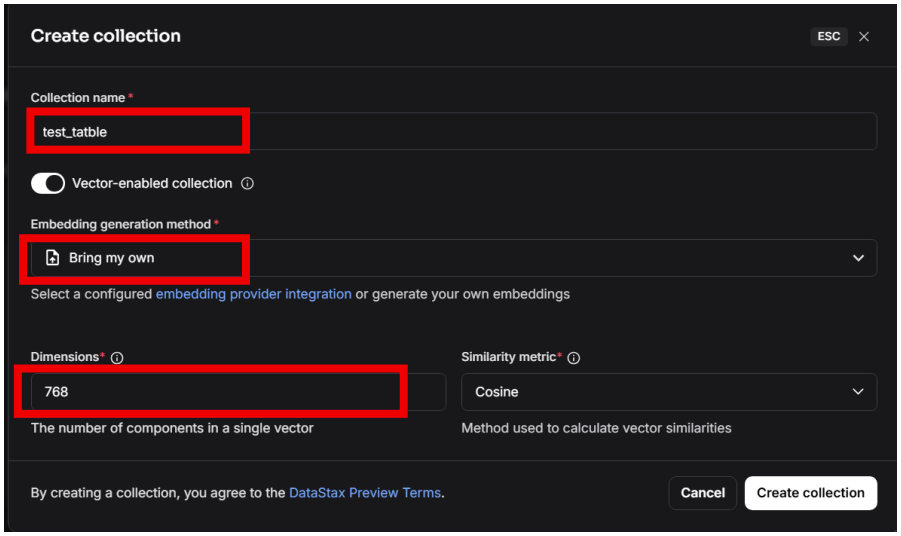
20. Now come to langflow and and that token in the Astra DB Application Token



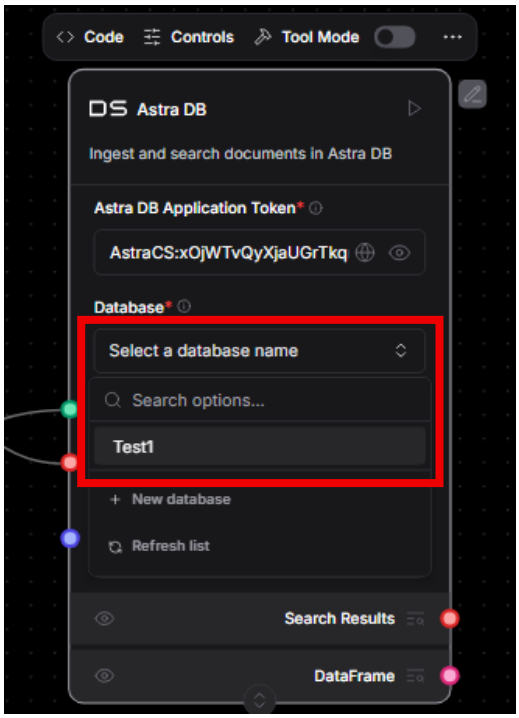
21. Once its done come back to Astra DB and now create one collection over here for this click on create collection button



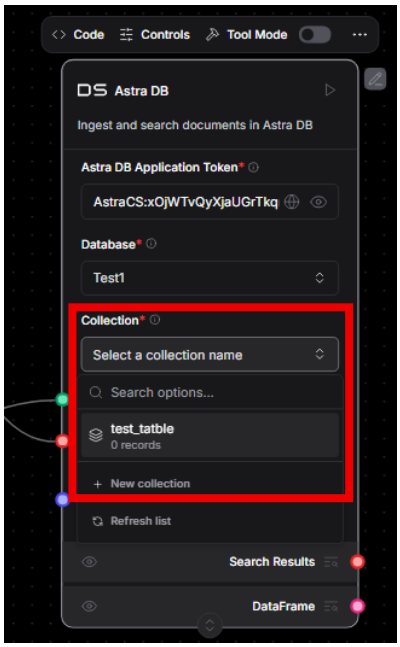
22. Give the name of **collection** and embedding generation method as **bring my own** and dimensions in **768** and click on create collection

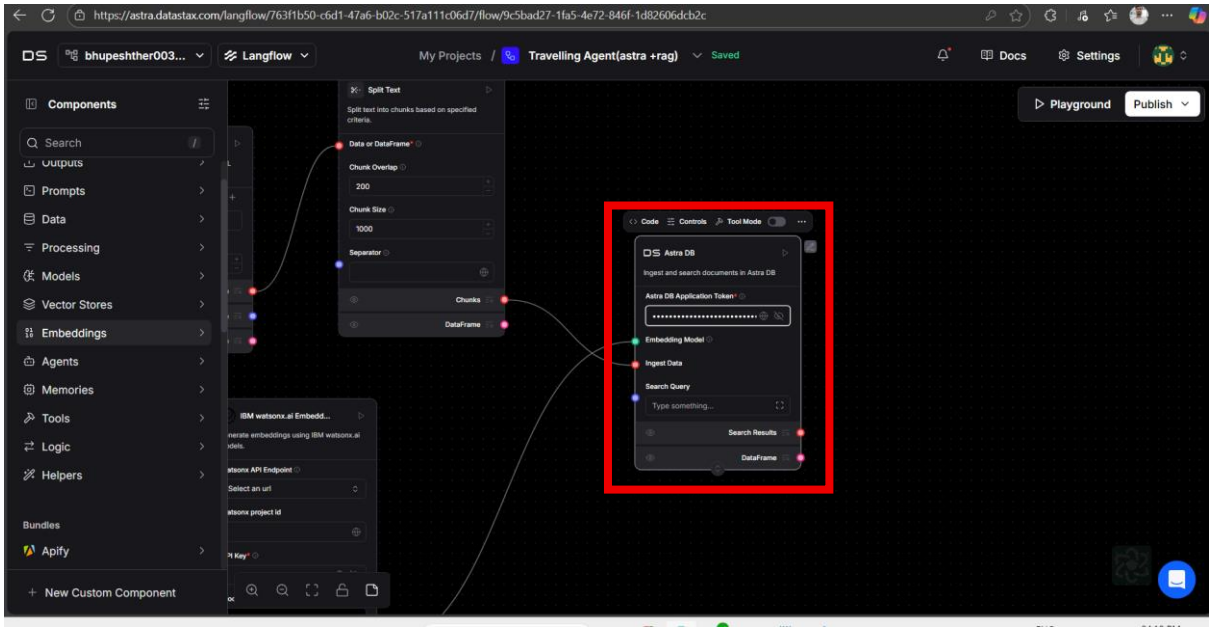


23. Once its done Now go back on langflow and select the database and after that select the Collection and you created

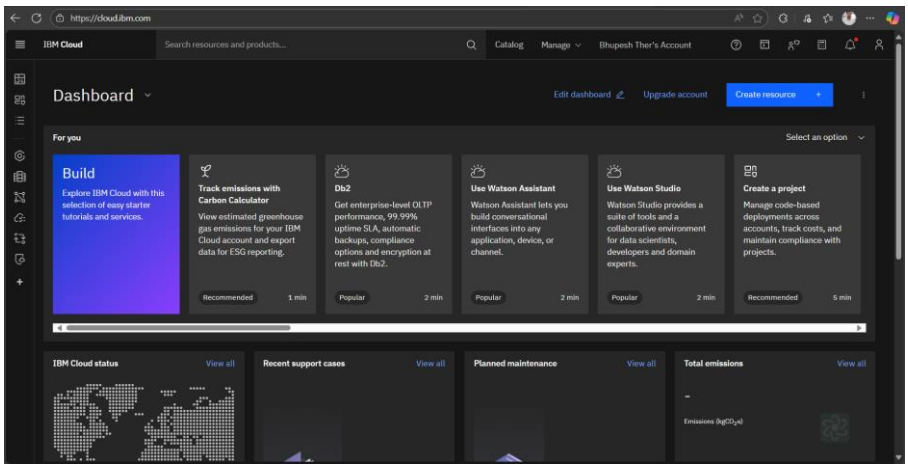


Final setup looks like below image

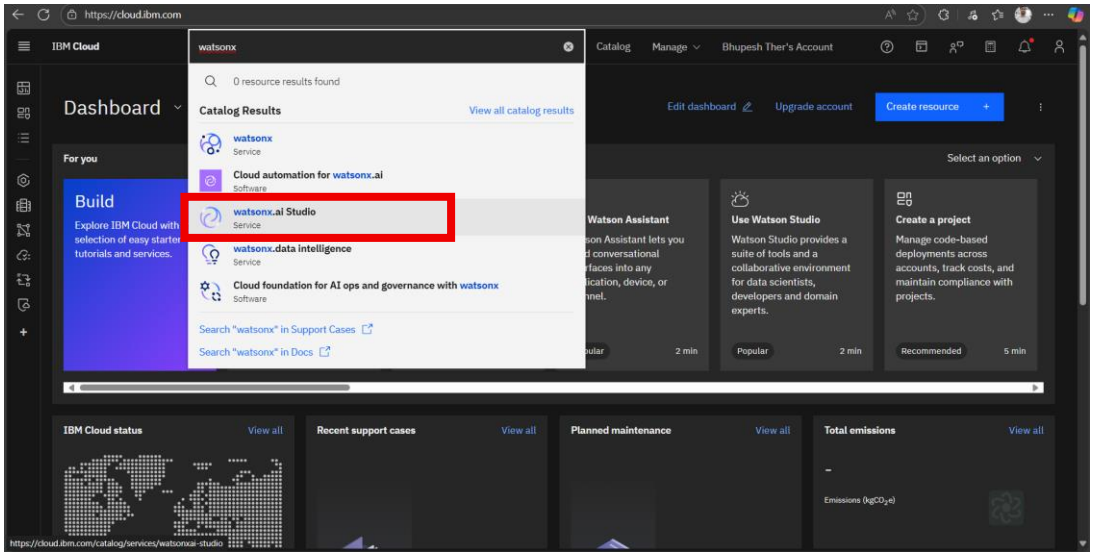




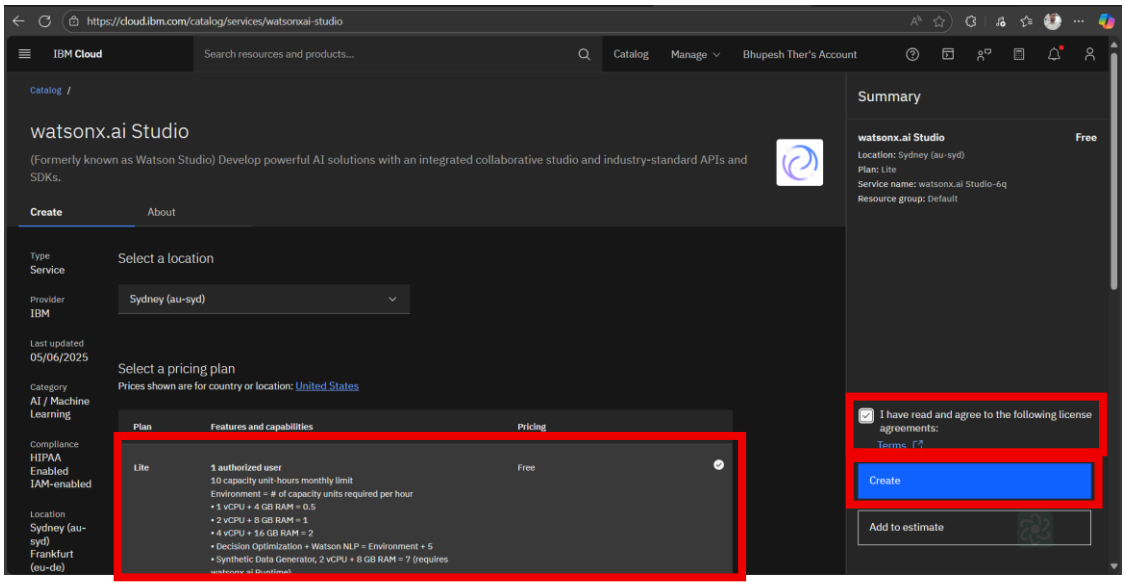
24. Now next for the embedding model we need project id and api for this go to ibm cloud



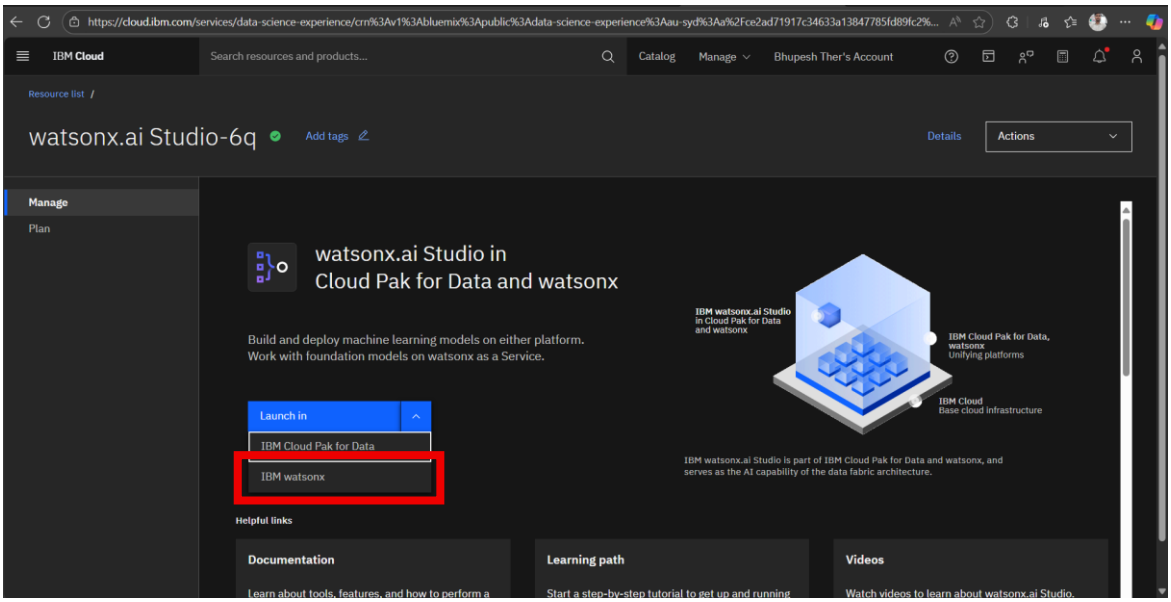
25. Go to search bar and search for the IBM watsonx.ai and click on that



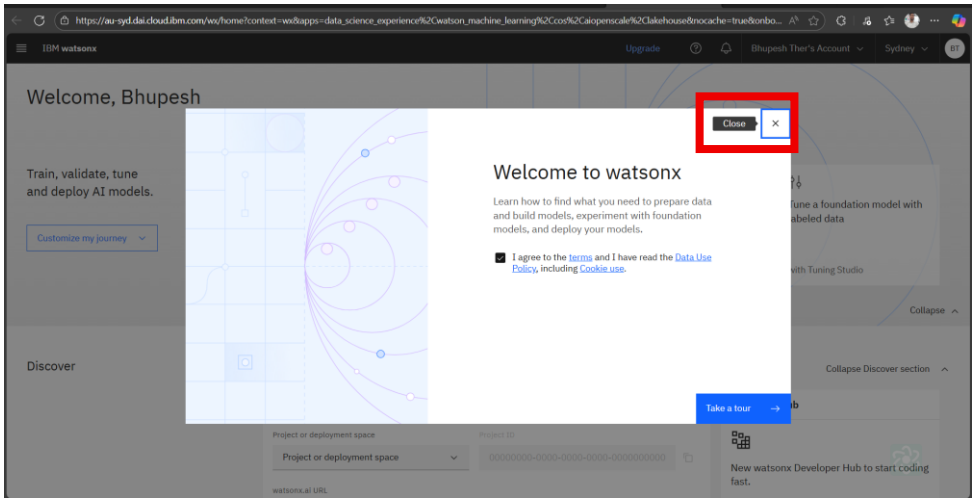
26. Now you need to create IBM watsonx service so make right side [] square box and click in the create



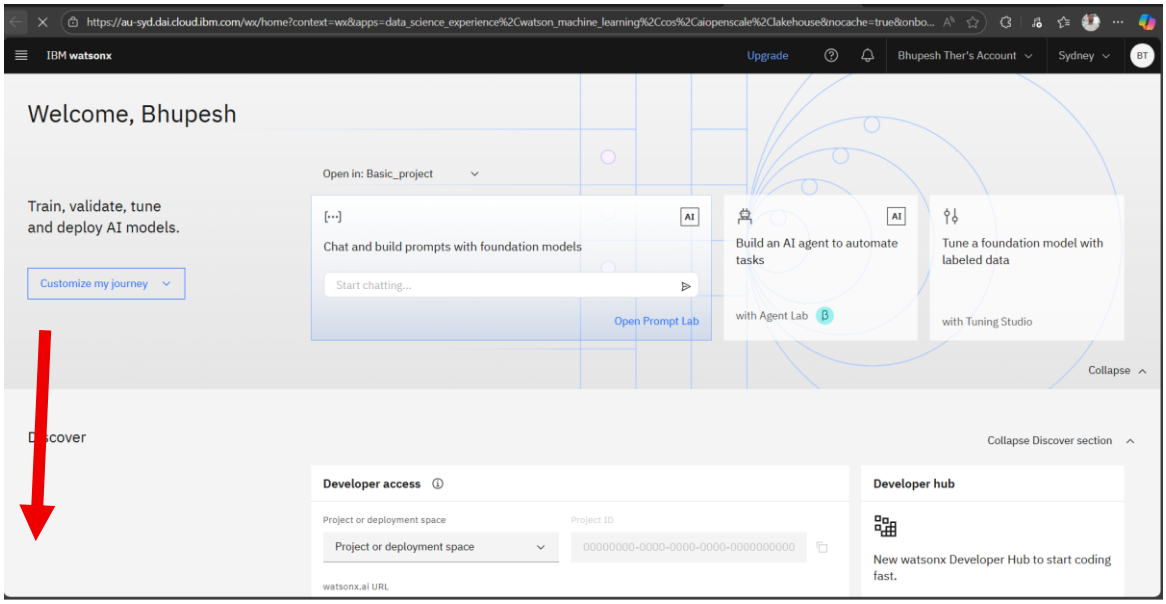
27. Once its done they will redirect you on the lunching page and click on the dropdown and select IBM watsonx

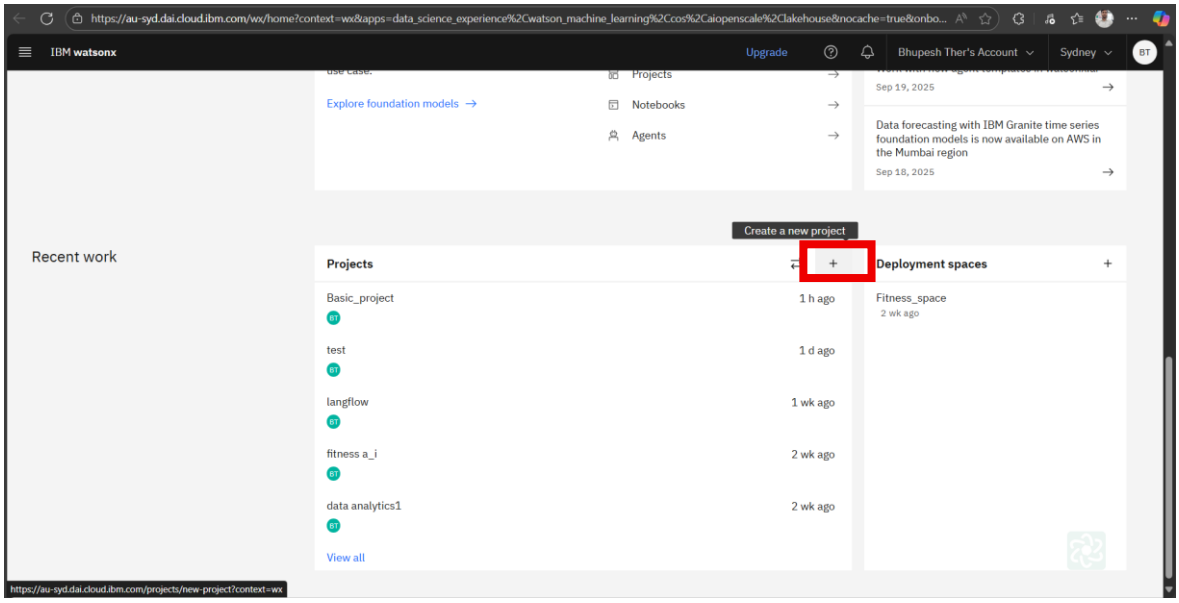


28. Now you will this on this page click on the X cross arrow and close this tour

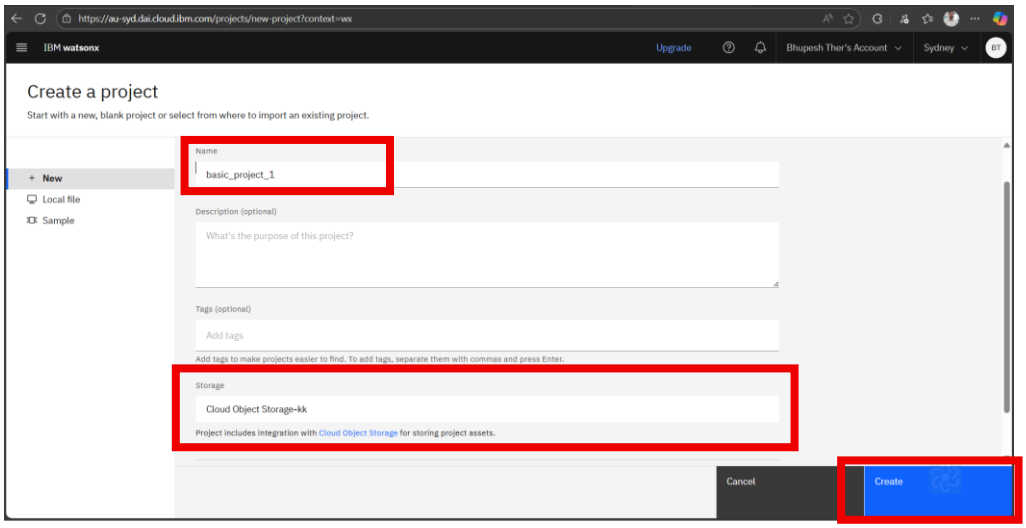


29. Scroll up this page little bit more and come to projection section and click on the + icon and create new project

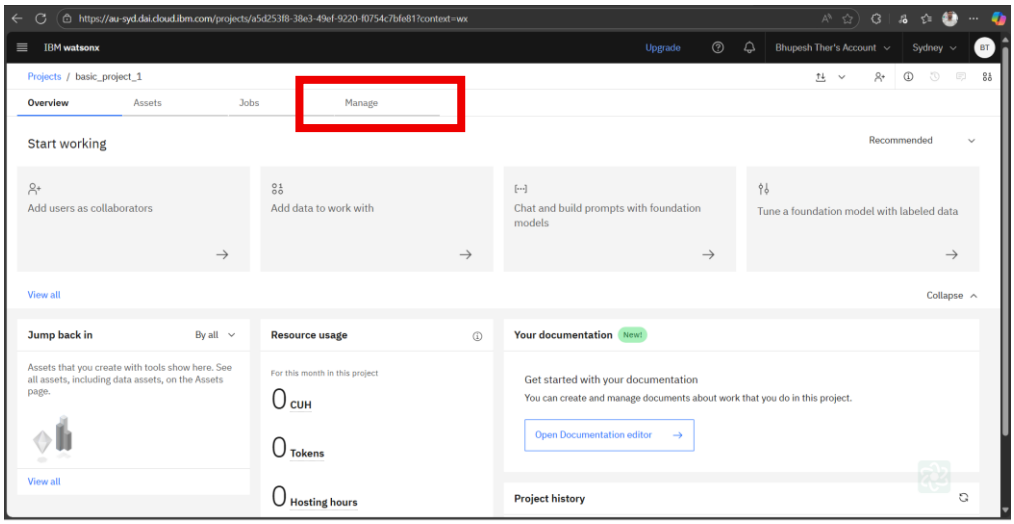




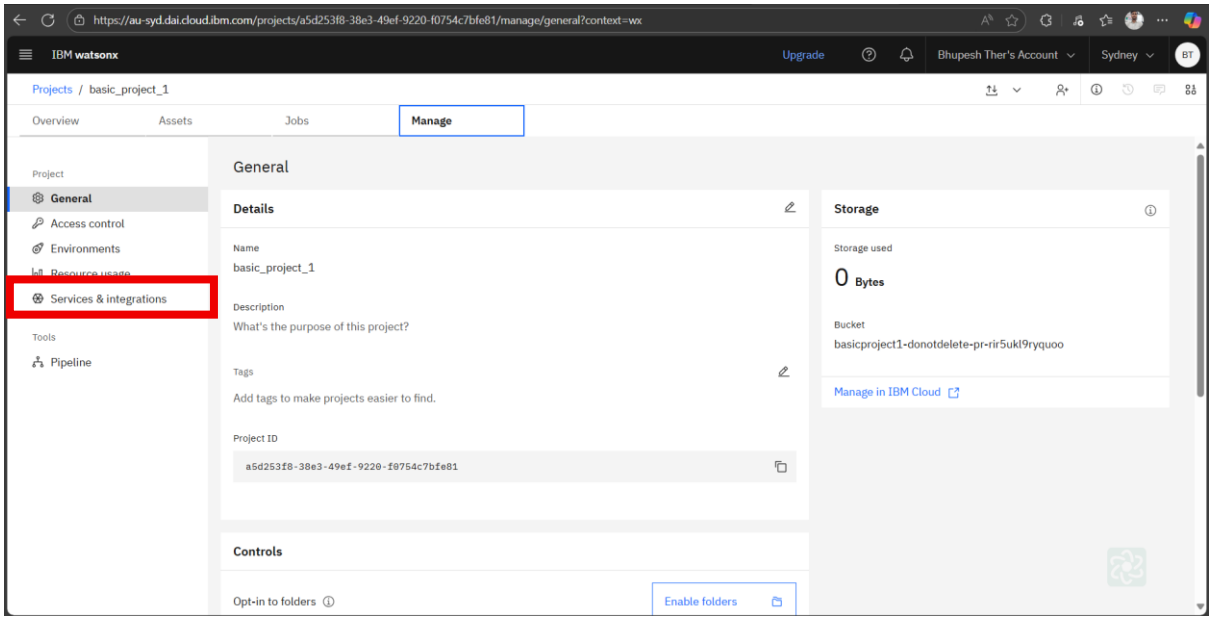
30. Give the project name and click on the create



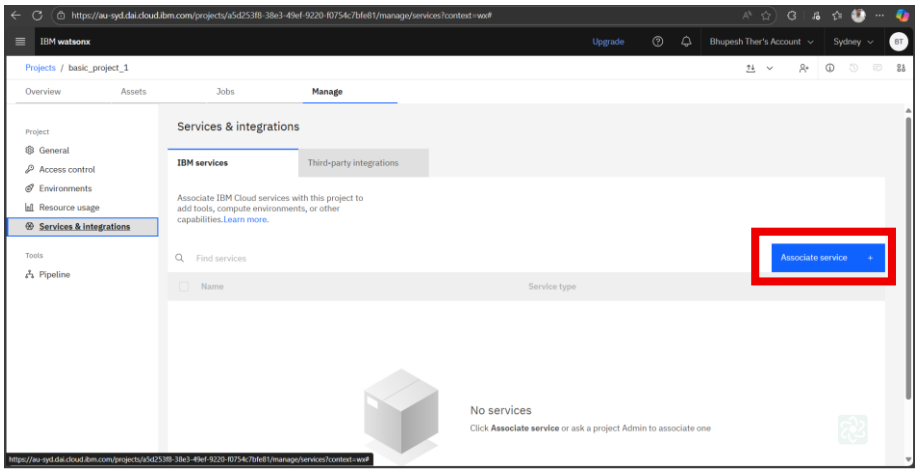
31. Now you need to go manage



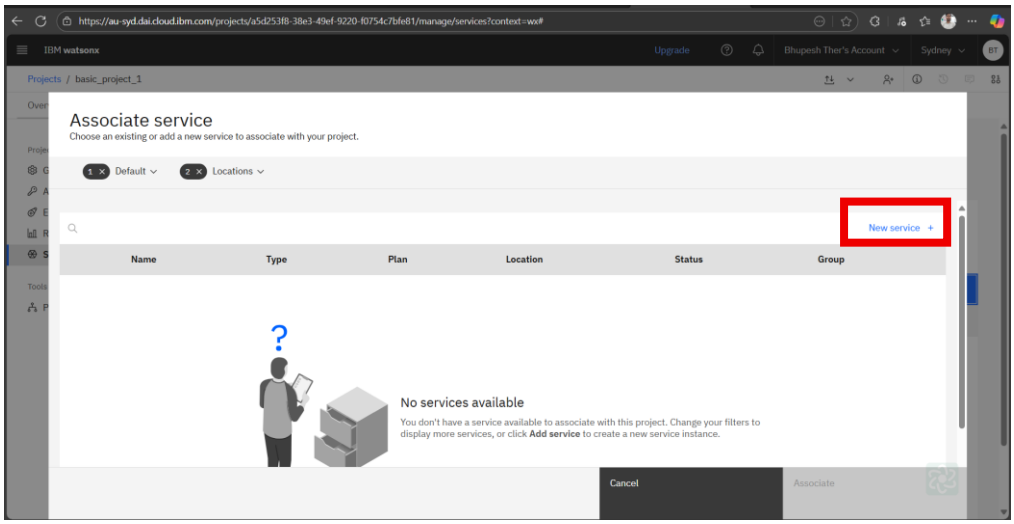
32. Now click on services & integrations



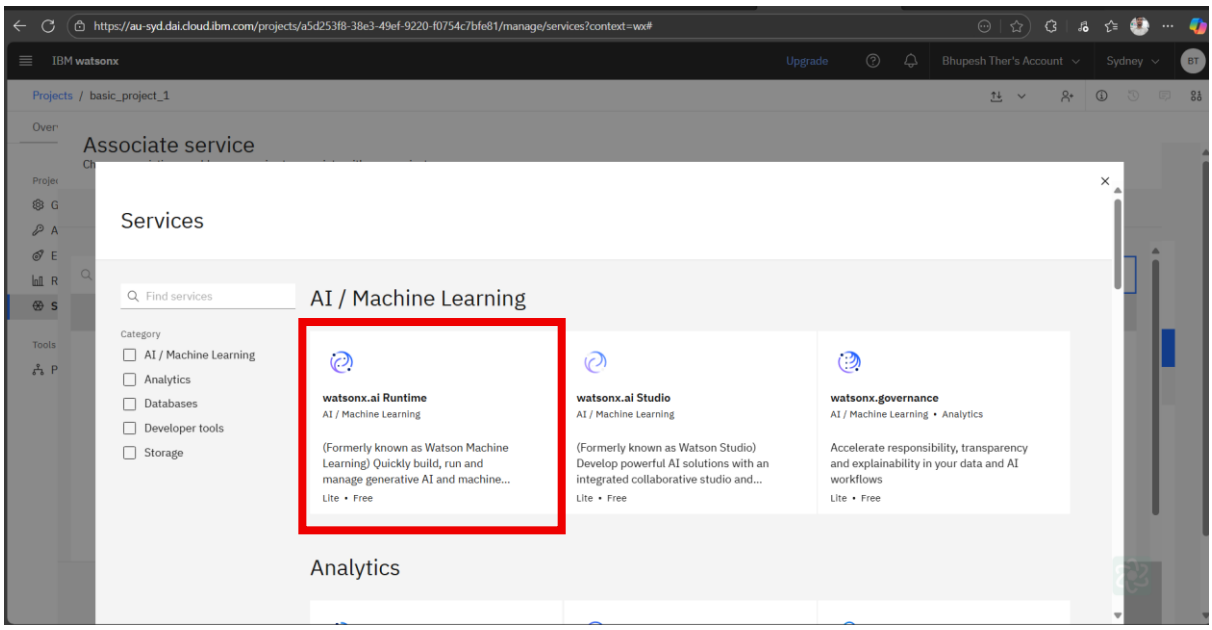
33. Click on the associate service



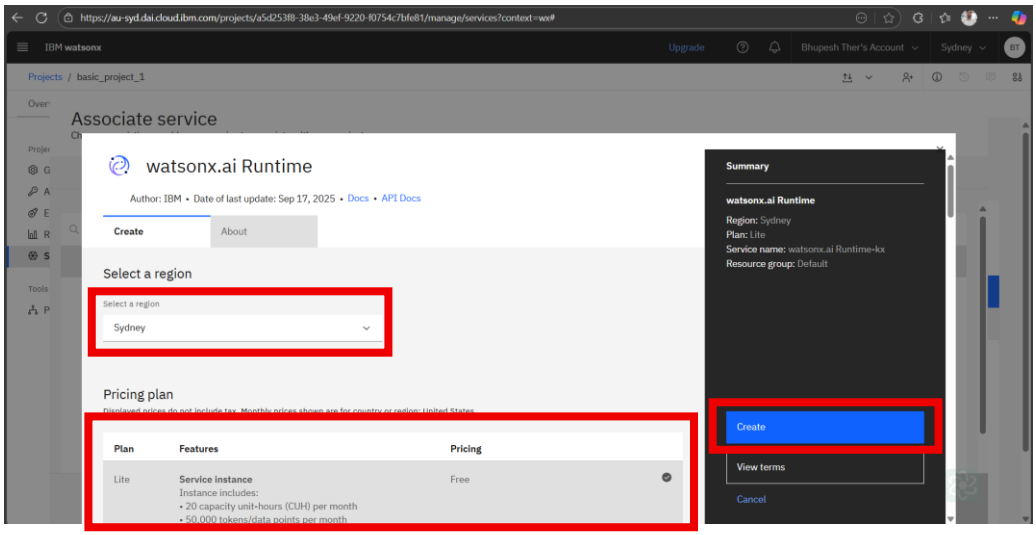
34. Click on new Service



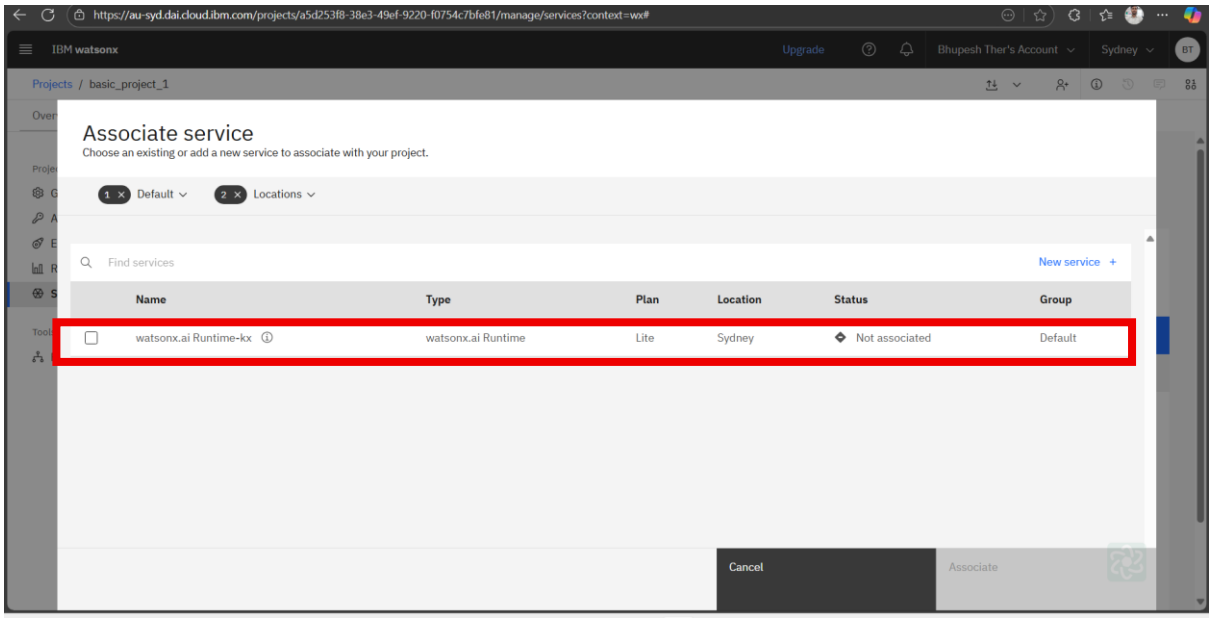
35. Now click on watsonx.ai runtime

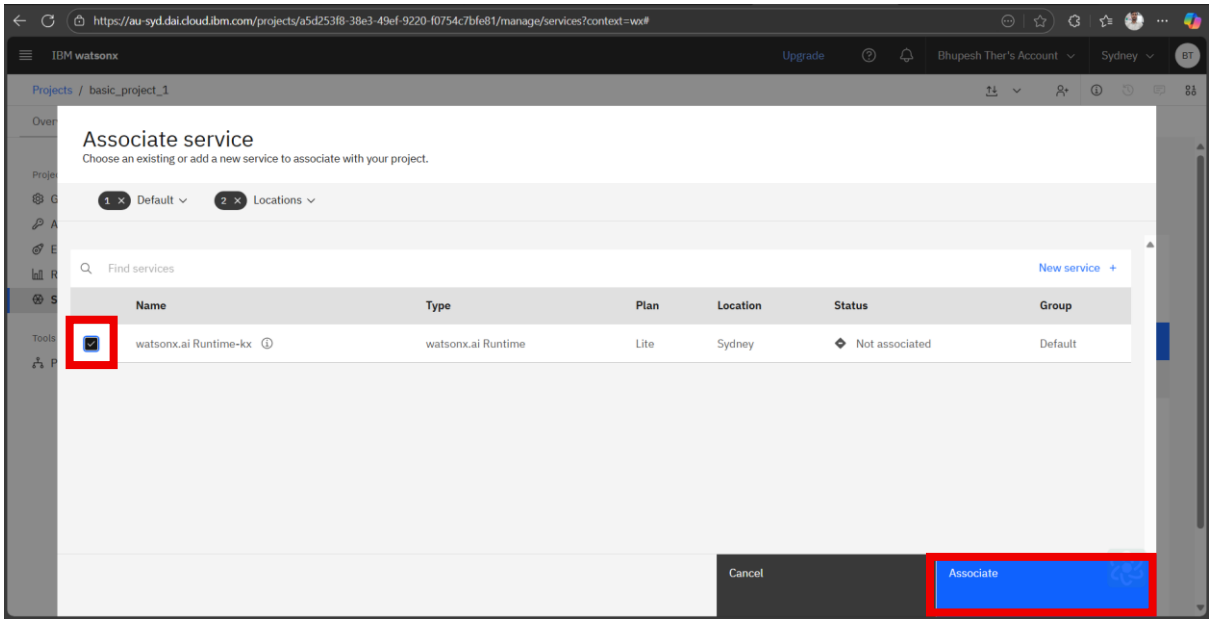


36. Assing the location and select lite plan and click on create

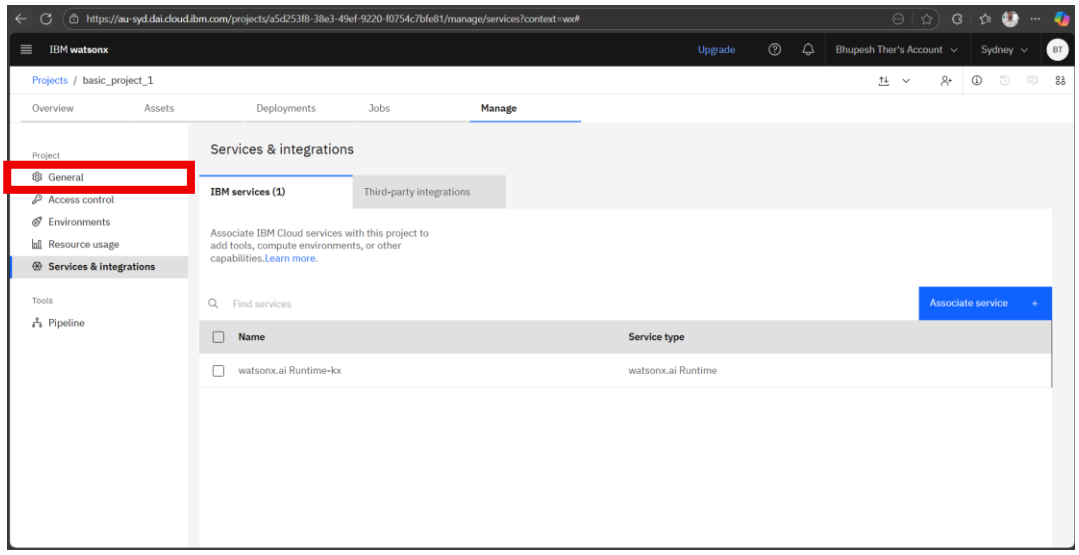


37. Once service created mark check box and click on associate

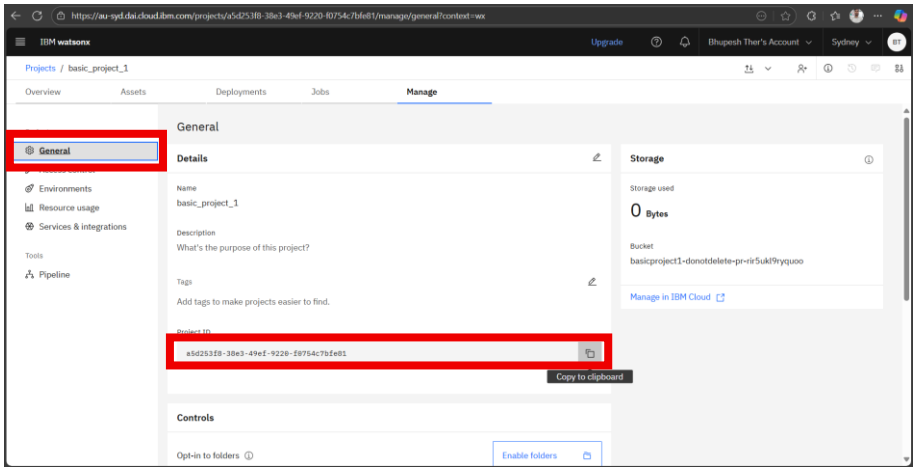




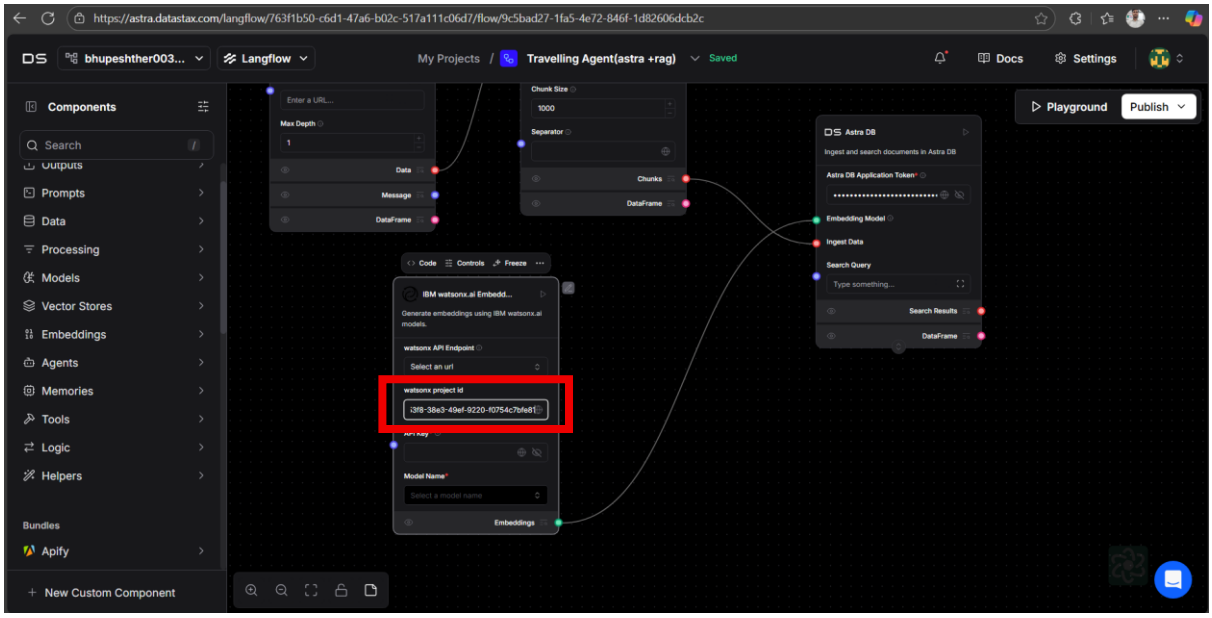
38. Once its done u will be able to see in services & integrations section



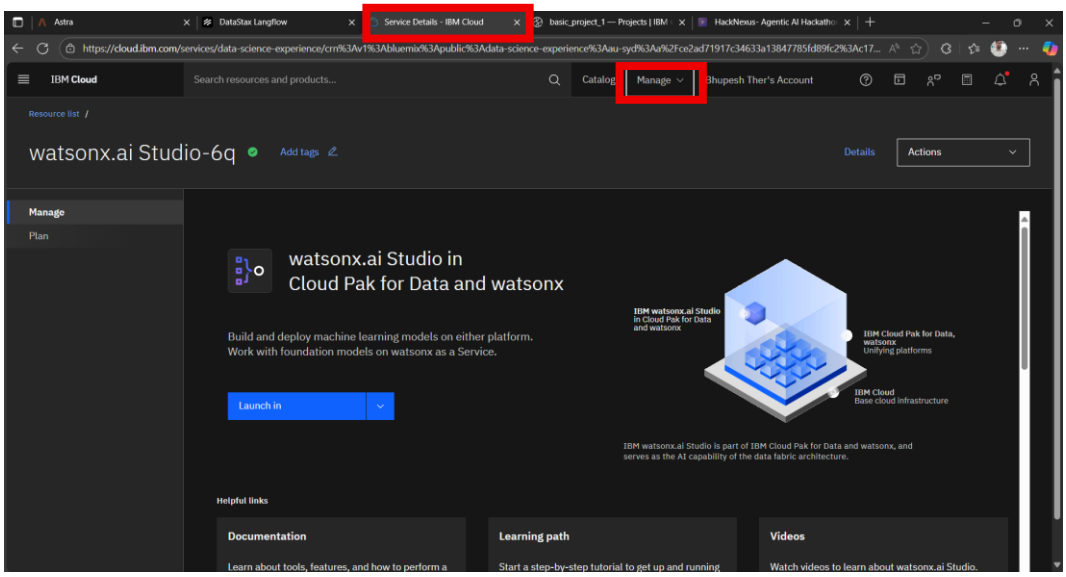
39. Now click on the general here you getting your project id copy your project ID its import so please paste somewhere



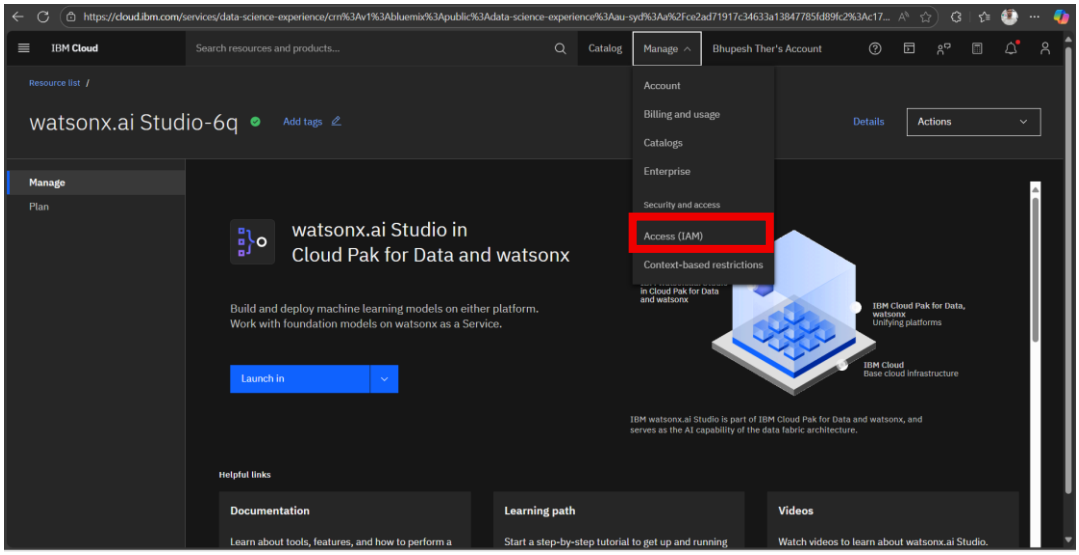
40. Now come back to langflow and add your project ID in Watsonx project ID



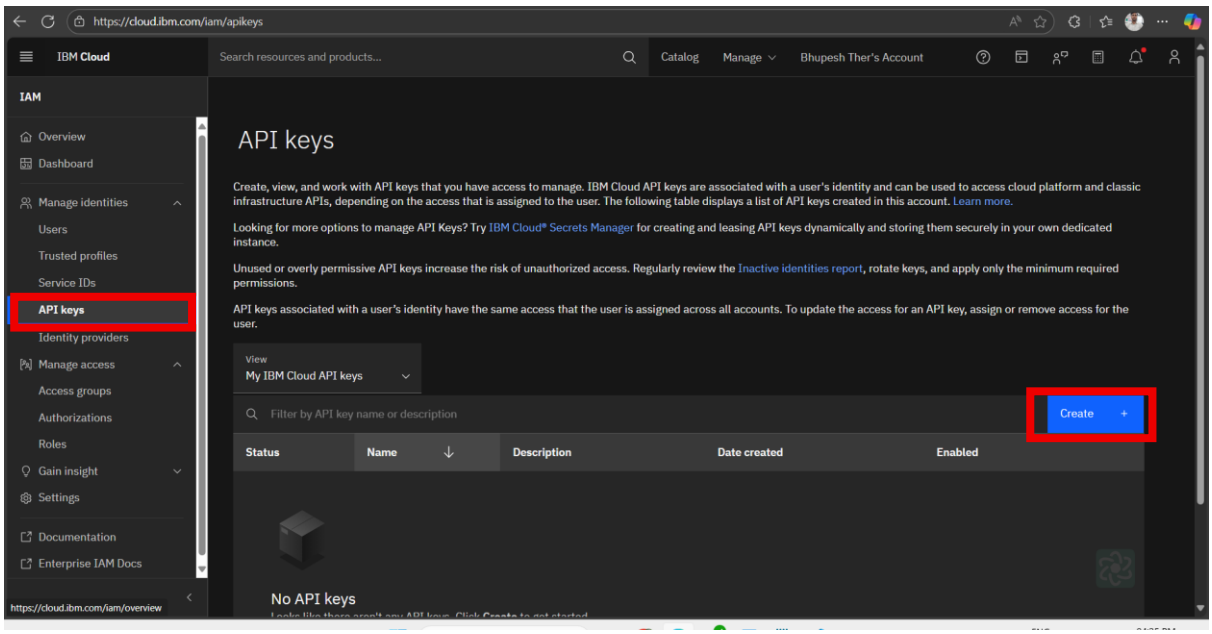
41. Once its done go back to your previous tab that name service details (launching page)



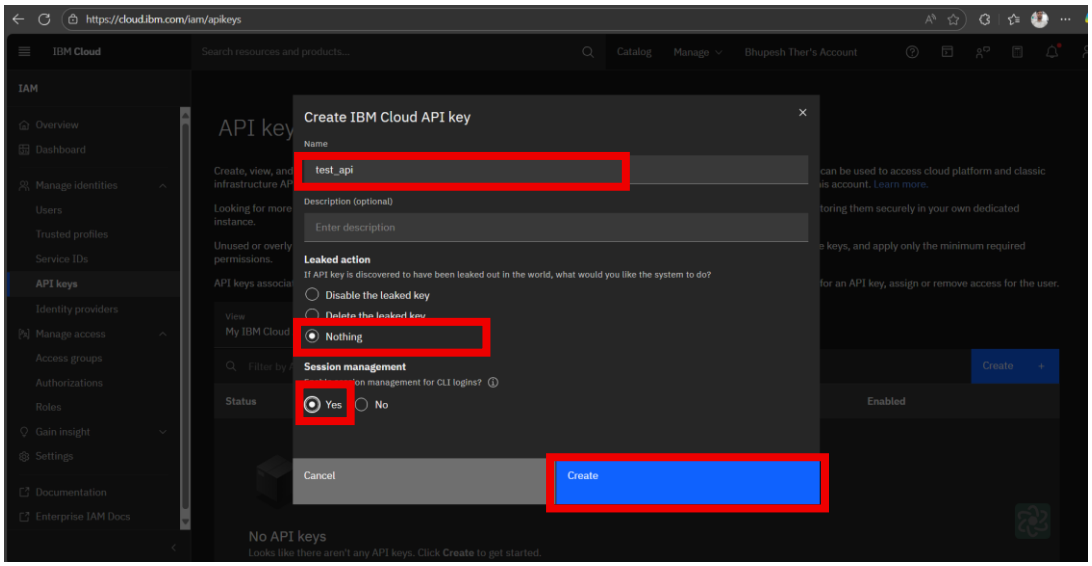
42. Click on manage and next Access(IAM)



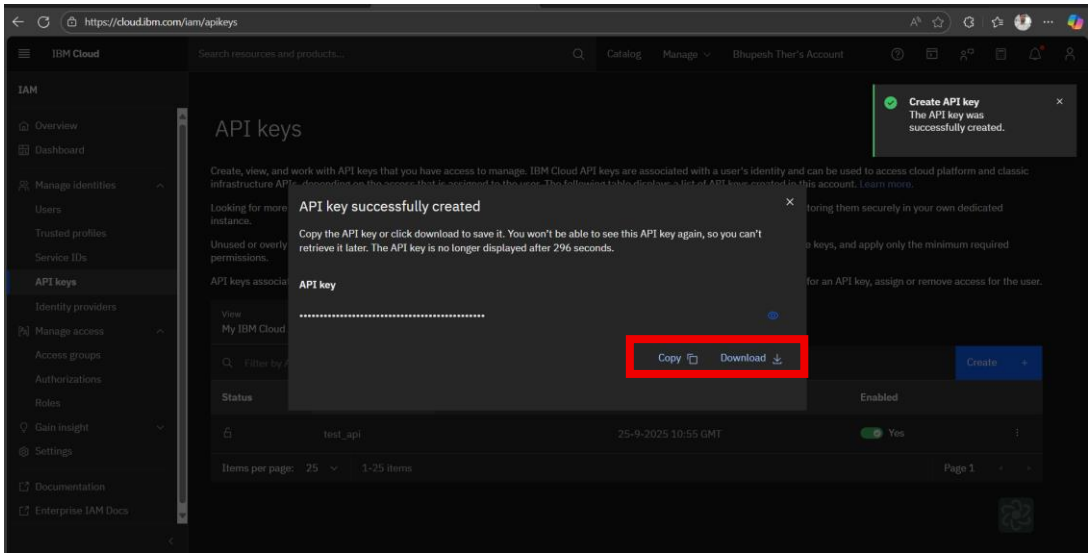
43. Now left side you getting the option of API Key click on that and after that on right side you will be getting option create click in that



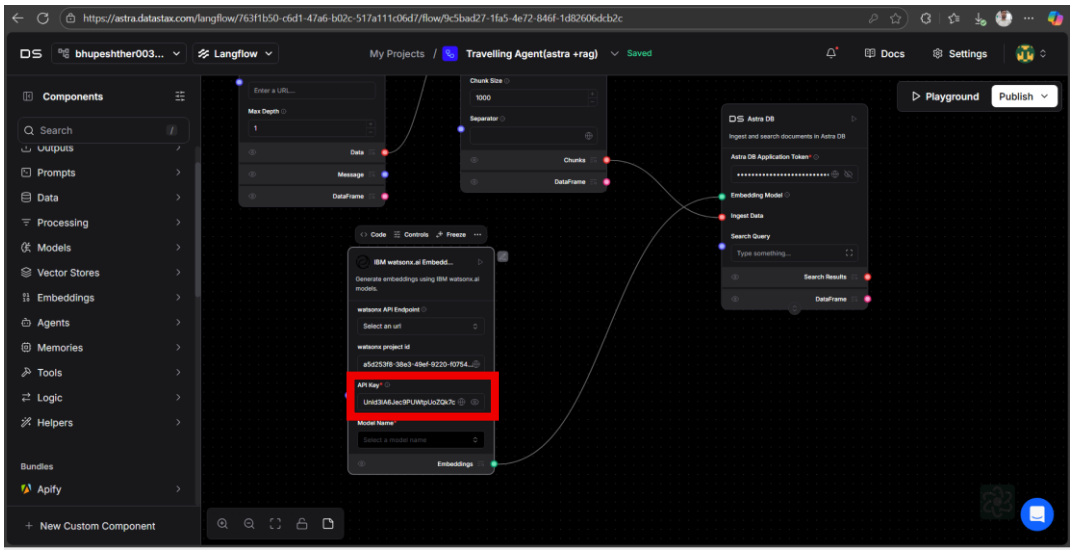
44. Give the name to your API key select leaked action nothing and session management yes and click on create



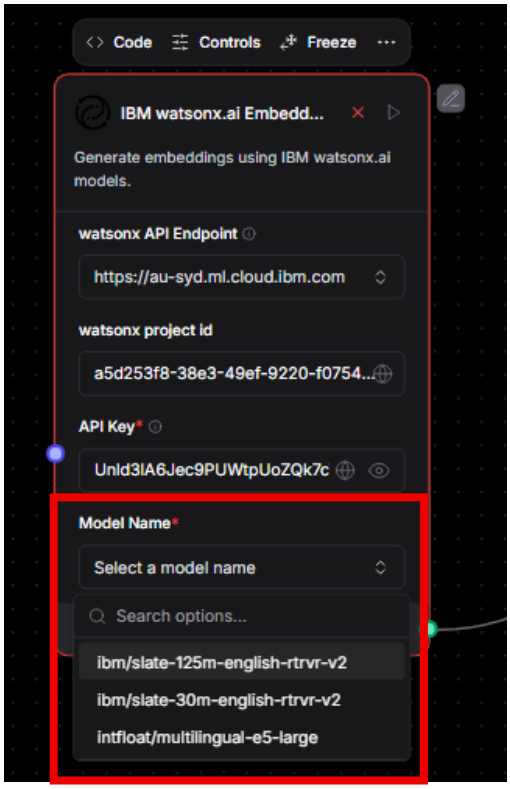
45. Once successfully created now you can copy your api also you can download in your system and its important so please paste somewhere

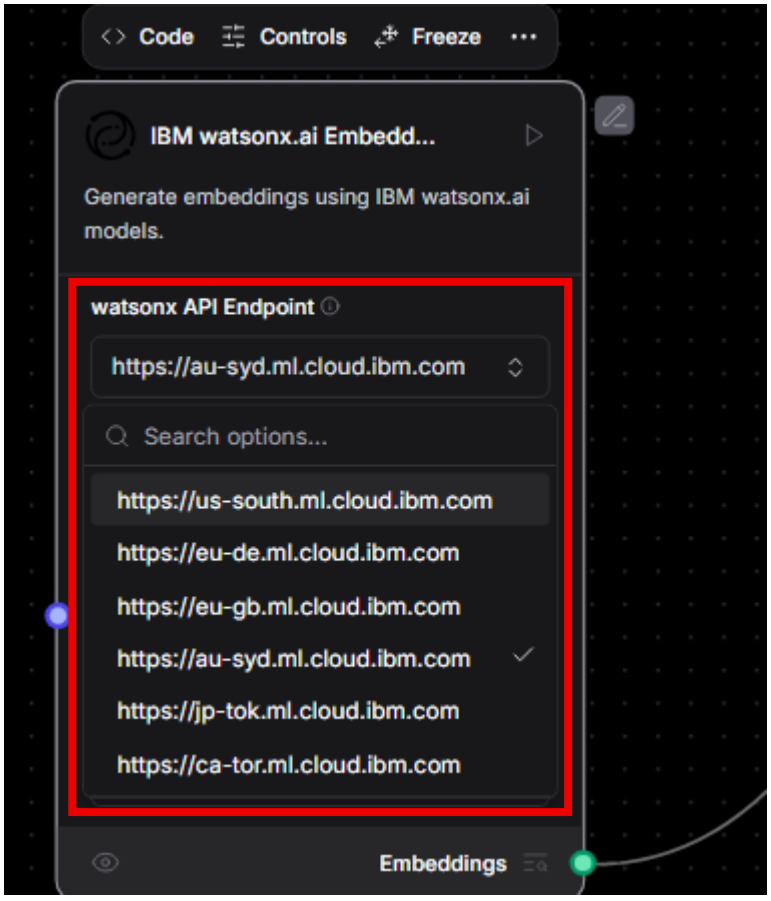


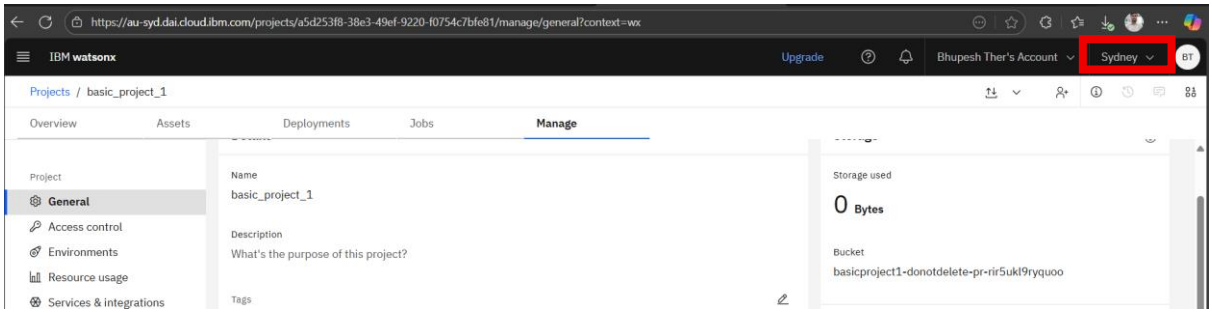
46. Once again back to langflow and now in Watsonx embedding and API key



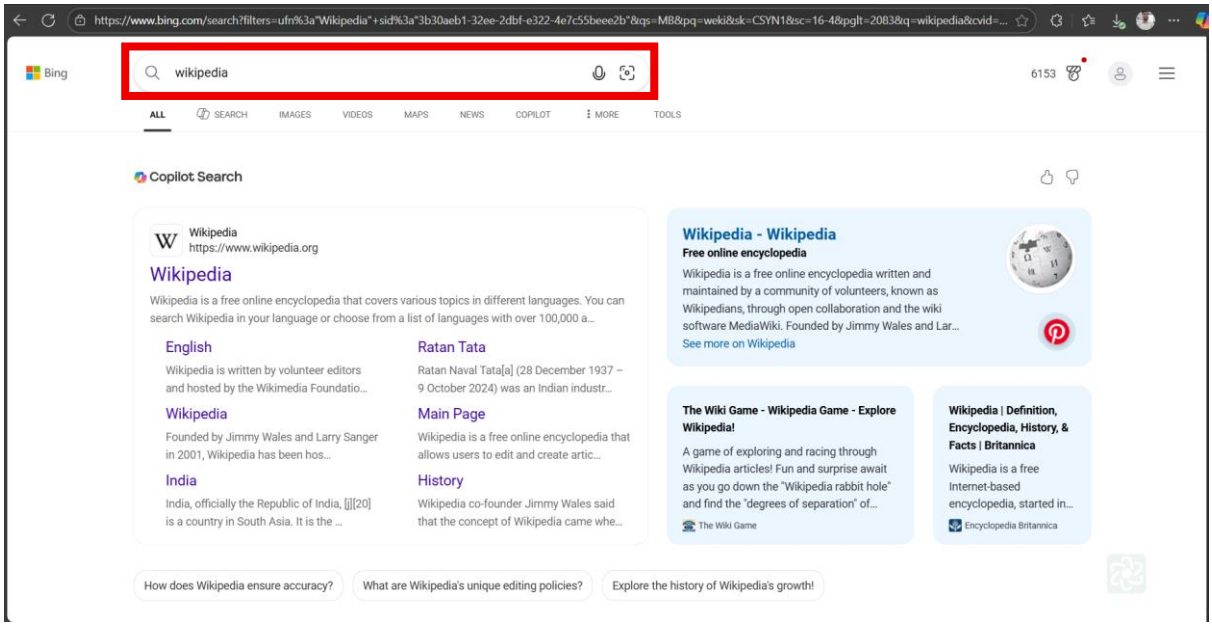
47. Once its done select the embedding model form the list select **ibm/slate-125m-english-rtrvr-v2** and in the watsonx API endpoint you need to select the location select https://au-syd.ml.cloud.ibm.com



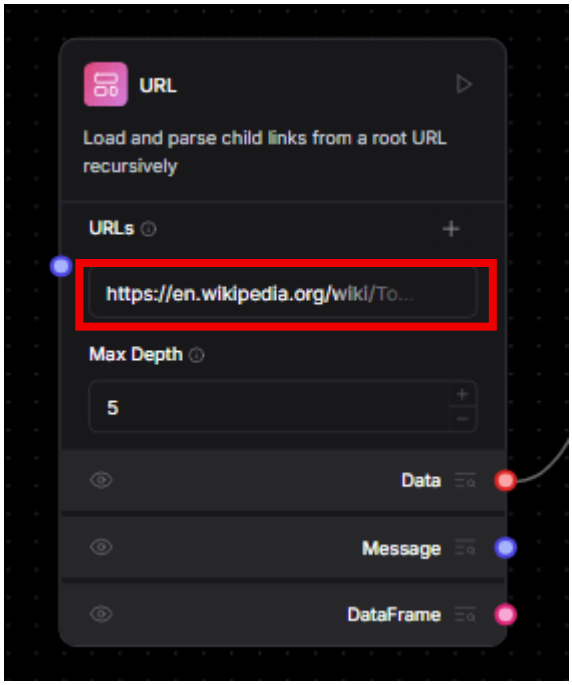




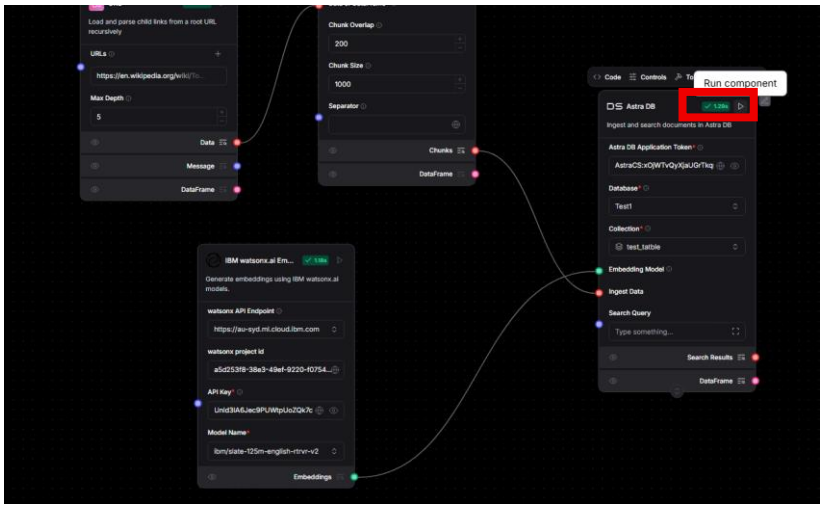
48. Once this all done open new tab and search Wikipedia



49. And copy Wikipedia page copy and paste in URLs

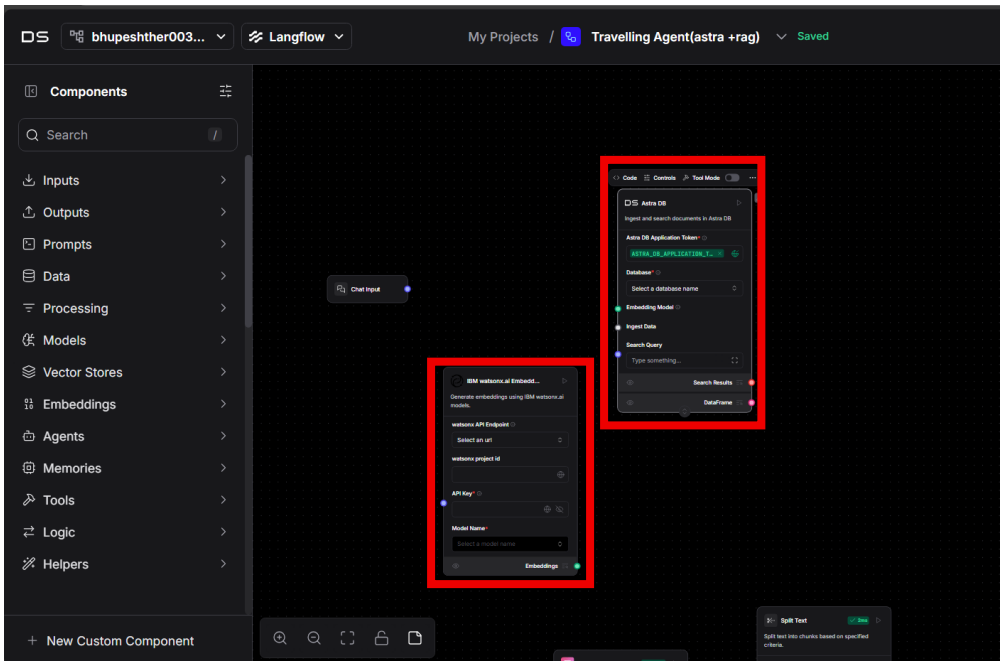


50. Once its done now time to run our code for this come to astra db and click on run component

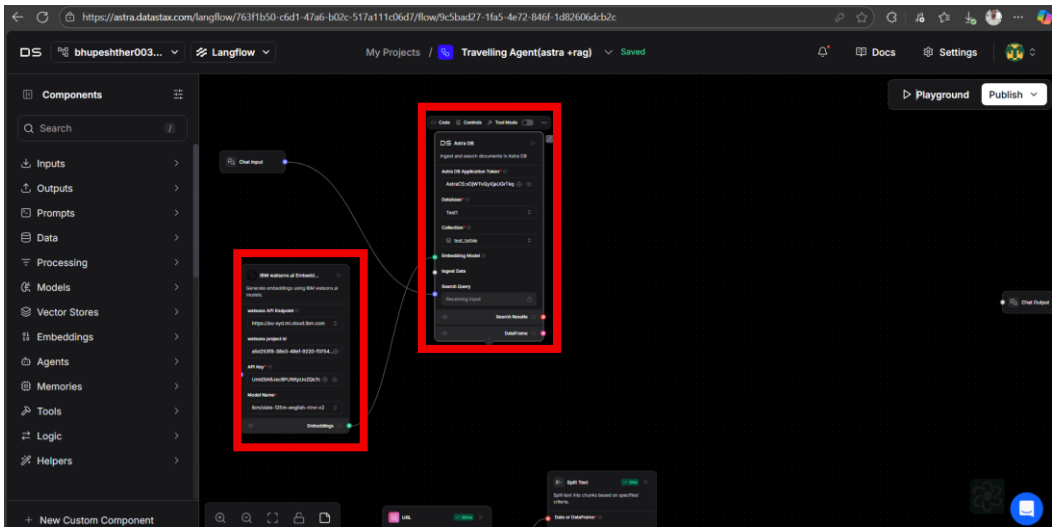


Our RAG done successfully works

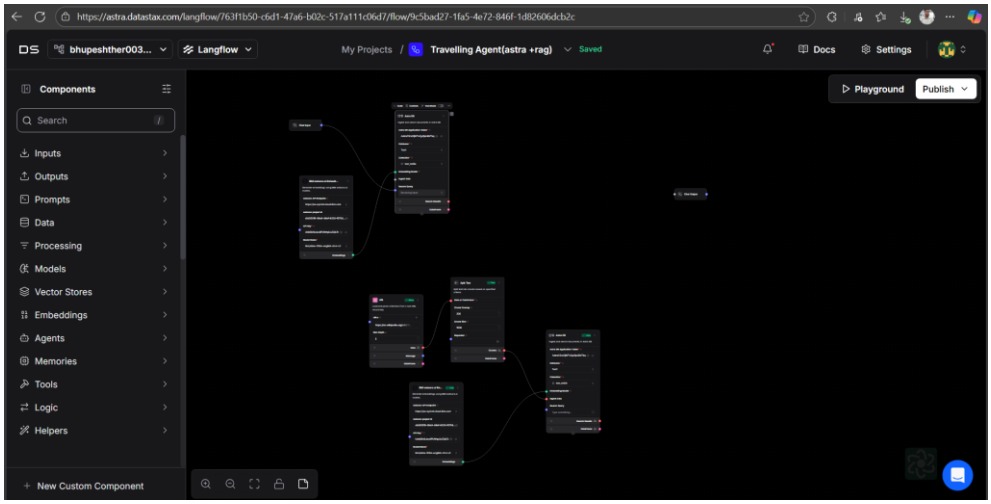
51. Now we need to create agent for this add once again Astra DB and IBM watsonx.ai embedding



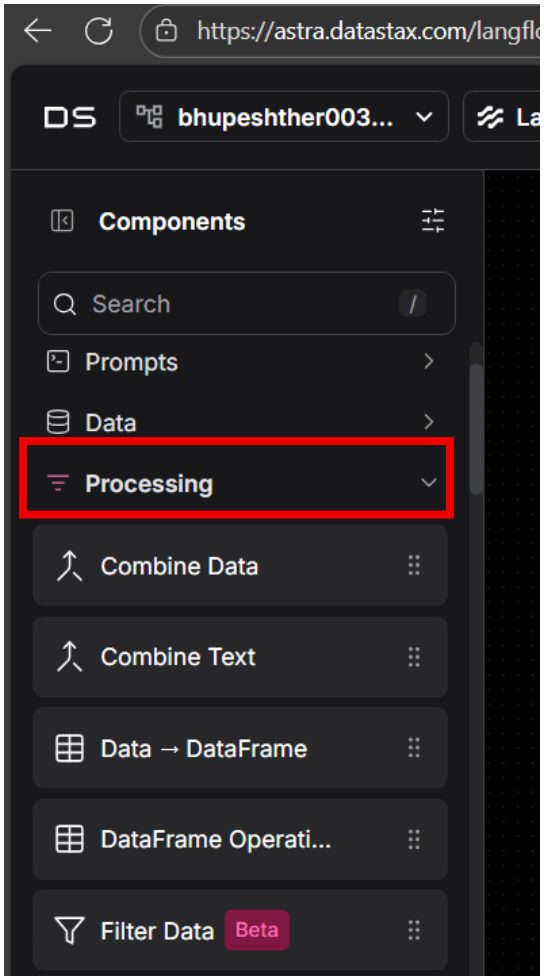
52. Make connection for this take reference form below image and also fill Token ID ,select database , select collection and Same for watsonx.embedding select endpoint, add project id ,api key and select same model



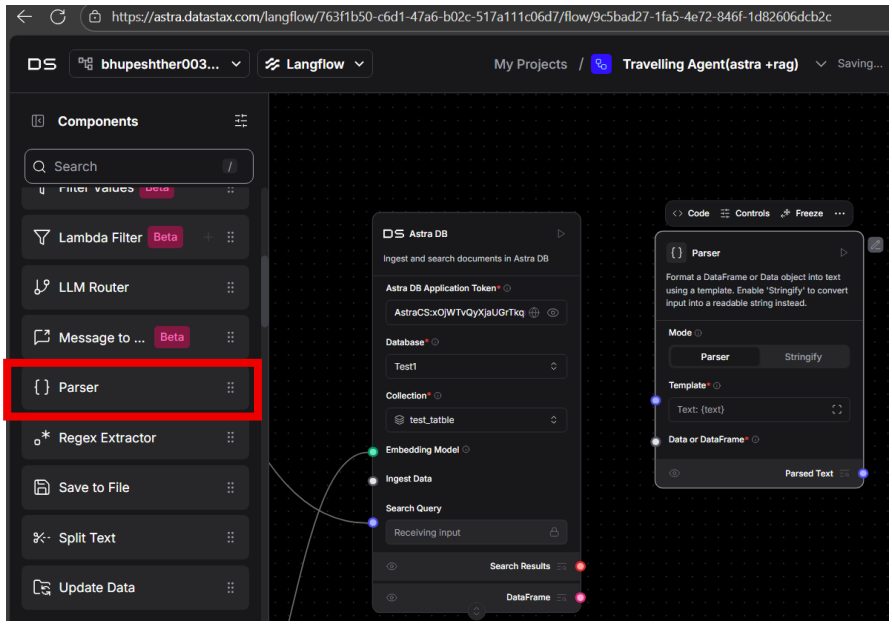
Full workspace diagram



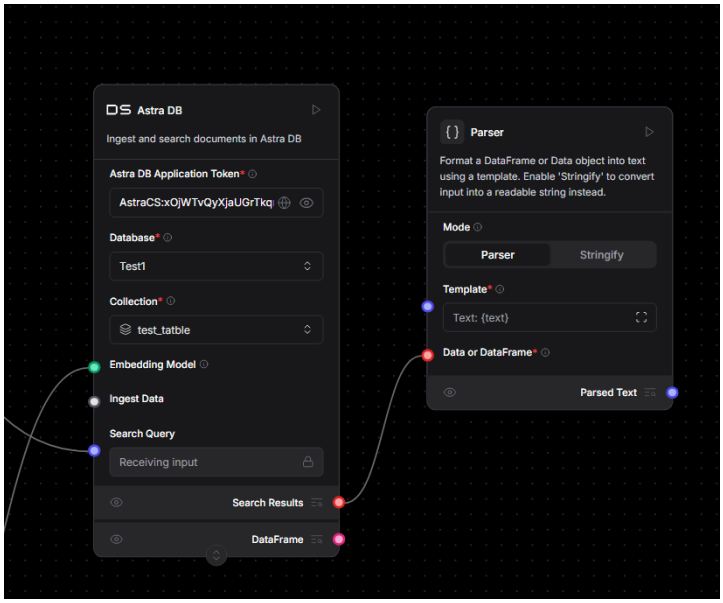
53. Now need to add parse for this go to preprocessing



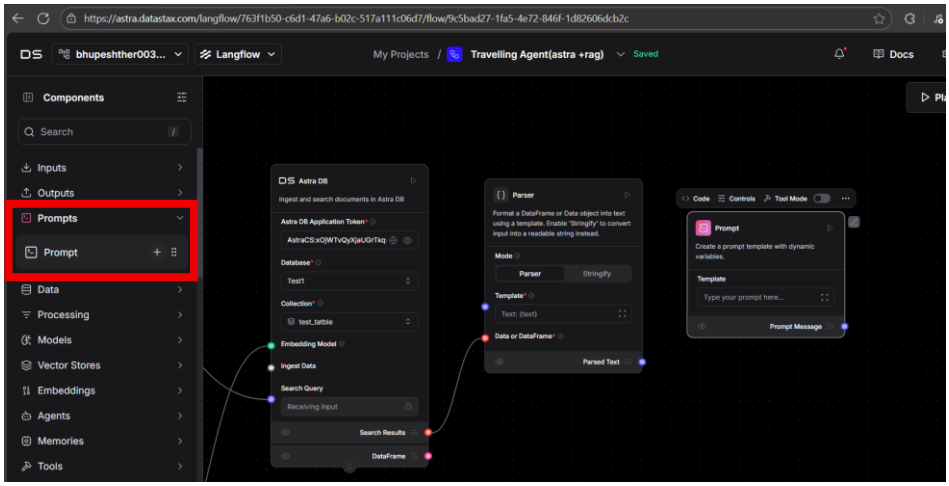
54. And select parse and add in your workspace.



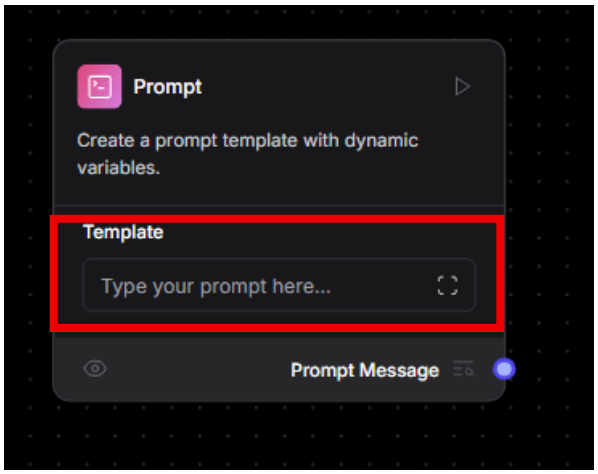
55. Once its done make connection of Astra DB to parse



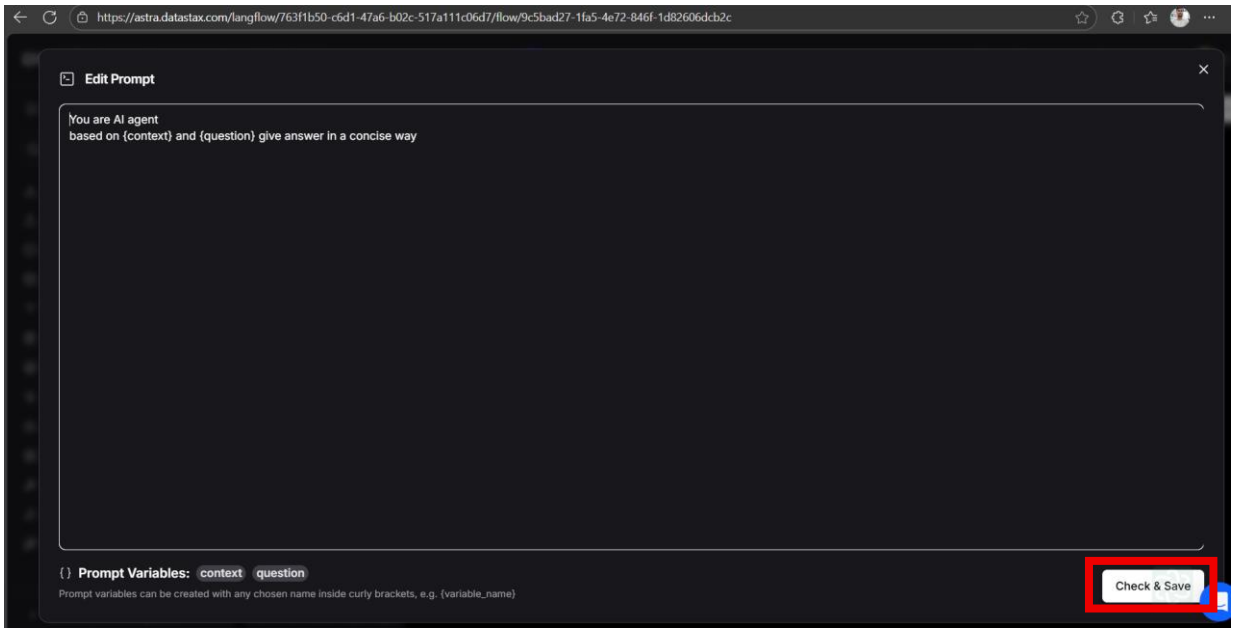
56. Now next go to prompt and add one prompt in your workspace



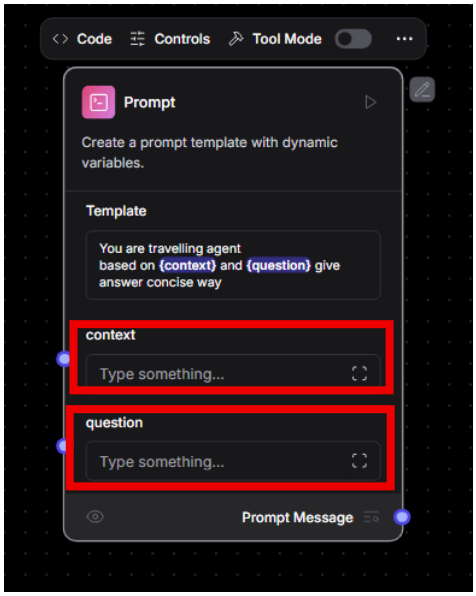
57. Now click on template and write simple template



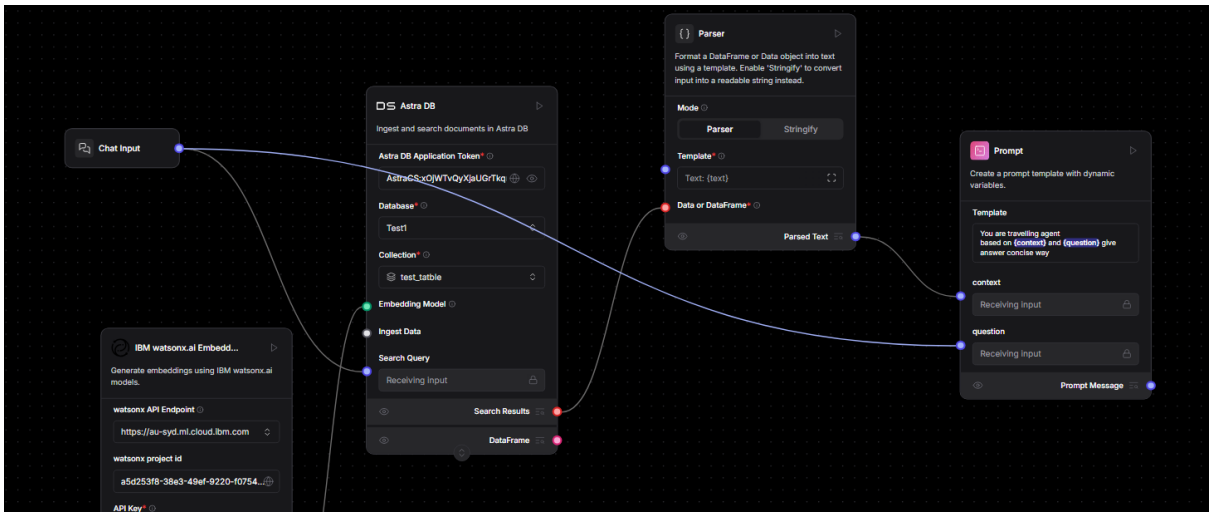
58. Give simple template and click on check and save



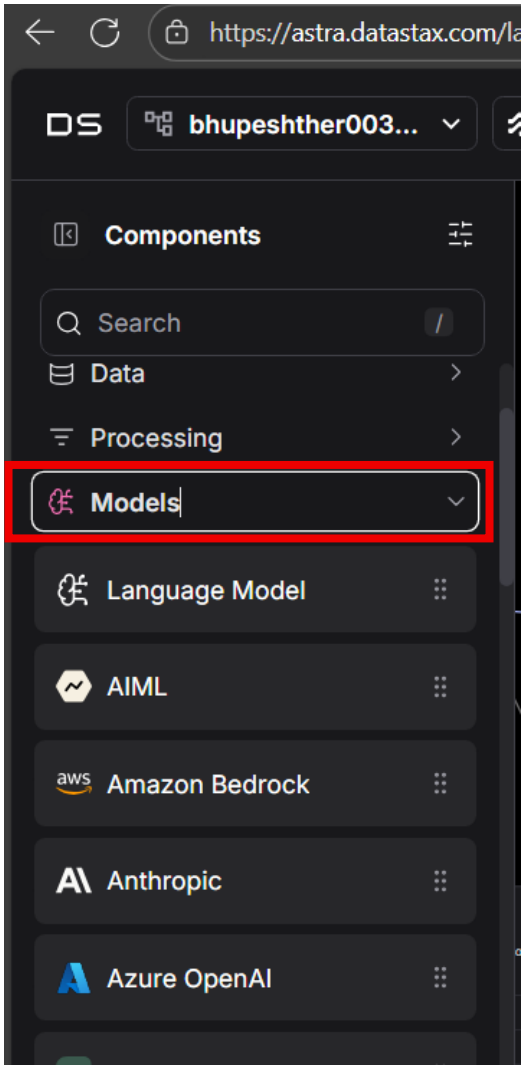
59. Once you done you will be getting two more option so make connection using below diagram



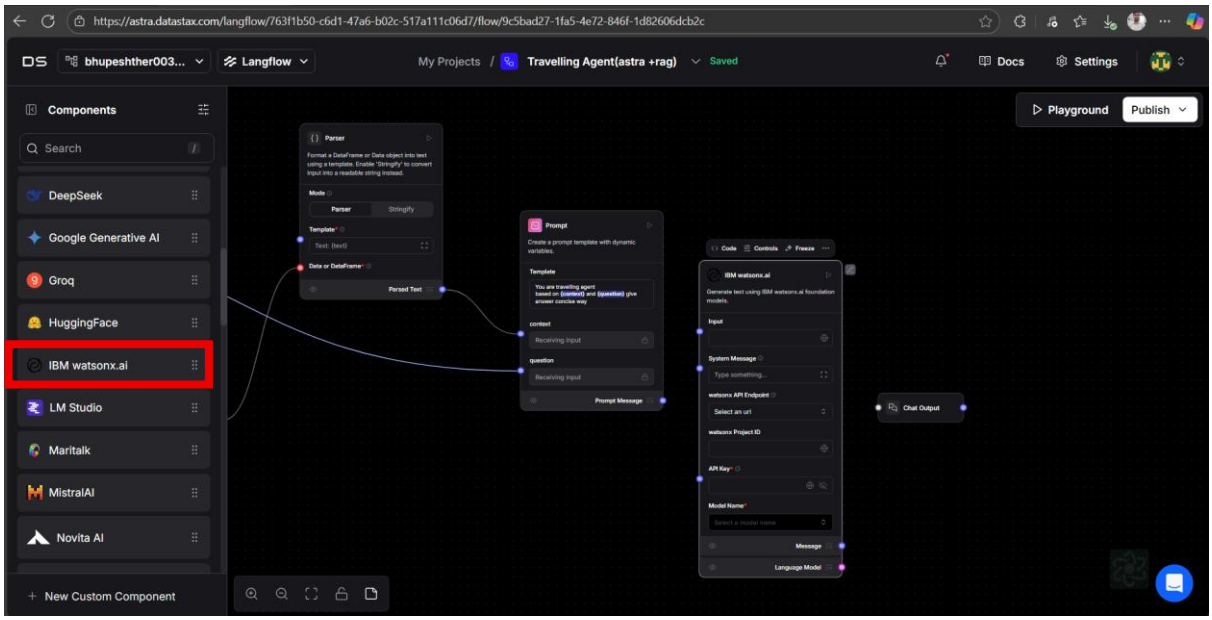
Make connections like below image



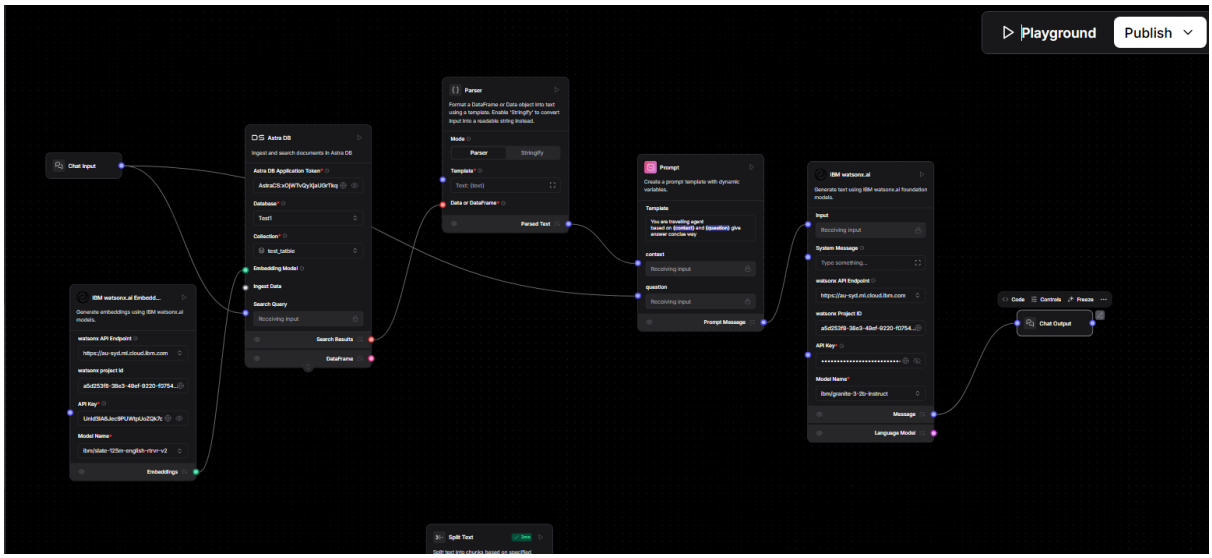
60. Now next we need to select model for this go to models.



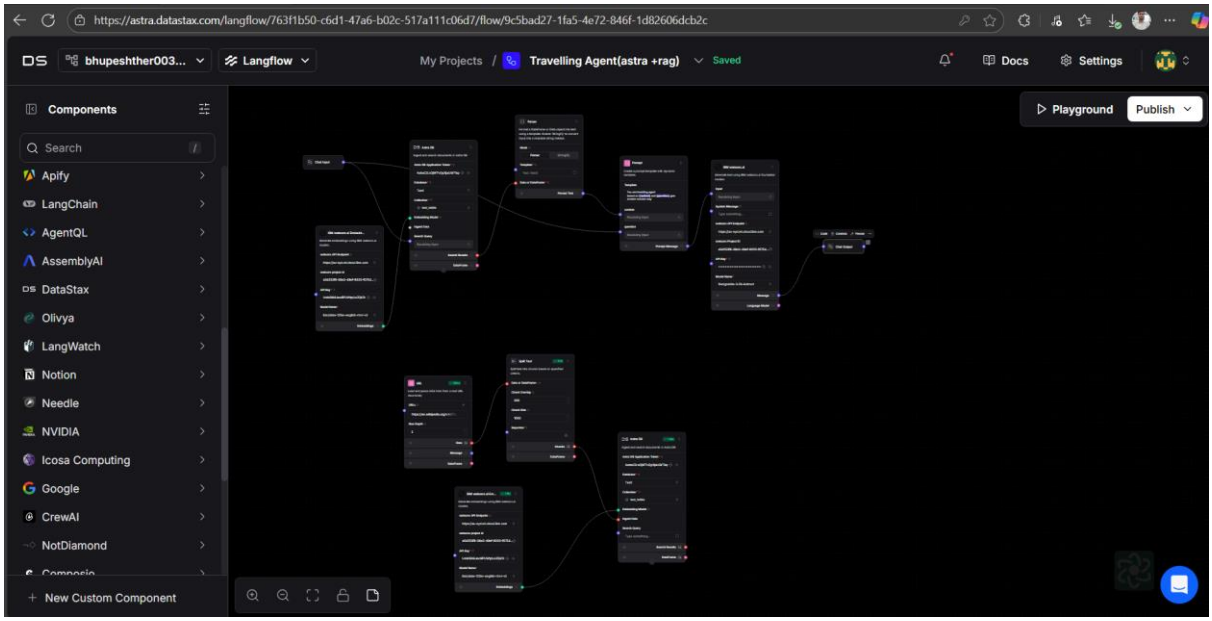
61. Now add model IBM watsonx.ai



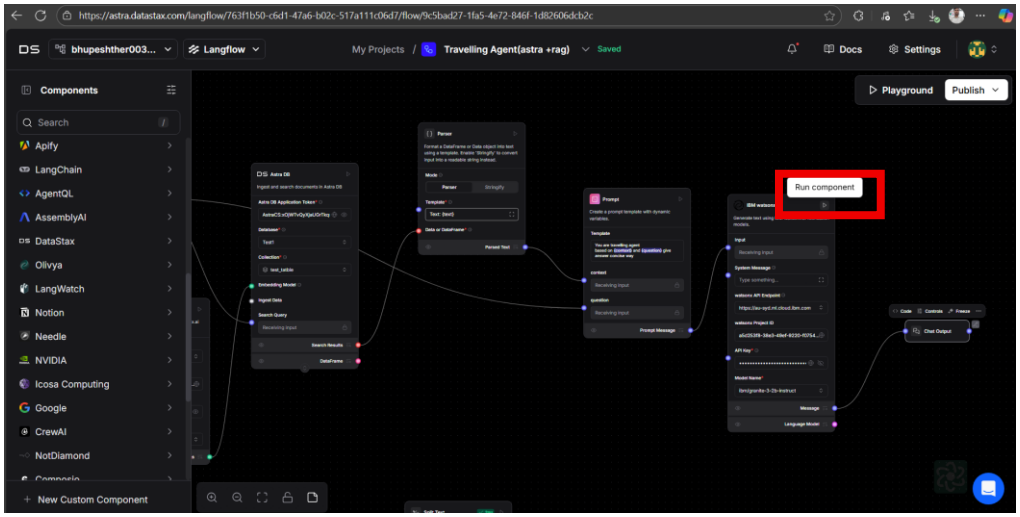
62. Make connection with parse and output with output refer below image for this



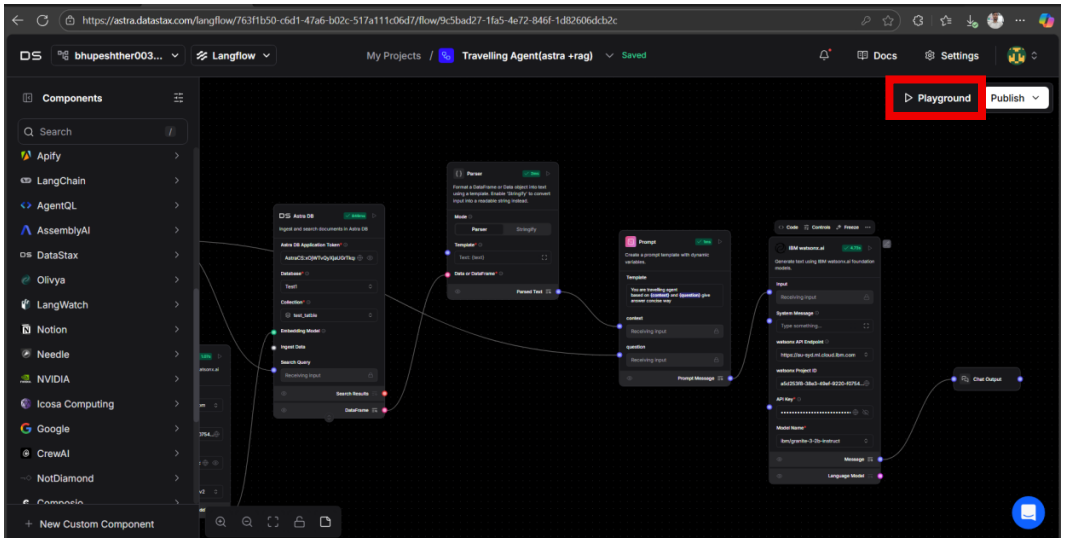
63. Final workspace output



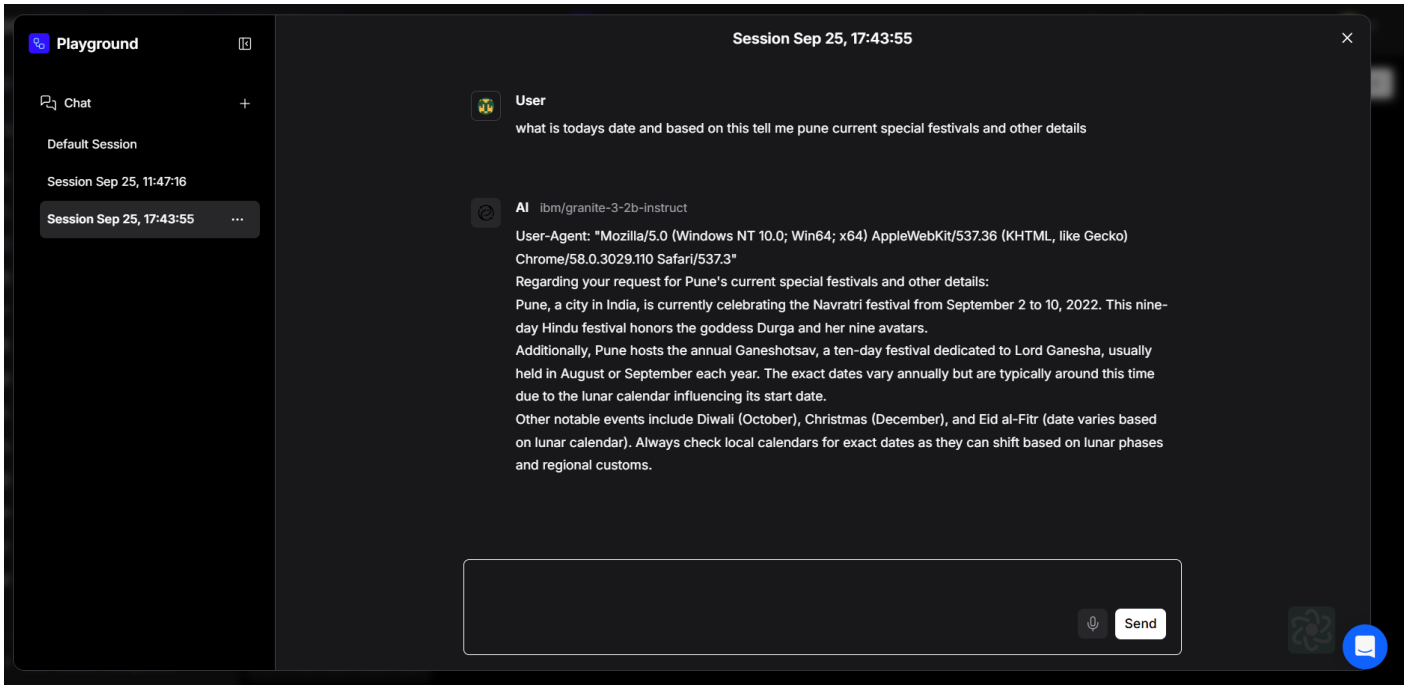
64. Now test your agent for this click on the run component



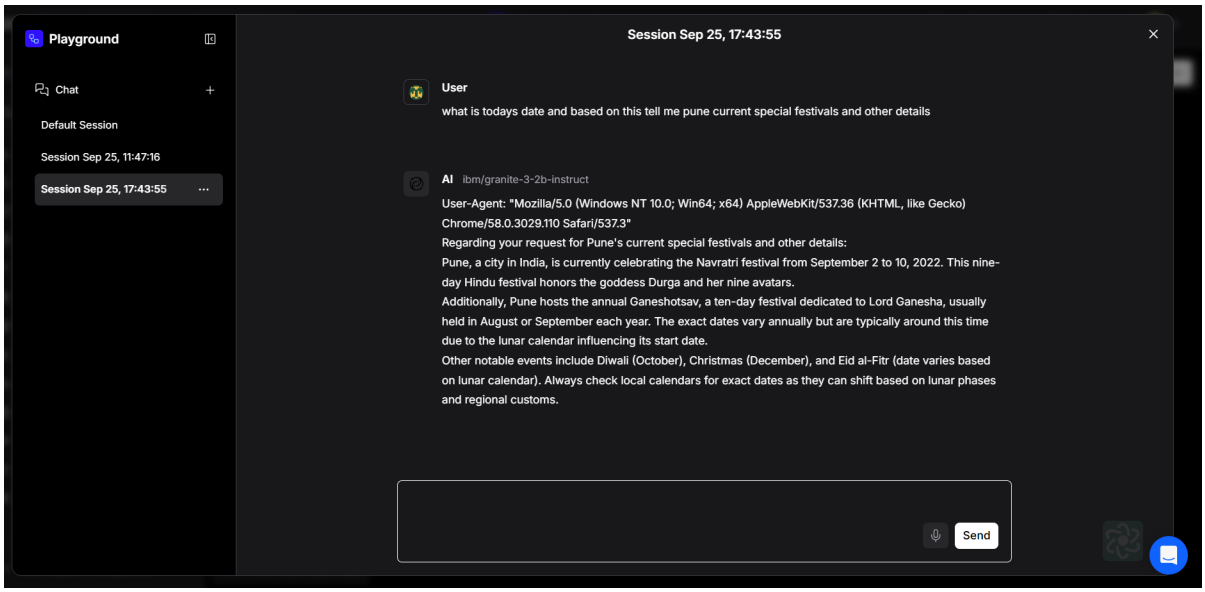
65. Once its successfully run now click on the playground and test your model



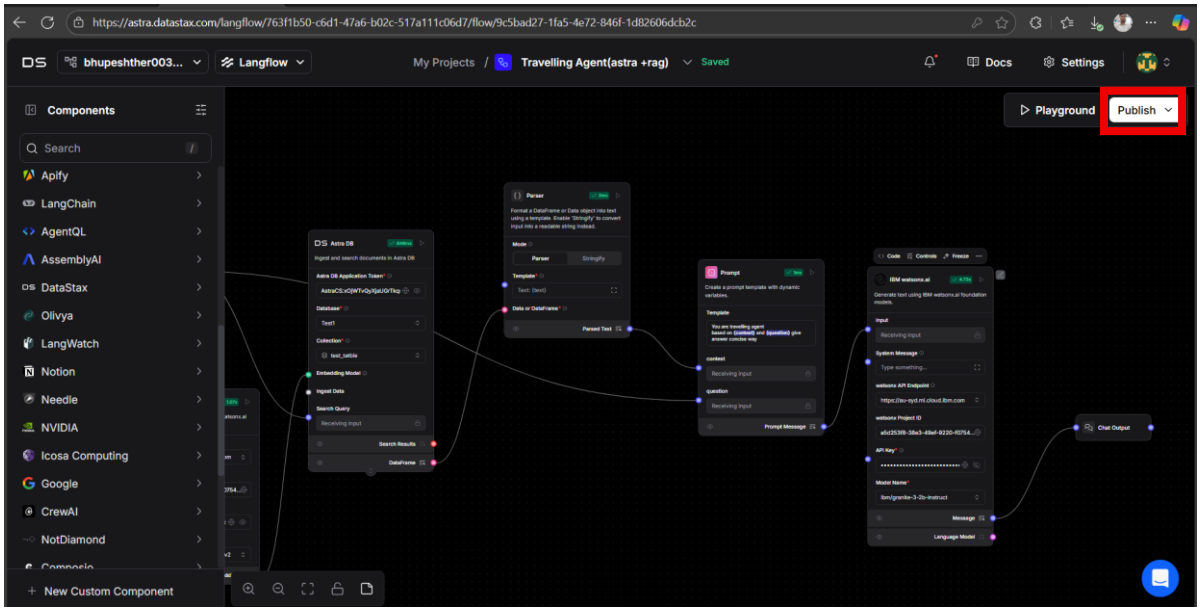
66. And now test your model



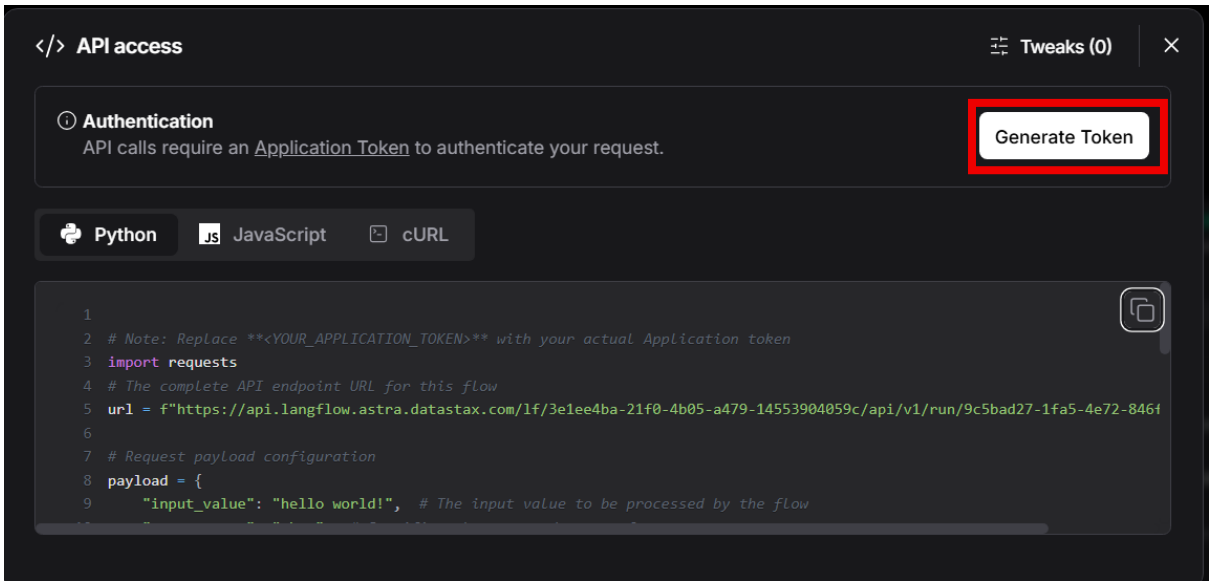
67. Once its done now deploy on sreamlit for this close playground



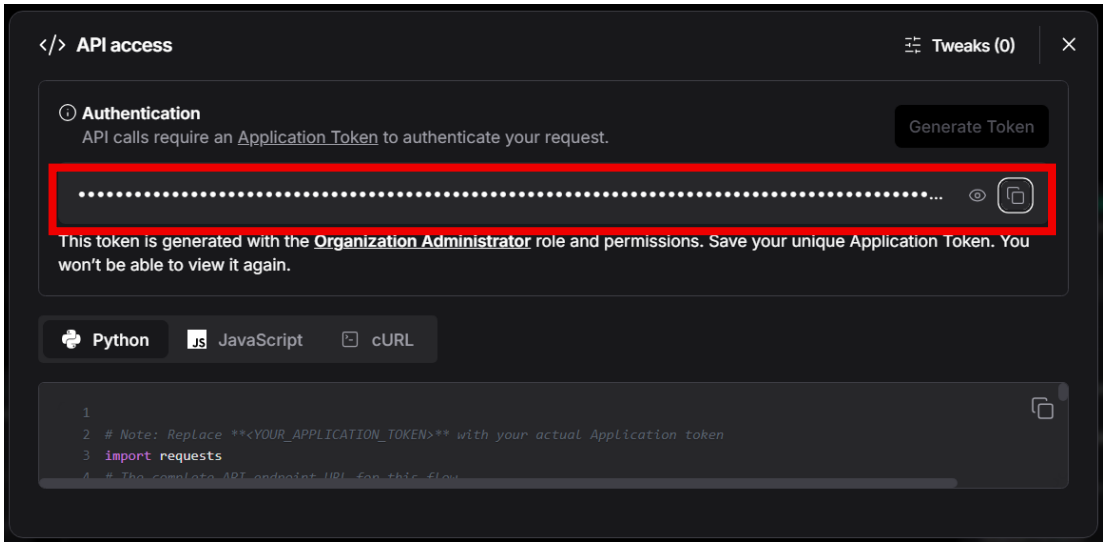
68. Now for the deploy click on publish



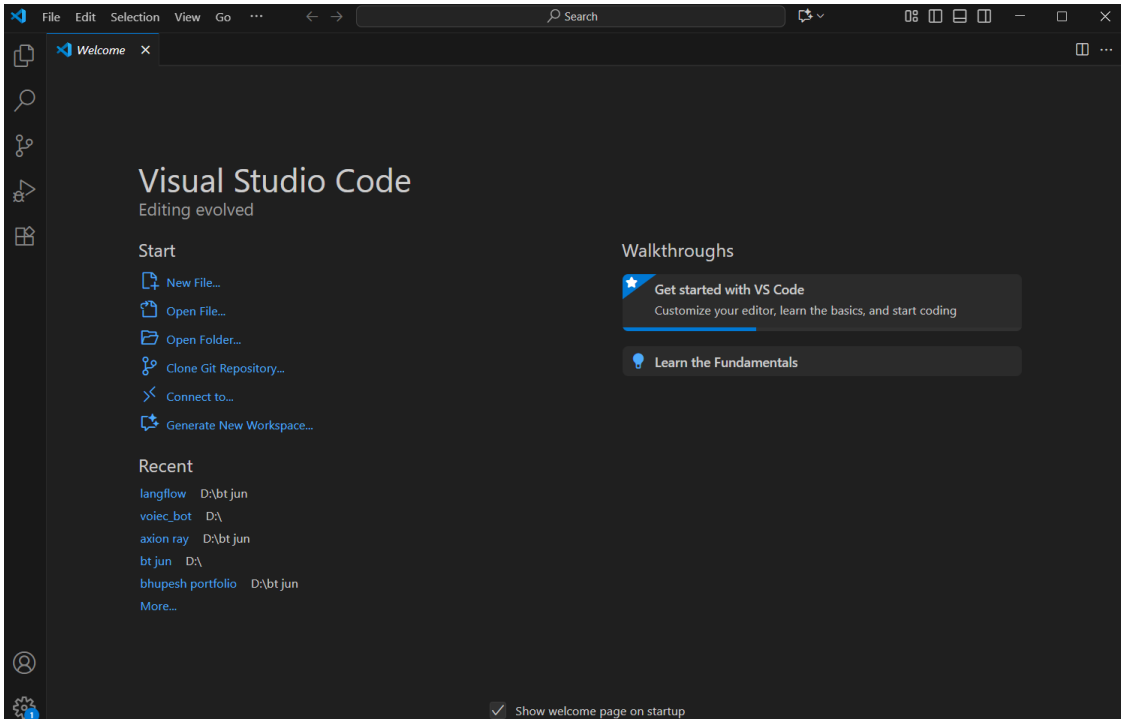
69. Now click on generate to token and generate token



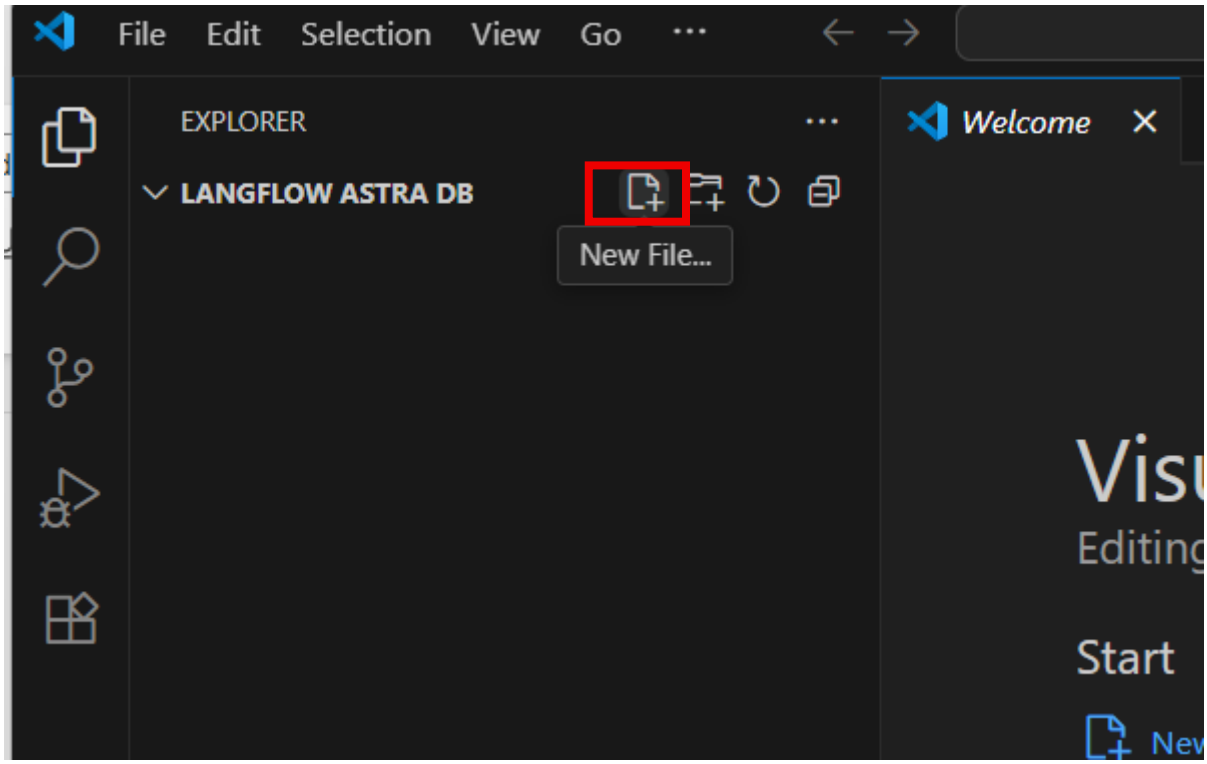
70. Copy token

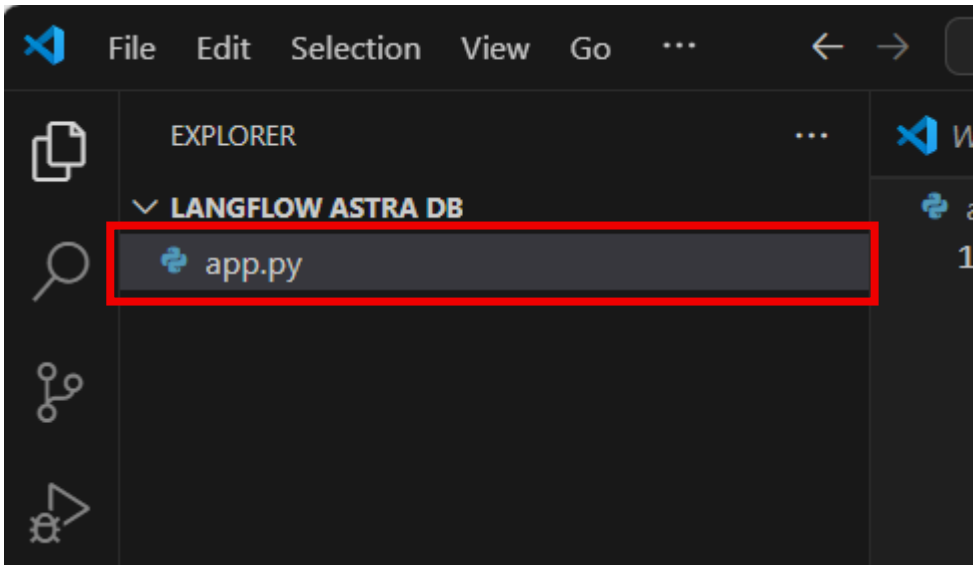


Open vs Code



Create app.py file



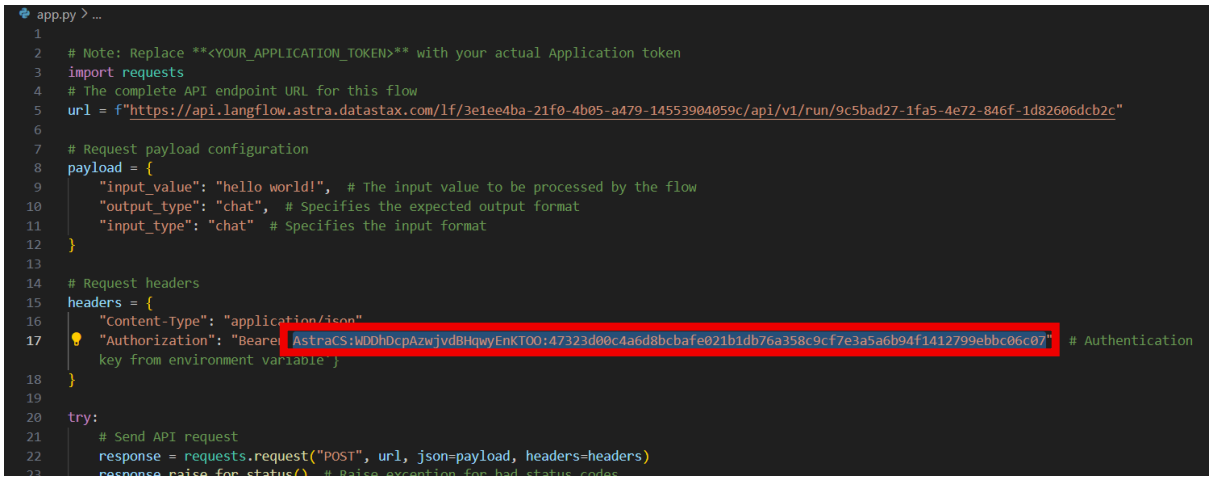


Paste bellow code in app.py file

```
import streamlit as st
import requests

# Streamlit App Title
st.title("🚀 Langflow Astra API - Trip Planner")

# User input
user_input = st.text_input("💬 Enter your request:", "give me 1 week pune trip plan according to todays date")


# API endpoint & headers
```

```
url = "https://api.langflow.astra.datastax.com/lf/3e1ee4ba-21f0-4b05-a479-14553904059c/api/v1/run/9c5bad27-1fa5-4e72-846f-1d82606dcb2c"
headers = {
    "Content-Type": "application/json",
    "Authorization": "Bearer AstraCS:WDDhDcpAzwjvdBHqwyEnKTOO:47323d00c4a6d8bcbafe021b1db76a358c9cf7e3a5a6b94f1412799ebbc06c07"
}

# Button to send input
if st.button("Send to Langflow"):
    payload = {
        "input_value": user_input,
        "output_type": "chat",
        "input_type": "chat"
    }

    try:
        response = requests.post(url, json=payload, headers=headers)
        response.raise_for_status()
        result = response.json()

        # Extract only the "text" field from nested JSON
        try:
            text_output = result["outputs"][0]["outputs"][0]["results"]["message"]["data"]["text"]
            st.success("✅ Trip Plan Generated")
            st.write(text_output)  # Only show text
        except (KeyError, IndexError):
            st.error("⚠️ Could not extract text from response")
            st.json(result)  # fallback: show full JSON

    except requests.exceptions.RequestException as e:
        st.error(f"❌ API request error: {e}")
    except ValueError as e:
        st.error(f"❌ Response parsing error: {e}")
```

71. replace the bellow highlighted code with the generated token on langflow



72. Once its done open terminal and create virtual environment for this type bellow command on terminal

```
PS D:\bt jun\langflow> py -m venv myenv
```

73. On activate the virtual environment run bellow command on terminal

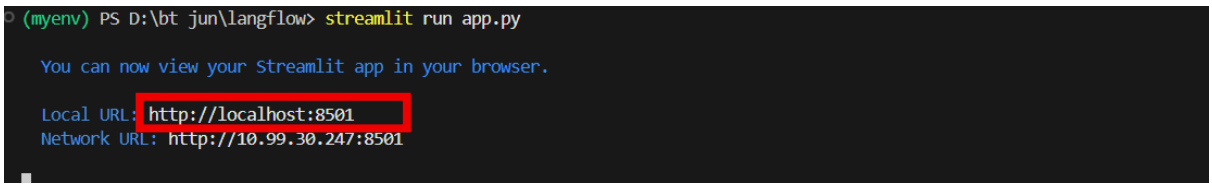
```
PS D:\bt jun\langflow> .\myenv\scripts\activate
```

Once its activated its will looks like this

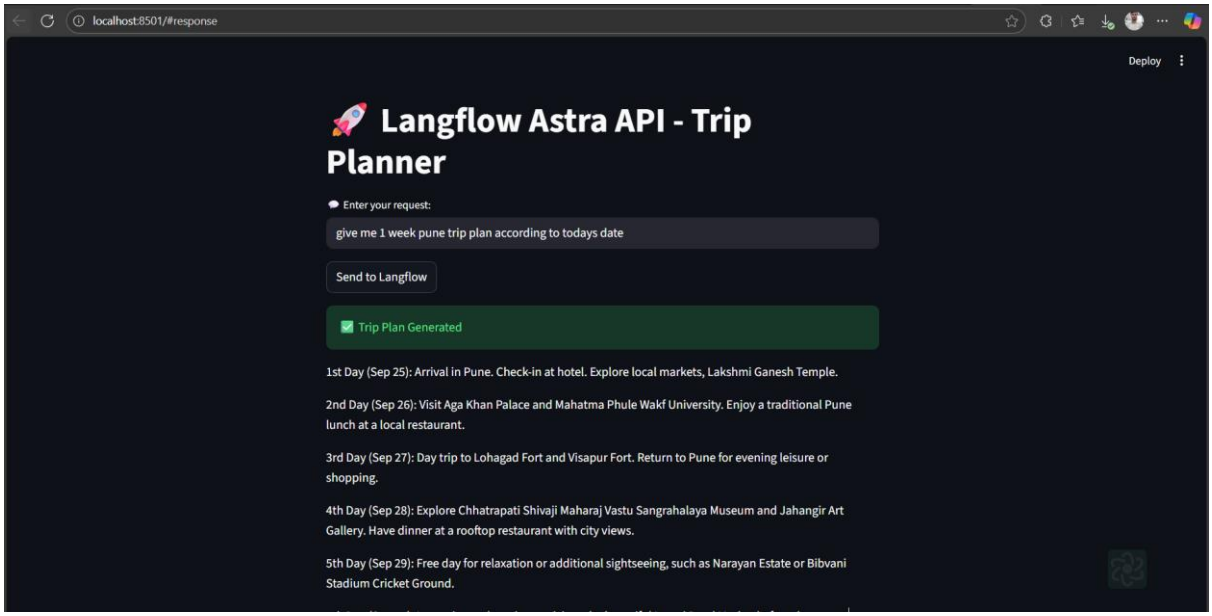
```
(myenv) PS D:\bt jun\langflow>
```

74. Now install streamlit in your system

75. Once its done run your code

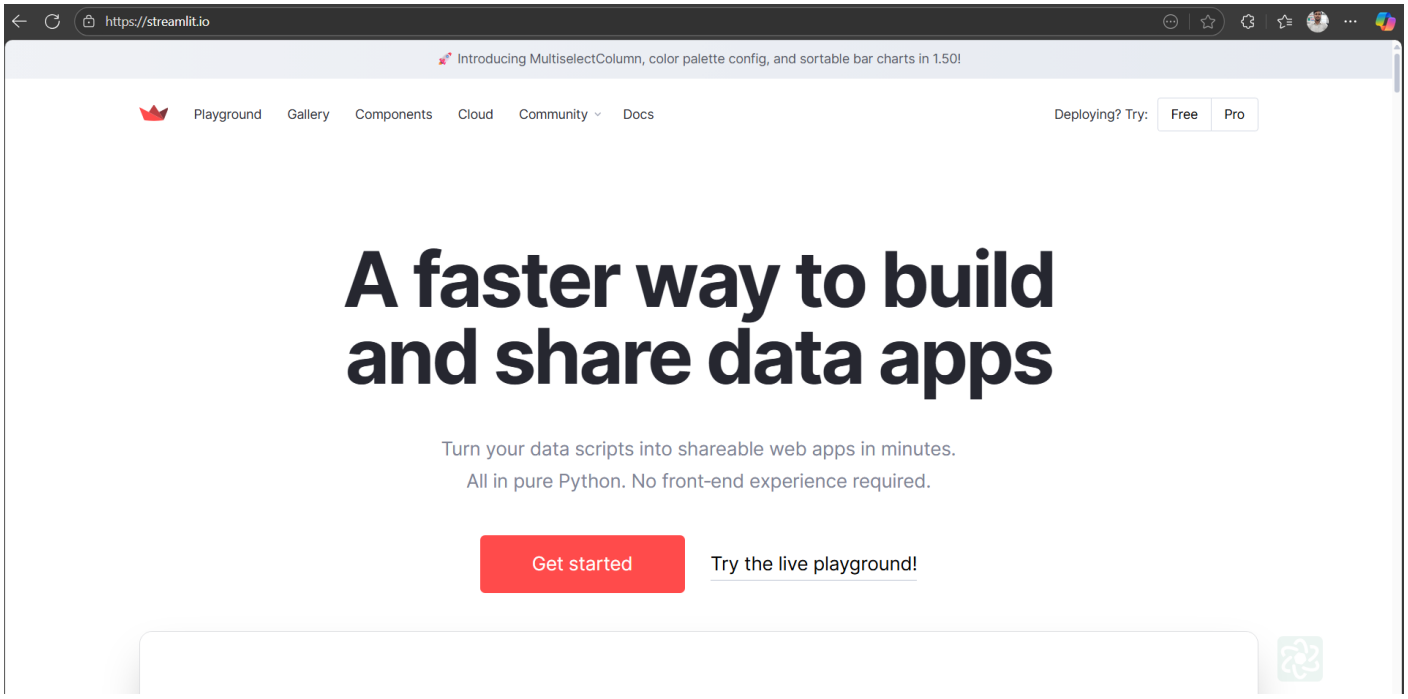


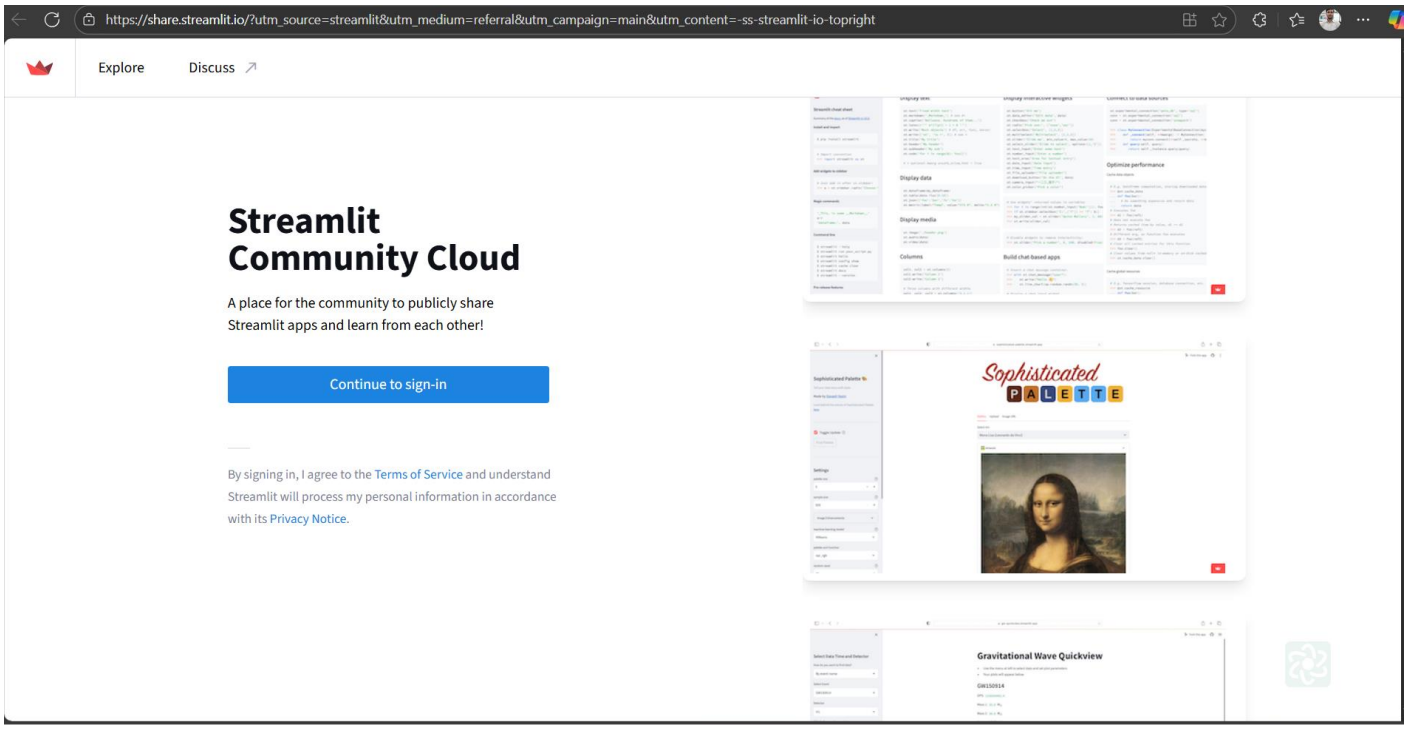
76. Click on localhost link and run your agentic ai

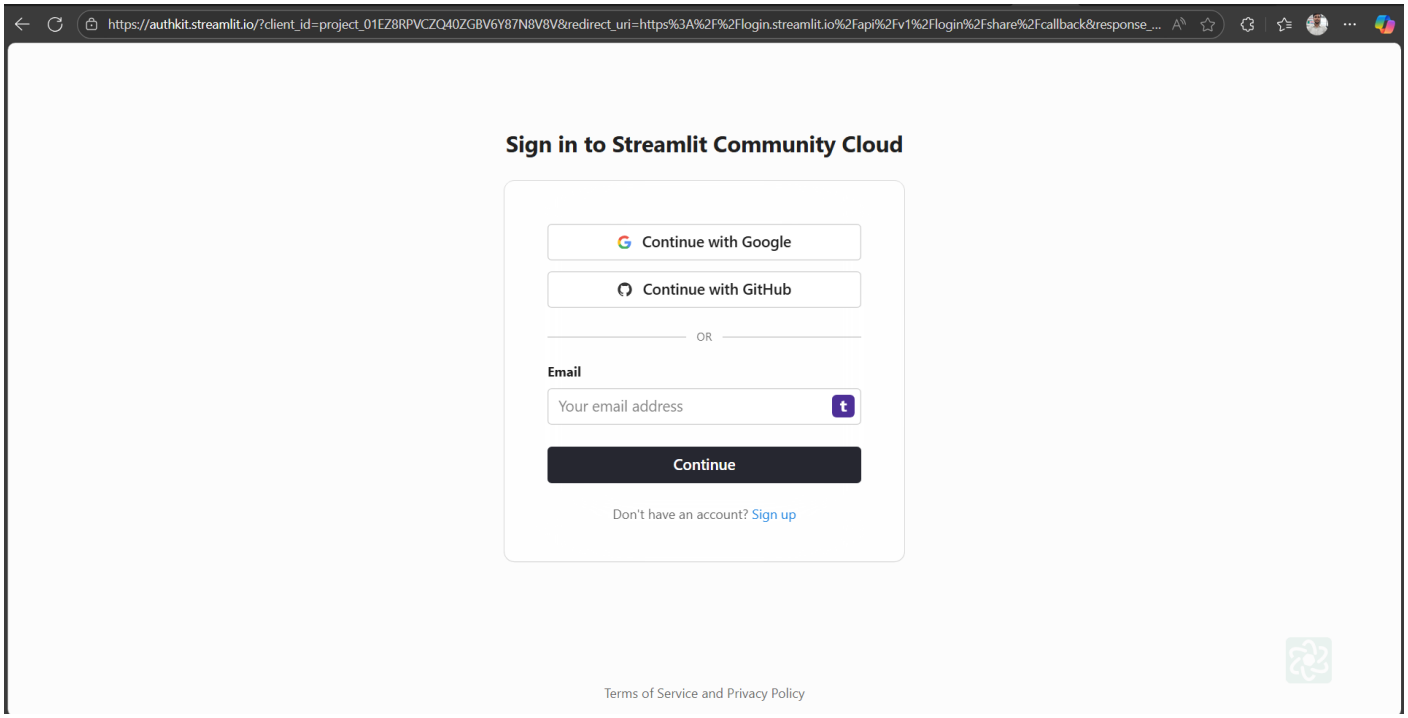


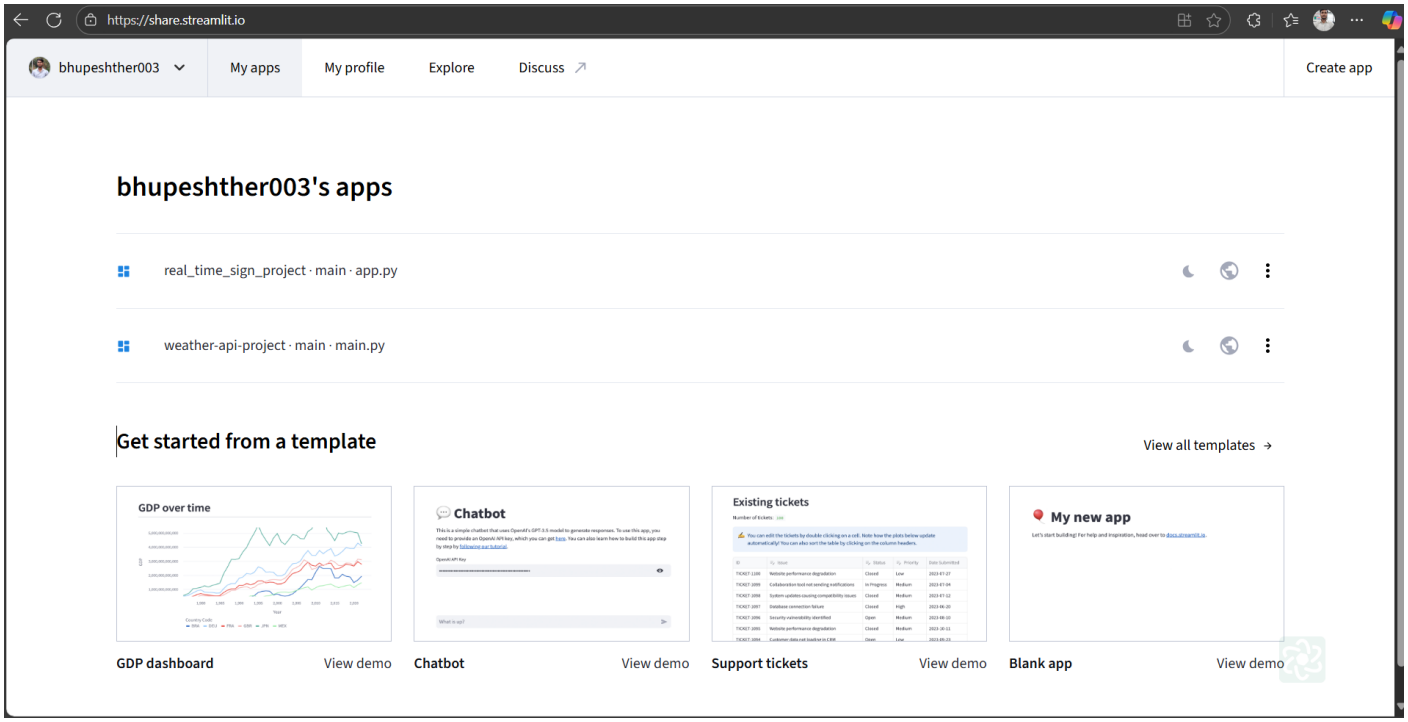
Deployment on Streamlit Cloud

[Streamlit]



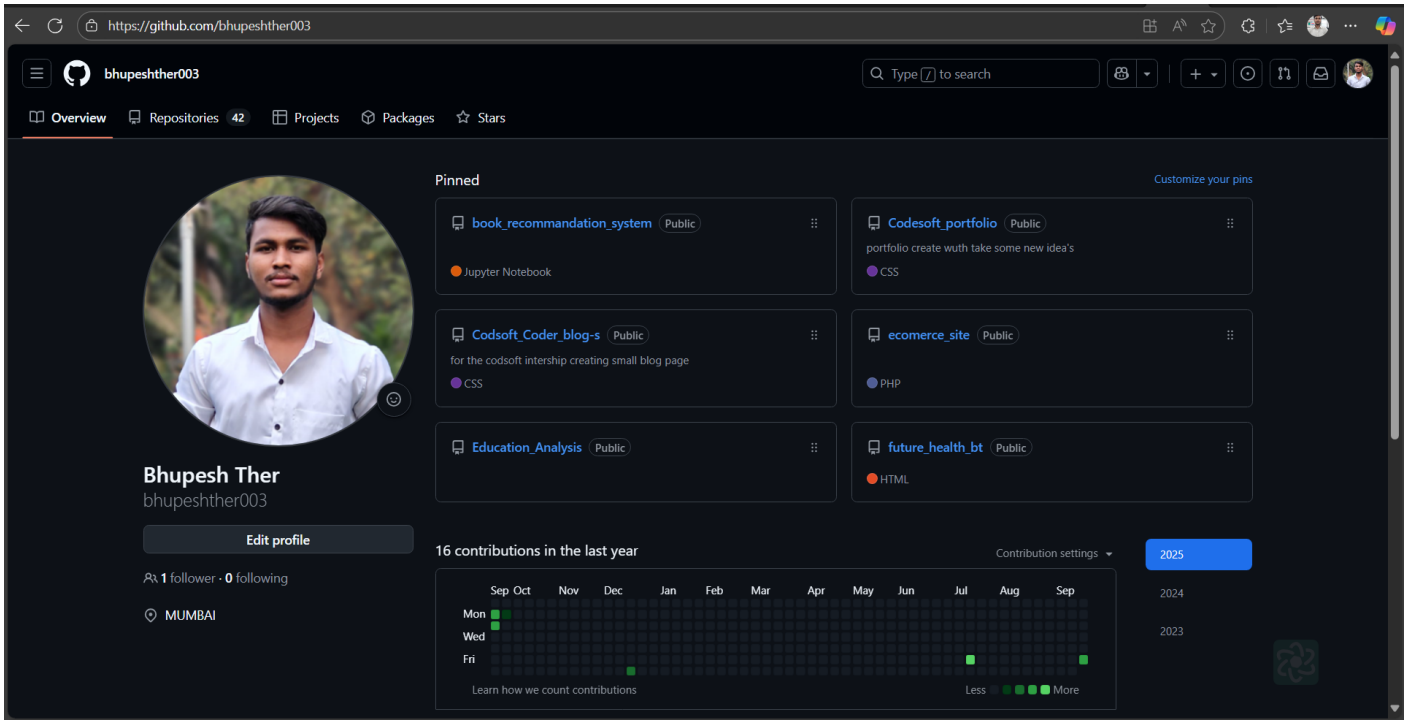


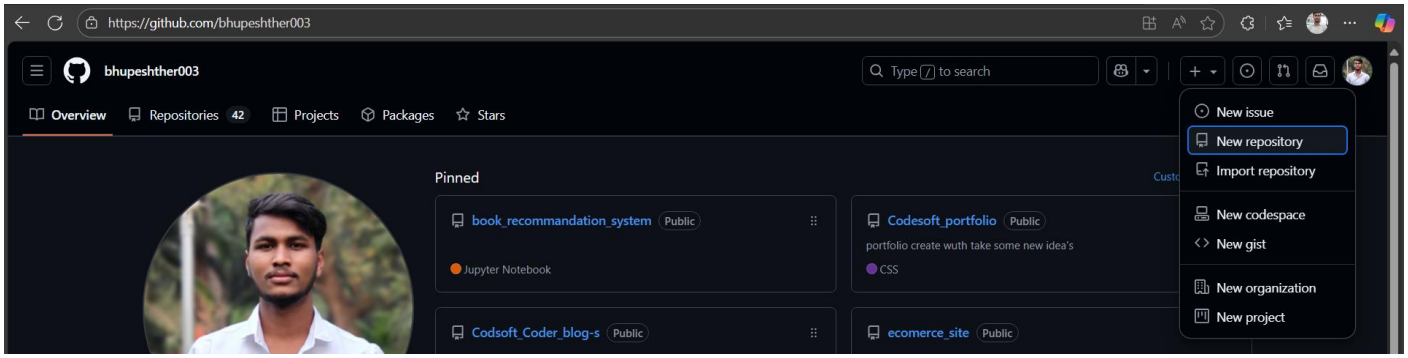


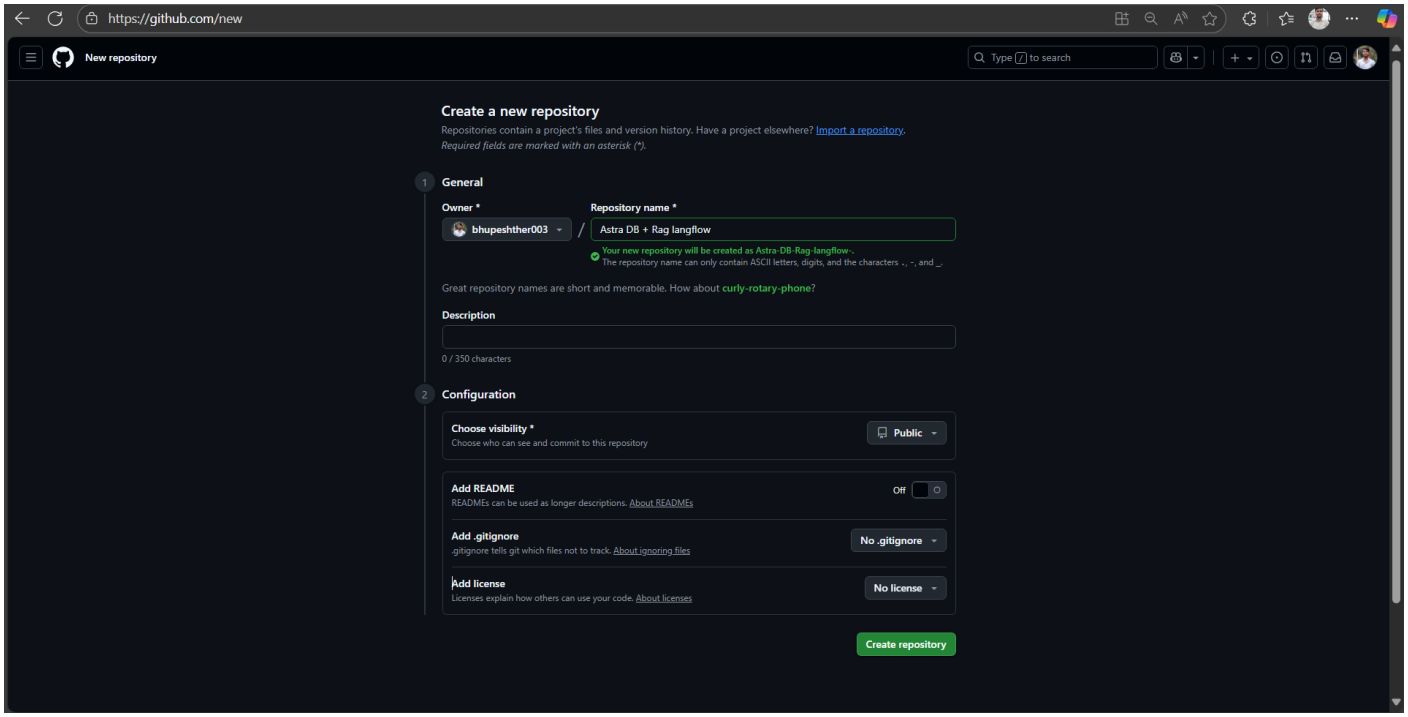


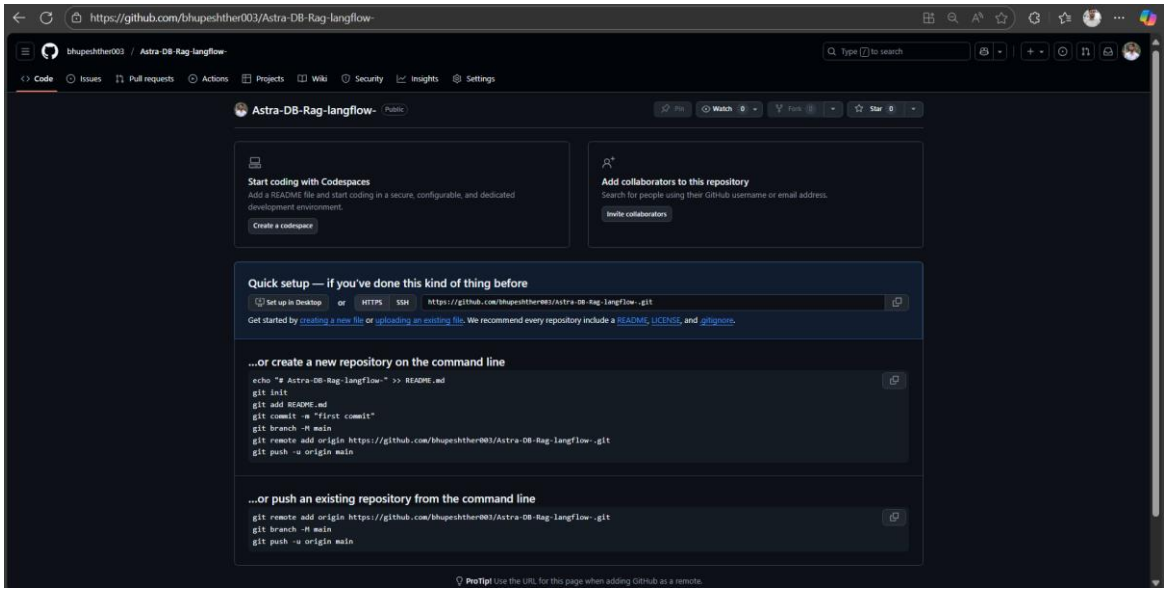
Now upload your code on github

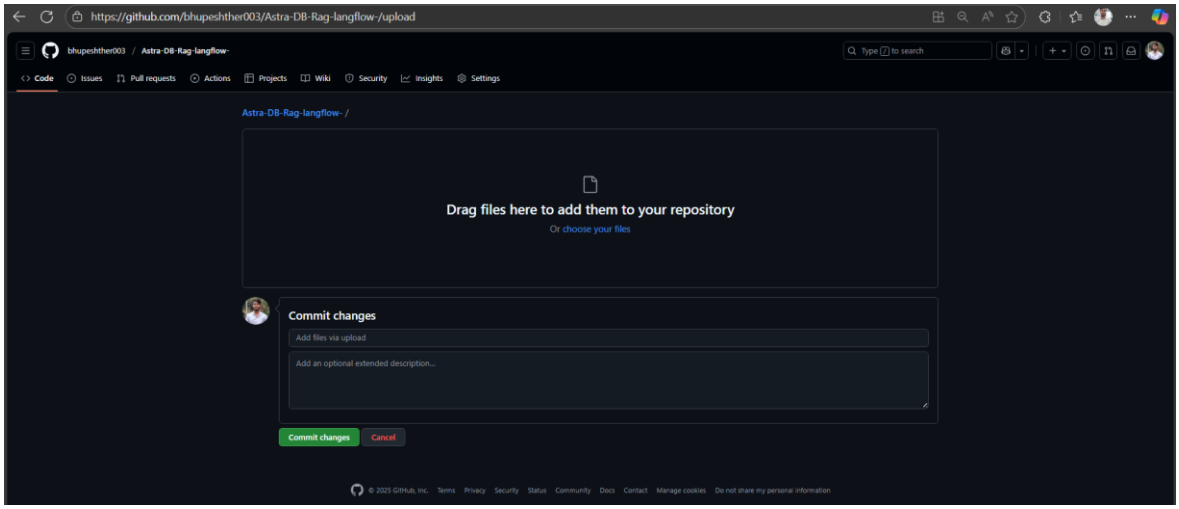
Open github

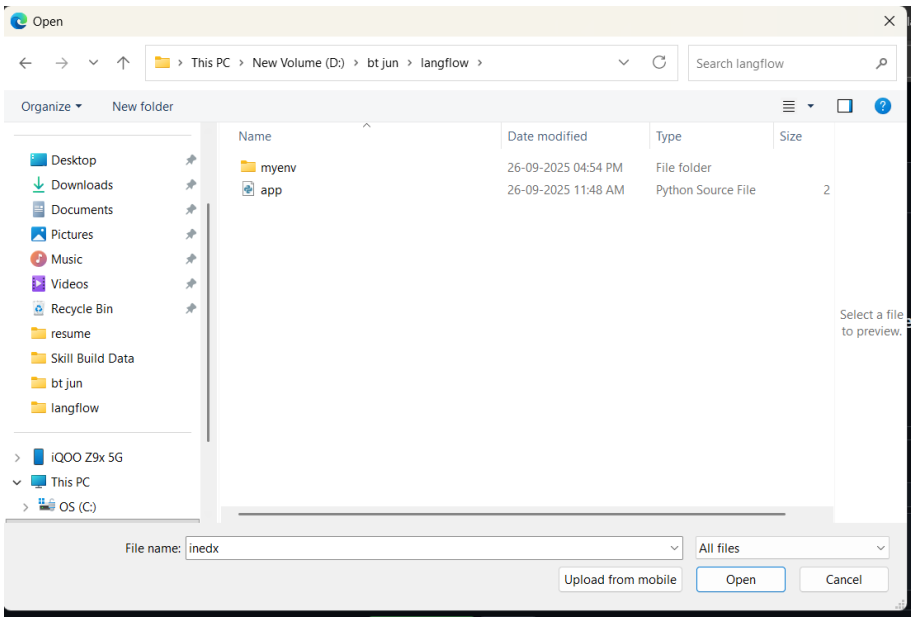

https://github.com/bhupeshther003
bhupeshther003
Type / to search
Overview
Repositories 42
Projects
Packages
Stars
Pinned
Customize your pins
book_recommandation_system Public
Jupyter Notebook
Codesoft_portfolio Public
portfolio create wuth take some new idea's
CSS
Codsoft_Coder_blog-s Public
for the codsoft intership creating small blog page
CSS
ecomerce_site Public
PHP
Education_Analysis Public
future_health_bt Public
HTML
Bhupesh Ther
bhupeshther003
Edit profile
1 follower · 0 following
MUMBAI
16 contributions in the last year
Contribution settings
2025
2024
2023
Sep Oct Nov Dec Jan Feb Mar Apr May Jun Jul Aug Sep
Mon
Wed
Fri
Learn how we count contributions
Less More

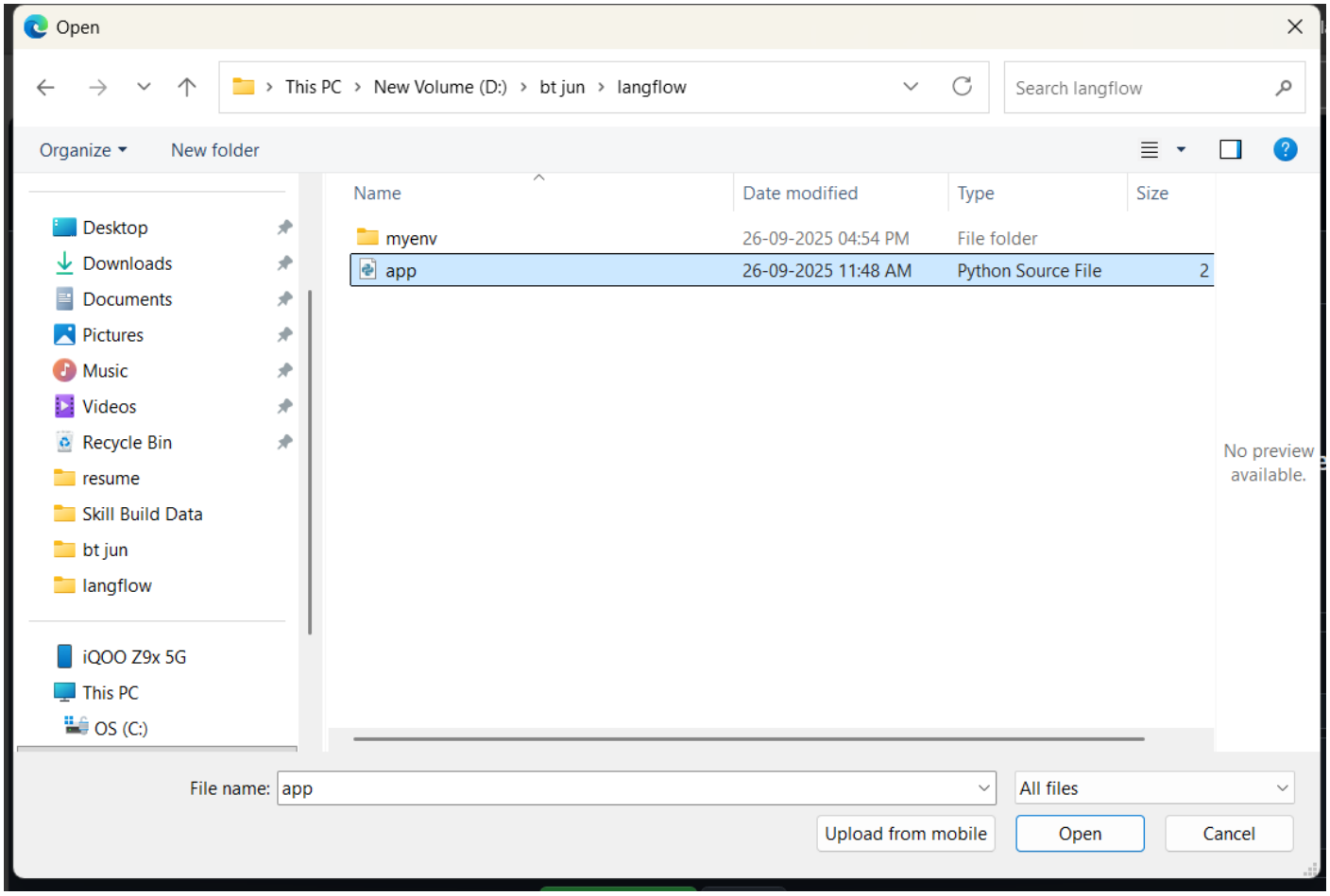

https://github.com/bhupeshther003
bhupeshther003
Type / to search
Overview
Repositories 42
Projects
Packages
Stars
New issue
New repository
Import repository
New codespace
New gist
New organization
New project
Pinned
book_recommandation_system Public
Jupyter Notebook
Codesoft_portfolio Public
portfolio create wuth take some new idea's
CSS
Codsoft_Coder_blog-s Public
ecomerce_site Public

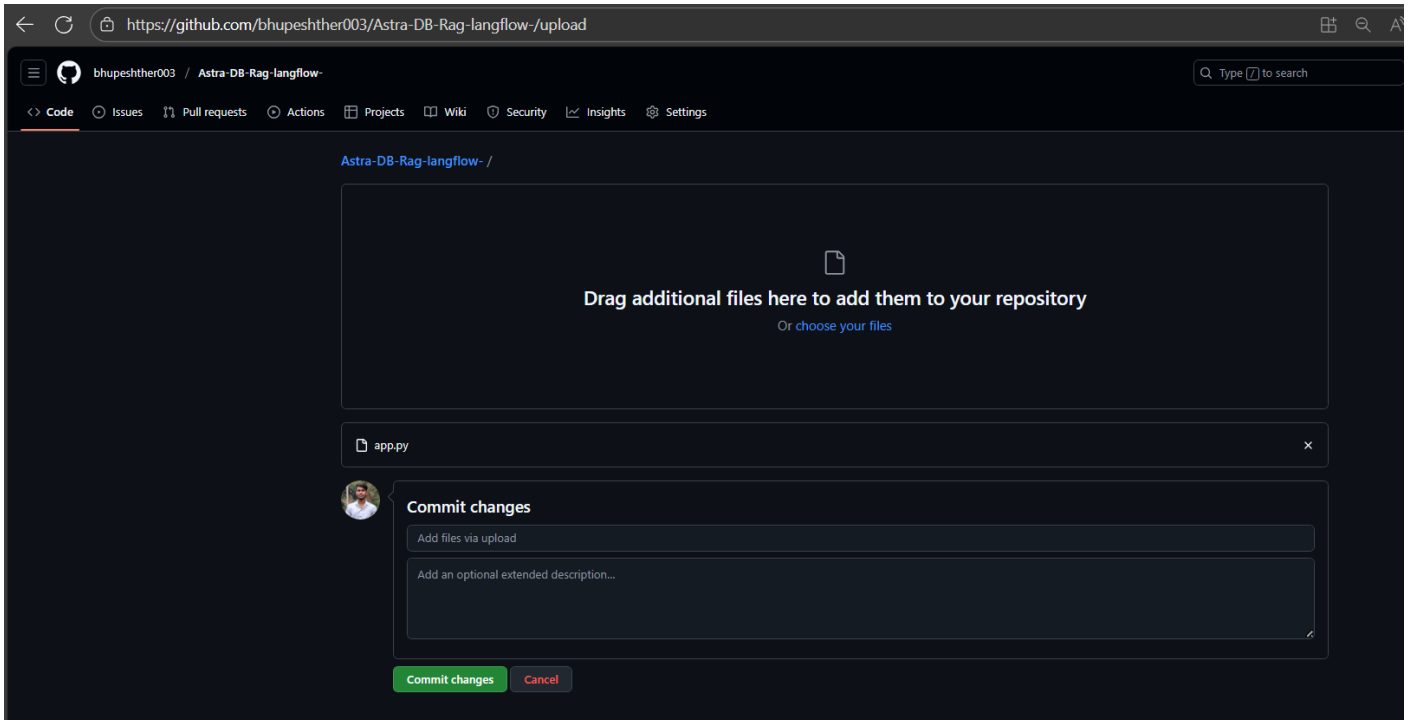

https://github.com/new
New repository
Type / to search
Create a new repository
Repositories contain a project's files and version history. Have a project elsewhere? Import a repository.
Required fields are marked with an asterisk (*).
1 General
Owner *
Repository name *
bhupeshther003
Astra DB + Rag langflow
Your new repository will be created as Astra-DB-Rag-langflow-.
The repository name can only contain ASCII letters, digits, and the characters ., -, and _.
Great repository names are short and memorable. How about curly-rotary-phone?
Description
0 / 350 characters
2 Configuration
Choose visibility *
Choose who can see and commit to this repository
Public
Add README
READMEs can be used as longer descriptions. About READMEs
Off
Add .gitignore
.gitignore tells git which files not to track. About ignoring files
No .gitignore
Add license
Licenses explain how others can use your code. About licenses
No license
Create repository

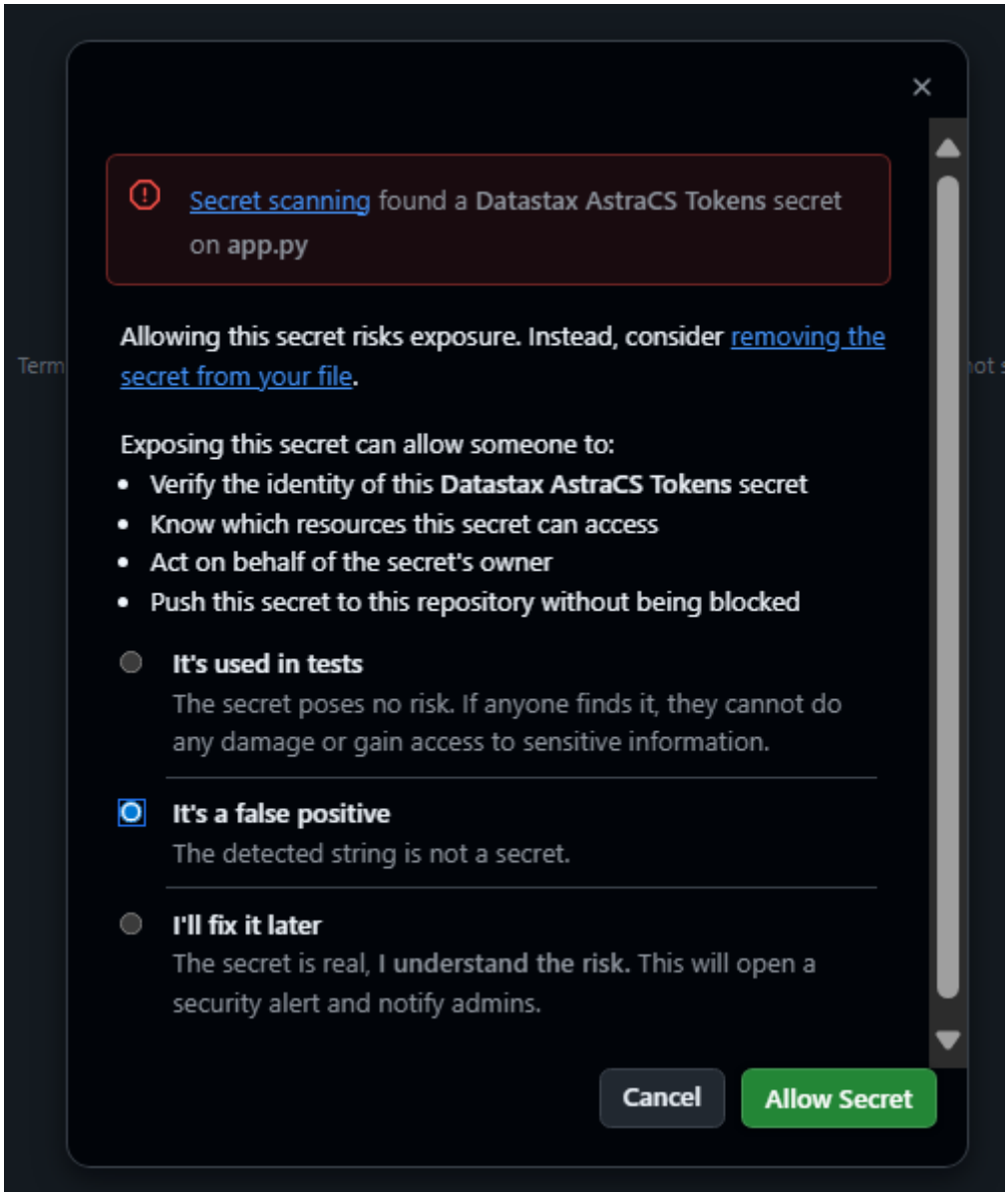

https://github.com/bhupeshther003/Astra-DB-Rag-langflow-
Astra-DB-Rag-langflow-
Start coding with Codespaces
Add collaborators to this repository
Quick setup — if you've done this kind of thing before
...or create a new repository on the command line
...or push an existing repository from the command line

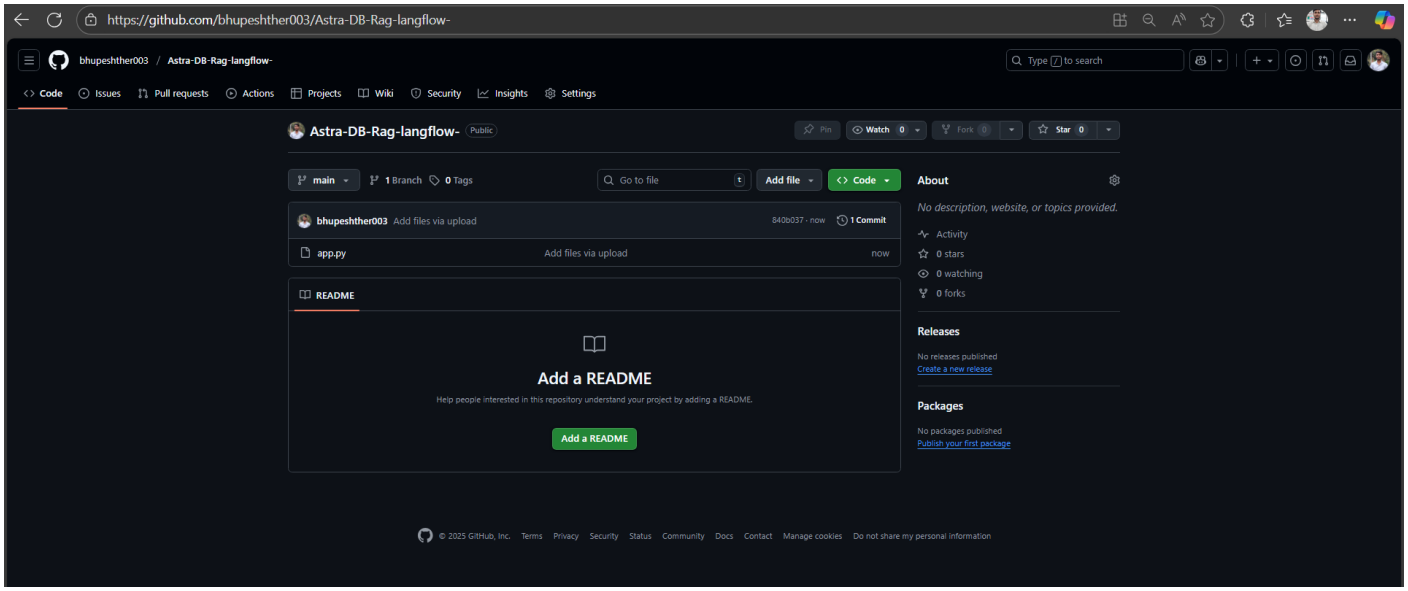

https://github.com/bhupeshther003/Astra-DB-Rag-langflow-/upload
Drag files here to add them to your repository
Commit changes
Commit changes
Cancel

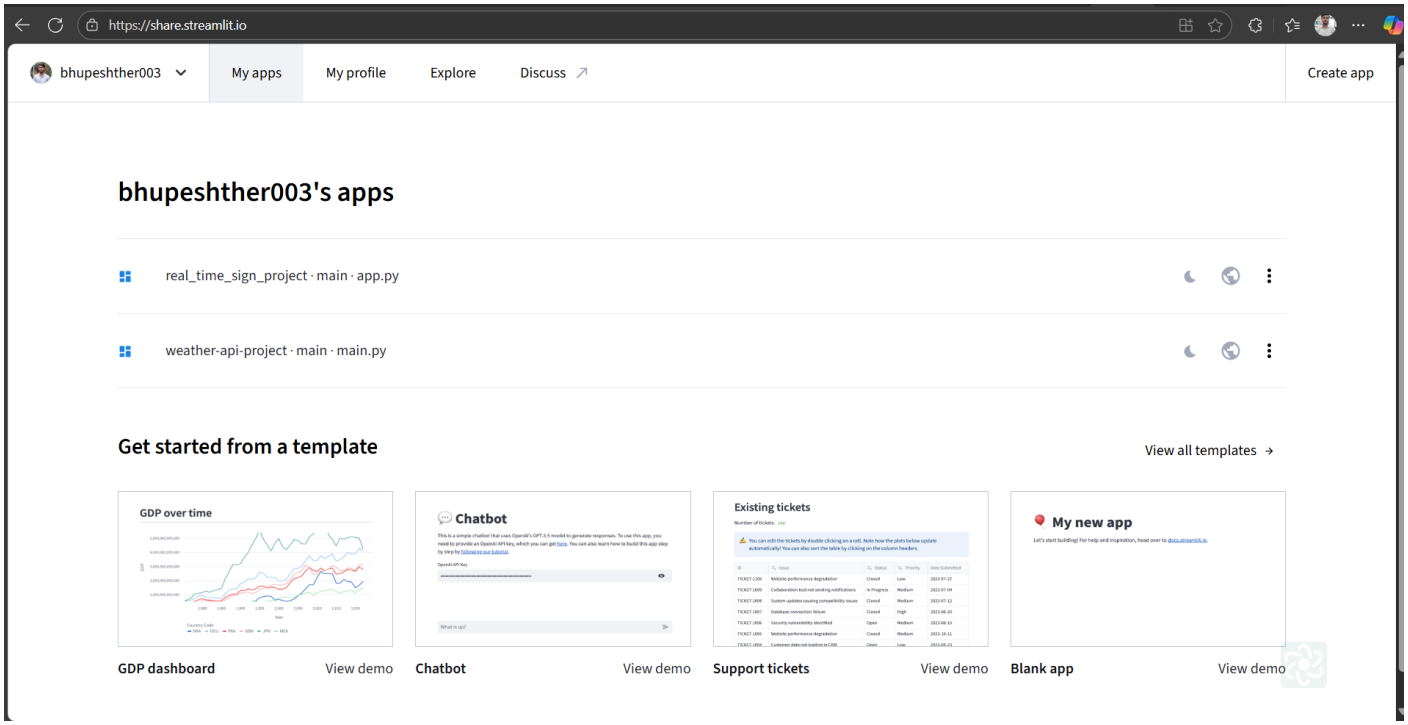

Open
This PC › New Volume (D:) › bt jun › langflow ›
Search langflow
Organize
New folder
Name
Date modified
Type
Size
myenv
26-09-2025 04:54 PM
File folder
app
26-09-2025 11:48 AM
Python Source File
Select a file to preview.
File name: inedx
All files
Upload from mobile
Open
Cancel


Open
This PC › New Volume (D:) › bt jun › langflow
Search langflow
Organize
New folder
Desktop
Downloads
Documents
Pictures
Music
Videos
Recycle Bin
resume
Skill Build Data
bt jun
langflow
iQOO Z9x 5G
This PC
OS (C:)
Name
Date modified
Type
Size
myenv
26-09-2025 04:54 PM
File folder
app
26-09-2025 11:48 AM
Python Source File
2
No preview available.
File name: app
All files
Upload from mobile
Open
Cancel


https://github.com/bhupeshther003/Astra-DB-Rag-langflow-/upload
bhupeshther003 / Astra-DB-Rag-langflow-
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
Astra-DB-Rag-langflow- /
Drag additional files here to add them to your repository
Or choose your files
app.py
Commit changes
Add files via upload
Add an optional extended description...
Commit changes
Cancel


Secret scanning found a Datastax AstraCS Tokens secret on app.py
Allowing this secret risks exposure. Instead, consider removing the secret from your file.
Exposing this secret can allow someone to:
Verify the identity of this Datastax AstraCS Tokens secret
Know which resources this secret can access
Act on behalf of the secret's owner
Push this secret to this repository without being blocked
It's used in tests
The secret poses no risk. If anyone finds it, they cannot do any damage or gain access to sensitive information.
It's a false positive
The detected string is not a secret.
I'll fix it later
The secret is real, I understand the risk. This will open a security alert and notify admins.
Cancel
Allow Secret


https://github.com/bhupeshther003/Astra-DB-Rag-langflow-
bhupeshther003 / Astra-DB-Rag-langflow-
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
Astra-DB-Rag-langflow-
Public
main
1 Branch
0 Tags
Go to file
Add file
Code
bhupeshther003 Add files via upload
1 Commit
app.py
Add files via upload
now
README
Add a README
Help people interested in this repository understand your project by adding a README.
Add a README
About
No description, website, or topics provided.
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package


https://share.streamlit.io
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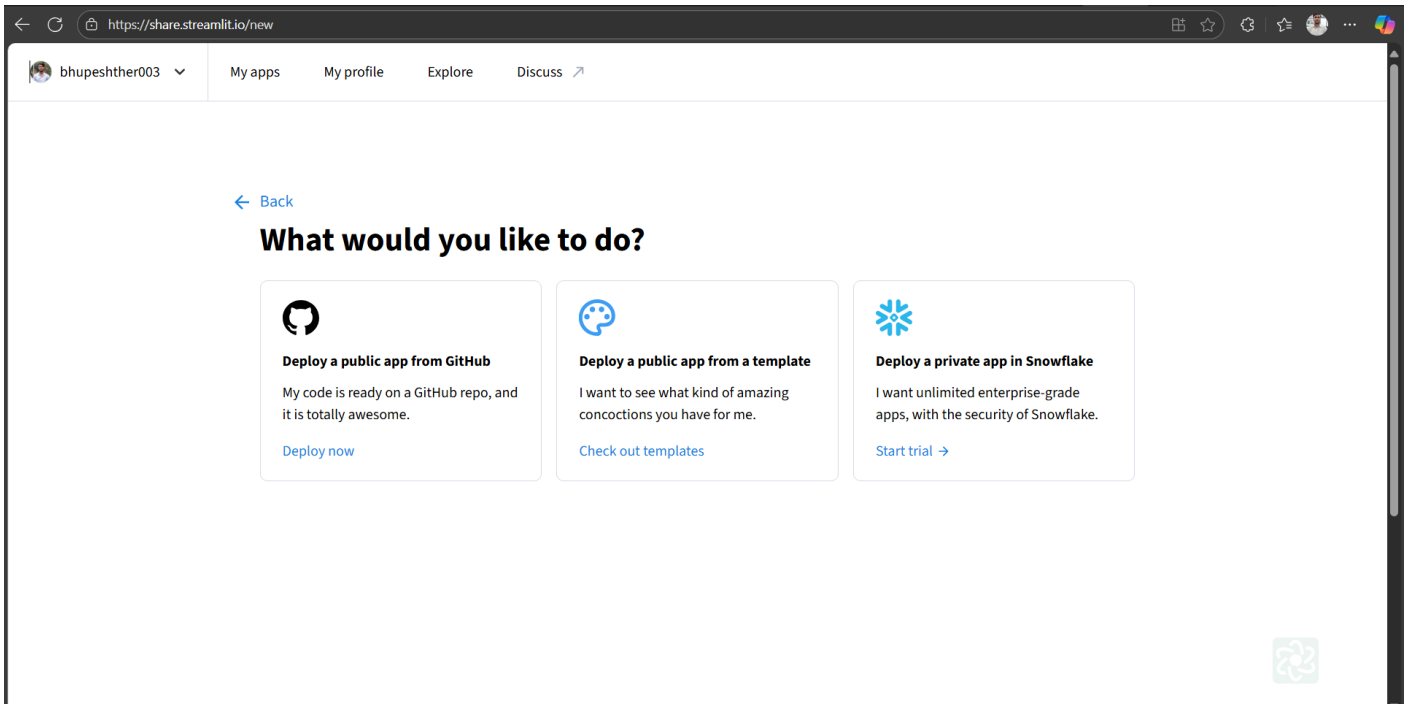
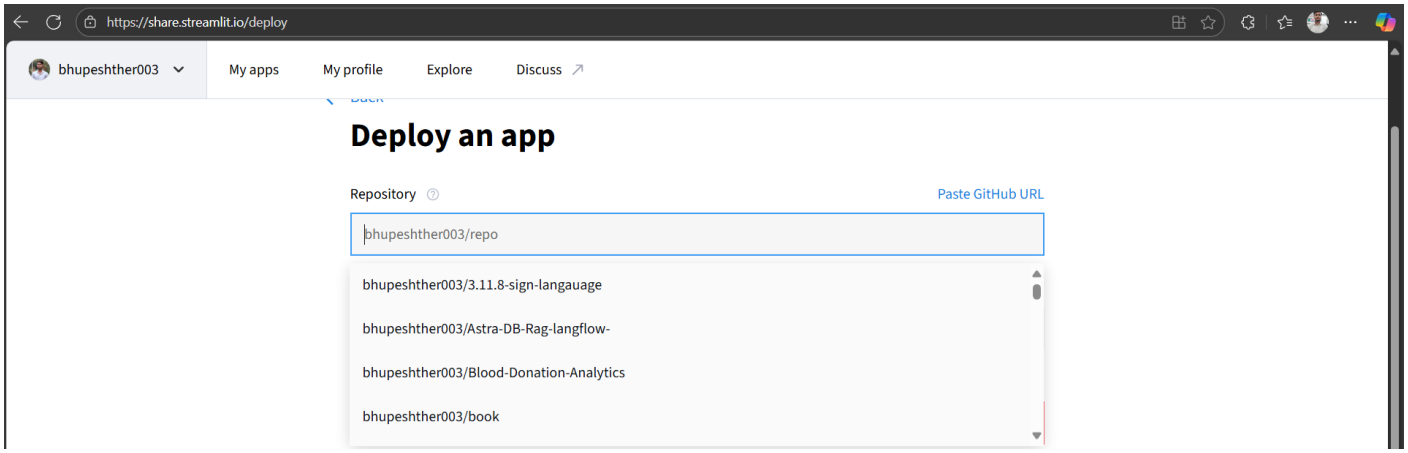
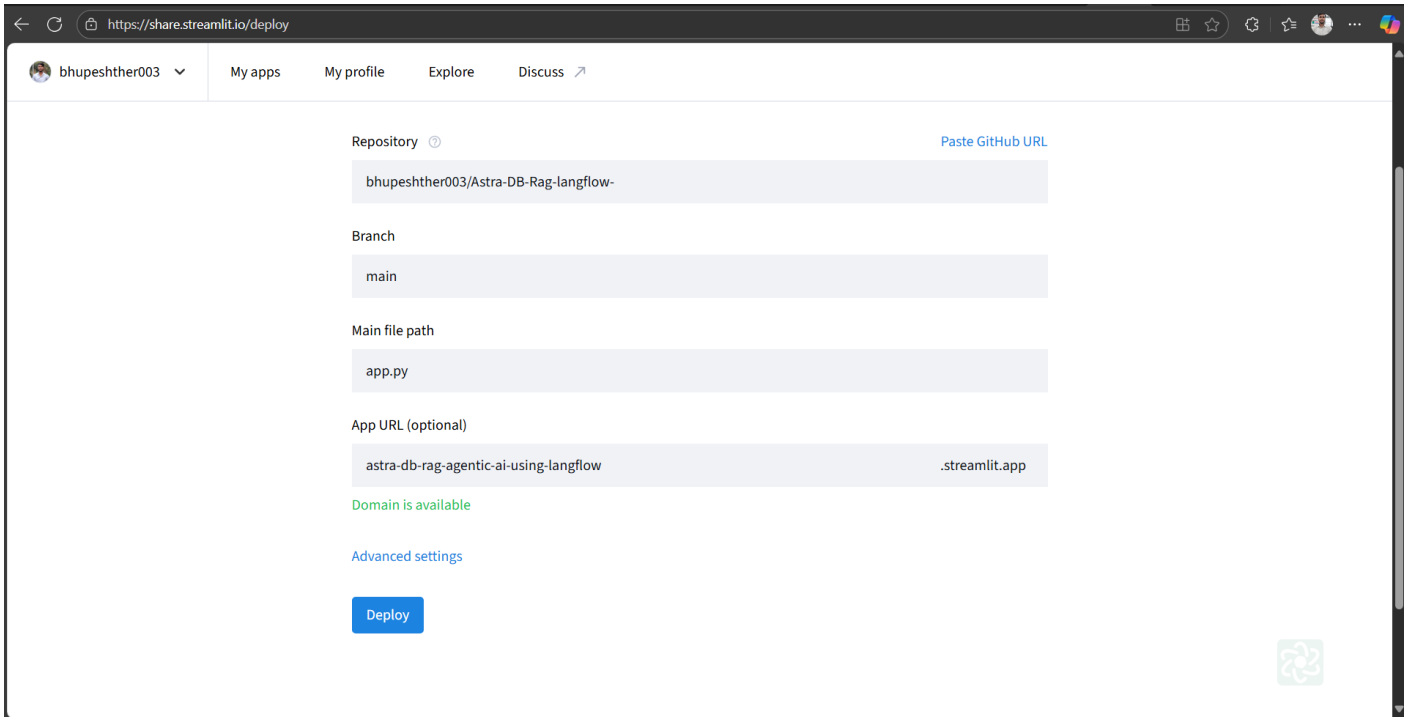
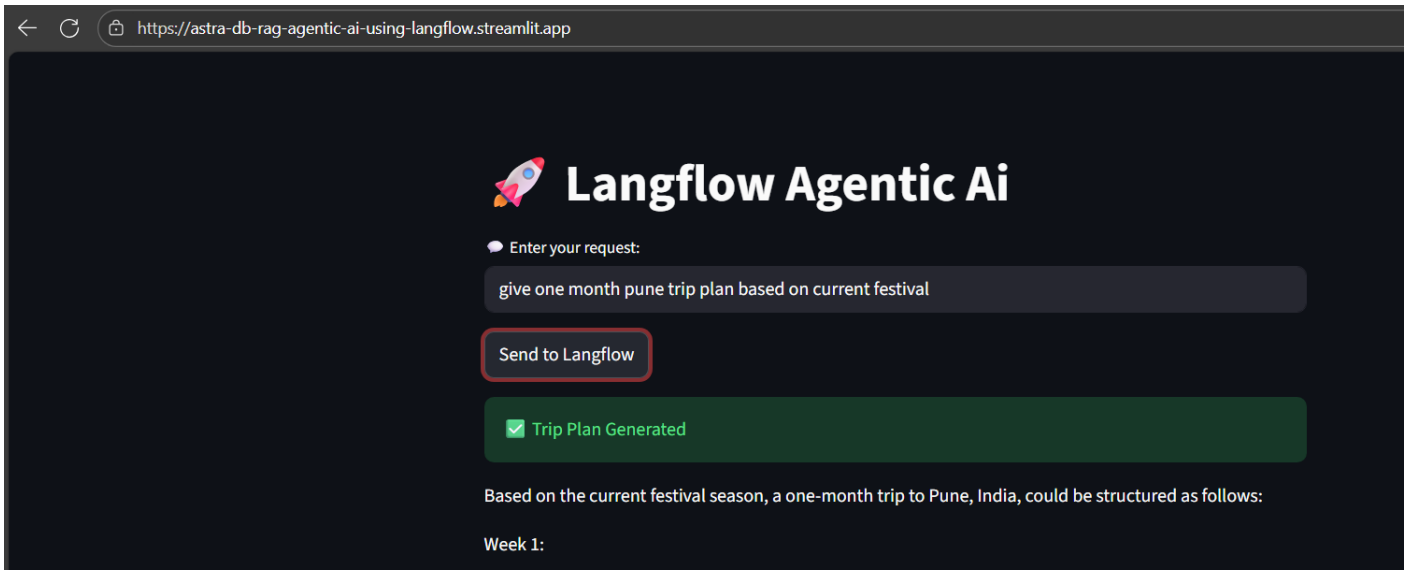
My apps
My profile
Explore
Discuss
Create app
bhupeshther003's apps
real_time_sign_project · main · app.py
weather-api-project · main · main.py
Get started from a template
View all templates
GDP over time
Chatbot
Existing tickets
My new app
GDP dashboard
View demo
Chatbot
View demo
Support tickets
View demo
Blank app
View demo


https://share.streamlit.io/new
bhupeshther003
My apps
My profile
Explore
Discuss
Back
What would you like to do?
Deploy a public app from GitHub
My code is ready on a GitHub repo, and it is totally awesome.
Deploy now
Deploy a public app from a template
I want to see what kind of amazing concoctions you have for me.
Check out templates
Deploy a private app in Snowflake
I want unlimited enterprise-grade apps, with the security of Snowflake.
Start trial


https://share.streamlit.io/deploy
bhupeshther003
My apps
My profile
Explore
Discuss
Deploy an app
Repository
Paste GitHub URL
bhupeshther003/repo
bhupeshther003/3.11.8-sign-langauage
bhupeshther003/Astra-DB-Rag-langflow-
bhupeshther003/Blood-Donation-Analytics
bhupeshther003/book


https://share.streamlit.io/deploy
bhupeshther003
My apps
My profile
Explore
Discuss
Repository
Paste GitHub URL
bhupeshther003/Astra-DB-Rag-langflow-
Branch
main
Main file path
app.py
App URL (optional)
astra-db-rag-agentic-ai-using-langflow
.streamlit.app
Domain is available
Advanced settings
Deploy


https://astra-db-rag-agentic-ai-using-langflow.streamlit.app
Langflow Agentic Ai
Enter your request:
give one month pune trip plan based on current festival
Send to Langflow
Trip Plan Generated
Based on the current festival season, a one-month trip to Pune, India, could be structured as follows:
Week 1: